DC/DC GEN3

BI-DIRECTIONAL DC/DC CONVERTER

Software and Programming Manual

for Power Electronics' staff
SW versions up to 2.0.0

Edition: July 2026
DCDCG3MSW02EEN Rev. E

ABOUT THIS DOCUMENT

PURPOSE

This manual contains important instructions for the setup and use of the Freemaq DC/DC GEN3 converters manufactured by Power Electronics. From now on, the term “product” will be used to refer to the Freemaq DC/DC GEN3 converters.

Throughout this document, in those tasks where access via Power Electronics remote management tool is indicated, the configuration and monitoring tool developed by Power Electronics must be used. The manuals for these tools can be found on the digital platform of Power Electronics.

Power Electronics reserves the right to modify product features.

TARGET AUDIENCE

This manual is intended for qualified personnel who have the current certification issued by Power Electronics who will install, configure and operate the product Freemaq DC/DC GEN3 converters manufactured by Power Electronics.

Only authorized technical personnel approved by Power Electronics may commission, configure and operate the products. The instructions assume that the installer is familiar with electrical installation rules and regulations.



NOTICE

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REVISIONS CONTROL

DATE (DD/MM/YYYY)	REVISION	DESCRIPTION
05 / 04 / 2024	A	First edition.
14 / 02 / 2025	B	Updated to software version 1.0.0. Add “Remote Control Functions” section.
16 / 04 / 2025	C	Updated to software version 2.0.0. Spanish version issued.
13 / 03 / 2026	D	Updated the sections “About this document” and “Important safety instructions”. Added notes related to error F447. New section added: “Best practices for Modbus communication”. Other improvements.
23 / 07 / 2026	E	Updated section “Important safety instructions”. Added F458 and F459 faults.

Note: changes in this manual, with respect to the previous version, are highlighted in blue.

The products and technical documentation are periodically updated. Power Electronics reserves the right to modify all or part of the contents of this manual without previous notice. The reproduction or distribution of the present manual is strictly forbidden, unless express authorization from Power Electronics.

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ACRONYMS

The terms commonly used in the documentation of Power Electronics' products of the *Solar & Storage* division are listed in the table below.

ACRONYM	MEANING
AASS	Auxiliary Services.
AC	Alternating Current.
AI	Analogue Input.
AO	Analogue Output.
BESS	Battery Energy Storage System.
BMS	Battery Manager Solution.
CP	Control Pilot.
CPU	Central Processing Unit.
dBA	Weighted decibel level.
DC	Direct Current.
DCL	Discharge Current Limit.
DI	Digital Input.
DSP	Digital Signal Processor.
DO	Digital Output.
EMS	Energy Management System.
FPGA	Programmable device (Field-Programmable Gate Array).
FRU	Field Replaceable Unit.
GFDI	Ground Fault Detector Interrupter.
GPRS	General Packet Radio Services, a data transmission system.
HVAC	Heating, Ventilation, and Air Conditioning.
HW	Hardware
IGBT	Insulated Gate Bipolar Transistor.
IMI	Insulation Monitoring Device.
IK	Degree of protection against mechanical impact.
IP	Degree of protection against solid/fluid objects.
IT	Grid system where the power supply is kept isolated and the electrical equipment system is grounded.
LOTOTO	Locking, tagging and verification procedure (Lock Out – Tag Out – Try Out).
LVR	Low Voltage Regulation.
MCB	Miniature Circuit Breaker.
MCCB	Moulded Case Circuit Breaker.
MPCS	Multi Power Conversion System.
MID	Measuring Instrument Directive.
MV	Medium Voltage. This term is used to refer to high voltage in general.
OF	Optical Fiber.

ACRONYM	MEANING
PCB	Printed Circuit Board.
PE	Ground connection (Protective Earth).
PI	Proportional and Integral.
PLC	Programmable Logic Controller.
POI	Point Of Interconnection.
PowerComms	Configuration and monitoring interface developed by Power Electronics.
PPE	Personal Protection Equipment.
PV	Photovoltaic energy.
RAL	Color code (Reichsausschuß für Lieferbedingungen und Gütesicherung).
RCD	Residual Current Device.
RCM	Residual Current Monitor.
SOC	State Of Charge – referred to battery.
SOH	State Of Health – referred to battery. It compares the actual state of the battery to its initial conditions. It is measured in percentage.
STO	Safe Torque Off.
SW	Software.
THD	Total Harmonic Distortion.
TN	Grid system where the power supply is grounded, and the electrical equipment system is brought to the same ground through the neutral connector.
TT	Grid system where both the power supply and the electrical devices are connected to the ground via separate connections.
UPS	Uninterruptible Power Supply.
Vac	Alternating current voltage.
Vdc	Direct current voltage.

IMPORTANT SAFETY INSTRUCTIONS

The user must read and understand all product-related instructions described below before performing any configuration task.

SAVE THESE INSTRUCTIONS

This manual contains important instructions for Freemaq DC/DC GEN3 converters that shall be followed during installation, operation, maintenance and troubleshooting of the converter.

Read carefully all documentation before manipulating the product and pay special attention to safety recommendations to maximize the performance of this product and to ensure its safe use and installation.

It is the responsibility of the user to follow the instructions provided in this manual and all product documentation, follow good electrical practices, and identify all warnings and recommendations before starting up, operating, or servicing the product.

Follow the health and safety regulations of each country / area and the good practice / recommendations of each industry. The most restrictive health and safety standards shall be followed in case of discrepancy. When adverse weather conditions exist, work restrictions shall be applied.

In this manual, safety messages are classified as follows:

 **WARNING**	Identifies potentially hazardous situations where dangerous voltage may be present, which if not avoided, could result in minor personal injury, serious injury or death. Be extremely careful and follow the instructions to avoid the risk of electrical shocks.
 **CAUTION**	Identifies potentially hazardous situations, which if not avoided, could result in product damage or minor or moderate personal injury. Read the message and follow the instructions carefully.
 **NOTICE**	Identifies important measures to take in order to prevent damages to the product.

Other symbols used in this manual for safety messages are:



Caution, risk of electric shock. Energy storage timed discharge. Wait for the indicated time to avoid electrical hazards.



WARNING

Keep all product documentation accessible throughout the life of the product for future reference.
Read all instructions and caution signs before the installation, operation, maintenance and troubleshooting of the product. Failure to comply with these warnings may result in serious electrical shock or death. Always pay attention to avoid possible accidents.

National/local restrictions for the product must be considered. The product must be used only for its intended application.
Power Electronics is not liable for damages resulting from improper use of the product or failure to comply with national or local regulations.

Do not apply voltage to a product that is damaged or with missing parts, even if the installation has been completed.

Ensure that the product is in the corresponding safety status before performing any action on it.

Follow the documentation of the product regarding the safety instructions to bring the product to the indicated safety status for the procedures that are going to be carried out.
Failure to comply with these warnings may result in serious electrical shock or death. Always pay attention to avoid possible accidents.
If the product is equipped with cutting devices, perform the maneuvers of these externally before accessing the product and/or performing any action.

Always use personal protective equipment (PPE) when opening a live door.

Always handle switches with dry hands and wear personal protective equipment (PPE). The required PPE must be in accordance with, but not limited to, what is indicated in this document. The safety instructions (risk assessment and method statements) of the user must specify what specific PPE is required and when and how it is to be used in accordance with the specific safety plan for the work to be performed.

All work on the product must be carried out by at least two operators. The product must never be operated by only one person. There must be a plant supervisor on the work team, both for technical and health and safety reasons.

Do not use cables with damaged insulation. Do not subject cables to abrasion, excessive stress, overloading or clamping.

If the product stops due to a loss of power, do not perform any work on it.
The automatic restart function may be active.



5 min

The product contains capacitors on the DC sides. The discharge time of the capacitors is 5 minutes. Wait until the capacitors have been discharged before carrying out any action.

In the case that any intervention or maneuver on the product is allowed, in the presence of voltage and with the appropriate PPE, consider that there is a potential risk of dropping a tool if work with metal tools inside the product is carried out. This can cause sparks or short circuits leading to an explosion.

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WARNING

In the cabinets where high voltage is present, the installer must add a warning signal “DANGER – HIGH VOLTAGE”. [1]

After the disconnection of the input voltage, the product may remain warm for several minutes. Be aware of the hot surfaces to avoid possible injury or burns.

Adjustments to the charging algorithm controls must be made according to the recommendations provided by the battery manufacturer for the particular battery type.



NOTICE

CYBERSECURITY DISCLAIMER

This product is designed to be connected to and to communicate information and data via a network interface. Access to the system is restricted to those employees who legitimately need it for reasons of maintenance and/or updating of the system.
It is sole responsibility of the user for providing and continuously ensuring a secure connection between the product and user network or any other network (as the case may be). User must establish and maintain any appropriate measures (such as but not limited to the installation of firewalls, application of authentication measures, encryption of data, installation of antivirus programs, etc.) to protect the product, the network, its system and the interface against any kind of security breaches, unauthorized access, interference, intrusion, leakage and/or theft of data or information.
Power Electronics and its affiliates are not liable for damages and/or losses related to such security breaches, any unauthorized access, interference, intrusion, leakage and/or theft of data or information.

[1] Medium or high voltage depending on the regulations of the country of installation.

1. USER ACCESS LEVEL

1

Software parameters are divided in groups, depending on their functionality:

- **Visualization and status parameters** cannot be modified and display useful information about the converter, such as: current and voltage, amount of energy generated by the converter, fault messages and many more. Please refer to section "Visualization parameters" for a detailed list of all parameters and their attributes.
- **Configuration parameters** allow setting and modifying the converter operating conditions and general parameters such as the current date and user password. They are explained in section, "Programming parameters".
- Parameters from the **service group** are reserved for Power Electronics' staff. They allow setting parameters for commissioning and configuring all the functionalities of the converter. They are explained in section, "Service group parameters".

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NOTICE

This document includes all login roles. Throughout this document, information highlighted in coral is that reserved to roles **PE Staff and Developer.**

The user modes are **Customer** and **Qualified Customer**, being the latter named in the document as Administrator (Admin, from now on). To prevent unintentional modifications that could lead to malfunctioning and even damage to the converter, access to parameters is password protected and managed by Power Electronics.

The available **user modes** are explained in parameter [G.1.3-User]. For further information refer to its description at "Programming parameters" section. By default, the user mode is Customer. If assistance is required, please contact Power Electronics.

2. STATUS MESSAGES, FAULTS, WARNINGS AND EVENTS

2

The software provides, at all times, information about the DC/DC converter status, warnings and faults:

- **STATUS** messages provide information about the current state of the converter (e.g. ready to start, operating…).
- A **FAULT** is a condition that, if satisfied, will force the converter to stop (e.g. an overtemperature condition has been detected).
- **WARNING** messages provide useful information about what is going on in the converter (e.g. derating, heating is active…). They should be checked periodically, and the corresponding actions should be taken (if required).

Also, the converter includes four status LEDs in the Main Control Cabinet door which show its operation status:

	COLOR	LED ON	LED OFF
BESS voltage	White	BESS voltage presence	No BESS voltage presence
Run	Green	Converter ON	Converter OFF
Fault	Red	A fault has occurred	No faults
PV voltage	White	PV voltage presence	No PV voltage presence

2.1. Faults

CODE	FAULT	DESCRIPTION
F31	Auto diagnosis	A fault has occurred during the auto-diagnosis process.
F40	Internal overtemperature	The internal temperature of the control PCB has exceeded the fixed threshold of 85°C.
F43	Emergency stop	The emergency stop pushbutton has been pressed.
F44	Drive-Select	One or more of the parameters of the drive select ([SG10: Drive select]) is out of range.
F46	Data Fault	There is a parameter out of range or the EEPROM memory has returned an error when trying to perform a writing or reading operation.
F47	Watchdog uP	The product has detected that one of the uP running tasks has not been executed on time and therefore a continuous blocking may have occurred in one of the tasks that does not allow to execute the following instructions. To get out of this uP study, execute a reset and this fault will appear once the reset is finished.
F51	PPC Comms	There has been a loss of continuous communication for longer than the time specified in the parameter "Comm fault timeout" of the subgroup [G6.5-PPC].
F56	Remote emergency stop	A remote emergency shutdown has occurred via Power Electronics' remote management tool or via Modbus.
F64	Invalid MAC	The product has detected an invalid MAC address.
F65	Inductance overcurrent	The inductance current has exceeded the protection threshold.
F83	Multiple desaturation	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F90	Tª IGBTS HF	Fault caused when the temperature of any IGBT exceeds 100ºC or when the temperature sensor of any IGBT fails.
F93	Main thermal LCL	If the inductance core overheats, it will open a bimetallic NC contact causing the fault.
F95	Source PCB	The internal/external power supply of the card does not provide the correct voltage.
F96	Tª PCB module	The internal temperature of the module PCB has exceeded 80ºC (non-configurable threshold).
F98	Idc derivation BESS	The power stage has detected a shunt in Idc on the BESS side.
F99	Idc derivation PV	The power stage has detected a shunt in Idc on the PV side.
F103	IGBTs temperature	The power stage fan is not working properly. It can be due to different causes: there is a contact fault on the PWM cable pins, the fan has no power supply or the fan has stopped working. Fault occurs when a power stage IGBT exceeds the threshold set in the Drive Select.
F104	PCB temperature	The temperature measured on the module control PCB has exceeded the configured maximum allowable value.
F126	Tº deviation IGBTs	This fault appears if a power stage exceeds the "IGBT derating" (maximum difference between the highest IGBT temperature and the average of the highest IGBT temperatures per power stage) and also exceeds the set temperature at the starting point for IGBT temperature derating.
F161	LOTO activated	The LOTO DC is activated and the product must be disconnected.
F200	SPD main circuit	The feedback of the overvoltage protector and its protection fuse of the main battery circuit will be monitored. This signal, by seriating both devices by means of NC contacts, reaches SW notifying the fault in the absence of this signal.
F202	Aux Transformer overtemp	The auxiliary services transformer core has overheated; it will open a bimetallic NC contact causing the fault.
F210	Comms failure uP-FPGA	The FPGA is not receiving correct frames from the microcontroller.
F220	PT100 Inductance	One, or several, PT100 of the inductance is not connected in the product.
F222	NTC admission temperature	Admission temperature NTC resistor is not connected.
F224	Inductance overtemperature	The maximum limit set for the inductance temperature has been exceeded.
F225	Inductance critical overtemperature	The maximum critical temperature threshold for the inductance has been exceeded.
F227	Reserved	The product has triggered a fault with a code that is not reported among the expected faults.

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CODE	FAULT	DESCRIPTION
F255	BESS side is open	The battery disconnection fault is triggered when performing soft charge, both in conventional and emergency mode.
F401	Unbalance bus PV	There is an unbalance between the high and low side of the DC bus at PV.
F402	Unbalance bus BESS	There is an unbalance between the high and low side of the DC bus at BESS.
F403	Current sense PV	The product detects a full-scale reading on the PV current sensor when the product is not switching.
F404	Current sense BESS	The product detects a full-scale reading on the BESS current sensor when the product is not switching.
F405	Current sensing L1	The inductance current sensor L1 is not measuring correctly, there is an offset error.
F406	Current sensing L2	The inductance current sensor L2 is not measuring correctly, there is an offset error.
F407	Current sensing L3	The inductance current sensor L3 is not measuring correctly, there is an offset error.
F408	Input overvoltage PV	The full bus voltage measured by the power stage has exceeded the configured threshold value.
F409	Input overvoltage BESS	The full bus voltage measured by the power stage has exceeded the configured threshold value.
F410	Overvoltage emergency PV	The full PV bus voltage measured by the power stage exceeds 1600V.
F411	Overvoltage emerg. BESS	The full BESS bus voltage measured by the power stage exceeds 1600V.
F412	Undervoltage emergenc PV	The measured PV full bus voltage is lower than the configured value.
F413	Undervoltage BESS	The measured BESS full bus voltage is lower than the configured value.
F414	Undervoltage PV	The full bus voltage of PV is less than 830V.
F415	Undervoltage emerge BESS	The full bus voltage of BESS is less than 830V.
F416	Overvoltage halfbus PV	It is detected that the half-bus voltage measured by the PV power stage exceeds a marked threshold.
F417	Ov.volt halfbus PV emer	It is detected that the half-bus voltage measured by the PV power stage exceeds a marked threshold.
F418	Overvoltage halfbus BESS	It is detected that the half-bus voltage measured by the BESS power stage exceeds a marked threshold.
F419	OvVolt halfbus BESS emer	It is detected that the half-bus voltage measured by the BESS power stage exceeds a marked threshold.
F420	Desat . IGBT Q1 bridge 1	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F421	Desat . IGBT Q2 bridge 1	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F422	Desat . IGBT Q3 bridge 1	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F423	Desat . IGBT Q4 bridge 1	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F424	Desat . IGBT Q5 bridge 1	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F425	Desat . IGBT Q6 bridge 1	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F426	Desat . IGBT Q7 bridge 1	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F427	Desat . IGBT Q8 bridge 1	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F428	Desat . IGBT Q1 bridge 2	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F429	Desat . IGBT Q2 bridge 2	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F430	Desat . IGBT Q3 bridge 2	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F431	Desat . IGBT Q4 bridge 2	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F432	Desat . IGBT Q5 bridge 2	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.

CODE	FAULT	DESCRIPTION
F433	Desat . IGBT Q6 bridge 2	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F434	Desat . IGBT Q7 bridge 2	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F435	Desat . IGBT Q8 bridge 2	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F436	Desat . IGBT Q1 bridge 3	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F437	Desat . IGBT Q2 bridge 3	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F438	Desat . IGBT Q3 bridge 3	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F439	Desat . IGBT Q4 bridge 3	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F440	Desat . IGBT Q5 bridge 3	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F441	Desat . IGBT Q6 bridge 3	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F442	Desat . IGBT Q7 bridge 3	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F443	Desat . IGBT Q8 bridge 3	Fault caused because there is desaturation in one or several IGBT drivers, depending on the fault.
F444	Timeout softcharge PV	The voltage on the PV bus has not reached the set value within the defined time.
F445	Timeout softcharge BESS	The voltage on the BESS bus has not reached the set value within the defined time.
F446	Fb breaker PV	The PV DC switch disconnector is not closing or opening correctly when required. This fault appears when there is a discrepancy between the activation command and the received feedback.
F447	Fb breaker BESS	The BESS DC switch disconnector is not closing or opening correctly when required. This fault appears when there is a discrepancy between the activation command and the feedback received. **Note**: this fault also appears when the selector is disabled without stopping the DC/DC converter (i.e., the DC/DC is ON).
F448	External fault	The customer has performed an emergency stop via the terminals wired to the “external fault”.
F449	Fans dirty zone	There is a problem in the communications between the central control board (E01) and the dirty zone fans.
F450	Fans clean zone	There is a problem in the communications between the central control board (E01) and the clean zone fans.
F451	Spring loaded PV	The spring from the PV DC switch disconnector could not be loaded
F452	Spring loaded BESS	The spring from the BESS DC switch disconnector could not be loaded
F454	Discharge	The product fails to download within 5 minutes.
F456	Bus DC unbalance PV	There is a balance between the high and low side of the DC bus in PV.
F457	Bus DC unbalance BESS	There is a balance between the high and low side of the DC bus in BESS.
F458	Overcurrent inductances	The inductance current has exceeded the restrictive protection threshold.
F459	SW incompatible uP-FPGA	Fault caused by an incompatibility between the uP and FPGA versions. After product startup and the establishment of communication between both devices, the version numbers are exchanged and, if they are not compatible with each other, fault F459 is generated.

EN

2.2. Warnings

CODE	WARNING	DESCRIPTION
W5	No start conditions	The configured startup conditions are not met.
W7	P limits	The P limit has been reached.
W9	I limits CCL/DCL	The current limit has been reached.
W12	Derating thermal IGBTs	A derating has been activated due to overheating of IGBTs.
W13	Intake thermal derating	A derating has been activated due to high admission temperature or due to too low voltage.
W150	No SD	No SD card has been detected.
W152	PPC comms	No response received from SCADA/PPC.
W165	Fan motor status	Anomaly in the fan. The fan's own record must be consulted.
W166	No response from central PCB Temperature sensor	There is a problem with the temperature sensor of the central board.
W170	SPD main circuit	The main circuit SPD has triggered. It is necessary to replace the cartridge.
W171	No RTC	Communication with the RTC is failing.
W178	FTP open	The FTP session is open; no information is being written on the SD card.
W180	SD problem	Displayed when the SD card has a data problem.
W181	SPD auxiliary circuit	The auxiliary circuit SPD has triggered. It is necessary to replace the cartridge.
W186	Supply low	The input voltage of the main control board is not enough.
W401	Stop conditions	The configured automatic shutdown conditions have been reached.
W402	High voltage PV (limit)	The upper voltage limit is exceeded in PV.
W403	High voltage BESS (limit)	The upper voltage limit is exceeded in BESS.
W404	Low voltage PV (limit)	The lower voltage limit is exceeded in PV.
W405	Low voltage BESS (limit)	The lower voltage limit is exceeded in BESS.
W406	Thermal derating current (maximum current)	The converter limits power because the maximum current rating, depending on the ambient temperature, has been reached.
W407	Blackbox downloading	A Blackbox downloading is running.

3. VISUALIZATION AND STATUS PARAMETERS

3

Visualization and status parameters cannot be modified and display useful information about the converter, such as: current and voltage, amount of energy generated by the inverter, fault messages and many more.

3.1. Group SV1: Internal visualization

This group allows the visualization of the internal state of the system. It shows the status of the equipment, the starting conditions, the status of the switch disconnectors, the status of the digital inputs and outputs, temperatures of the electronic boards that make up the equipment and the temperatures of the inlet, as well as the limits and power of the equipment.

EN

3.1.1. Subgroup SV1.1: Status

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.1.1-Current fault = NO FAULT</td><td>NONE</td><td>Shows the current fault code in case there is any.</td></tr>
<tr><td>SV1.1.2-Current warning = NO WRN</td><td>NONE</td><td>Shows the current warning code in case there is any.</td></tr>
<tr><td>SV1.1.3-Current status = Idle</td><td>NONE</td><td>Shows the current status of the converter.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>OPTION</th><th>STATUS</th></tr>
<tr><td>0</td><td>Idle</td><td>5</td><td>Stop</td></tr>
<tr><td>1</td><td>Ready</td><td>6</td><td>Discharge</td></tr>
<tr><td>2</td><td>Softcharge</td><td>7</td><td>Polarization</td></tr>
<tr><td>3</td><td>Run</td><td>8</td><td>Auto diagnosis</td></tr>
<tr><td>4</td><td>Wait</td><td>9</td><td>Fault</td></tr>
</table></td></tr>
<tr><td>SV1.1.4-Softcharge status = Idle</td><td>NONE</td><td>Shows the current status of the softcharge.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Idle</td><td>The equipment is ready for initializing the softcharge</td></tr>
<tr><td>1</td><td>SC BESS</td><td>The BESS softcharge is initializing</td></tr>
<tr><td>2</td><td>SC BESS done</td><td>The BESS softcharge is done</td></tr>
<tr><td>3</td><td>SC PV</td><td>The PV softcharge is initializing</td></tr>
<tr><td>4</td><td>SC PV done</td><td>The PV softcharge is done</td></tr>
</table></td></tr>
<tr><td>SV1.1.5-Power Source = 0.00V</td><td>V</td><td>Shows the voltage measured on the supply and auxiliary transformer.</td></tr>
<tr><td>SV1.1.6-Working mode = Buck</td><td>NONE</td><td>Shows the working mode of the equipment.
<table>
<tr><th>OPTION</th><th>STATUS</th></tr>
<tr><td>0</td><td>Buck</td></tr>
<tr><td>1</td><td>Buck Boost</td></tr>
<tr><td>2</td><td>Boost</td></tr>
</table></td></tr>
</table>

3.1.2. Subgroup SV1.2: Start conditions

SCREEN	UNITS	DESCRIPTION
SV1.2.1-Start conditions all = No	NONE	Shows if there are start conditions.
SV1.2.2-Start conditions PV = No	NONE	Shows if there are PV voltage start conditions.
SV1.2.3 – Start conditions BESS = No	NONE	Shows if there are BESS voltage start conditions.
SV1.2.4-Start con spring BESS = No	NONE	Shows if the device has loaded spring breaker BESS.
SV1.2.5-Start con spring PV = No	NONE	Shows if the device has loaded spring breaker PV.

3.1.3. Subgroup SV1.3: Drive select

SV1.3.1: General

SCREEN	UNITS	DESCRIPTION
SV1.3.1.1-DCDC type = [SG10.1.4-Converter type]	NONE	Shows the application off the converter. OPTION: Standalone — Shows when the converter is controlled by a standalone. OPTION: Inverter — Shows when the converter is controlled by inverter.
SV1.3.1.2-Number of modules = 1	NONE	Shows the number of modules. **Note:** from version 2.0.0, this parameter is not available (it has changed to internal variable).
SV1.3.1.4-Max DC current = [SG10.1.6 SG10.1.6-Idc max] A	A	Shows the maximum DC current.
SV1.3.1.5-Nominal P = [SG10.1.7-Nominal P] kW	kW	Shows the nominal power.
SV1.3.1.7-Hardware version = [SG10.1.9 -Hardware version]	NONE	Shows the hardware versions (hexadecimal value).
SV1.3.1.8-Enable inductances = 0XX	NONE	Shows the enable inductances. OPTION 0 — STATUS L1 OPTION 1 — STATUS L2 OPTION 2 — STATUS L3

SV1.3.2: Protections

SCREEN	UNITS	DESCRIPTION
SV1.3.2.1-Overcurr prot = [SG10.2.1-Overcurrent threshol] A	A	Shows the overcurrent protection threshold.
SV1.3.2.2-Threshold temperature IGBTS = [SG10.2.2-IGBTs threshol Temperature] ºC	ºC	Shows the IGBTs temperature threshold. **Note:** from version 2.0.0, this parameter is not available.

SV1.3.3: Limits

SCREEN	UNITS	DESCRIPTION
SV1.3.3.1-Max I limit = [SG10.3.1-Max I limit] A	A	It allows to limit the current of the equipment.
SV1.3.3.2-Max P limit = [SG10.3.2-Max P limit] %	%	Shows the maximum limit of P with respect to the nominal value. They also limit the maximum value that the limit configuration parameters can take. This means that if a user attempts to set a value greater than the drive select limit it will be rejected and overridden by the drive select value.

3.1.4. Subgroup SV1.4: Equipment

SV1.4.1: DC equipment

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.4.1.1-BESS switch status = Unknown</td><td>NONE</td><td>Shows the status of BESS switch disconnector (status of its digital output).
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Open</td><td>Shows when the BESS switch disconnector is open.</td></tr>
<tr><td>Closed</td><td>Shows when the BESS switch disconnector is closed.</td></tr>
<tr><td>Unknown</td><td>Shows when the BESS switch disconnector is unknown.</td></tr>
</table></td></tr>
<tr><td>SV1.4.1.2-PV switch status = Unknown</td><td>NONE</td><td>Shows the status of PV switch disconnector (status of its digital output).
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Open</td><td>Shows when the BESS switch disconnector is open.</td></tr>
<tr><td>Closed</td><td>Shows when the BESS switch disconnector is closed.</td></tr>
<tr><td>Unknown</td><td>Shows when the BESS switch disconnector is unknown.</td></tr>
</table></td></tr>
<tr><td>SV1.4.1.3-BESS spring status = Unknown</td><td>NONE</td><td>Shows the status of the BESS spring switch disconnector (status of its digital output. The BESS switch disconnector of the equipment is motorized. It has an associated spring action which allows the switch disconnector to close. In this way, when the switch disconnector is commanded to close, the spring must be charged first. Once the switch disconnector is closed, the spring is unloaded. This parameter displays the status of the spring: charged, discharged or unknown. The unknown status is displayed when it is not possible to determine the status of the spring.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Discharged</td><td>Shows when the BESS spring switch disconnector is discharged.</td></tr>
<tr><td>Charged</td><td>Shows when the BESS spring switch disconnector is charged.</td></tr>
<tr><td>Unknown</td><td>Shows when the BESS spring switch disconnector is unknown.</td></tr>
</table></td></tr>
<tr><td>SV1.4.1.4-PV spring status = Unknown</td><td>NONE</td><td>Shows the status of the PV spring switch disconnector (status of its digital output. The PV switch disconnector of the equipment is motorized. It has an associated spring action which allows the switch disconnector to close. In this way, when the switch disconnector is commanded to close, the spring must be charged first. Once the switch disconnector is closed, the spring is unloaded. This parameter displays the status of the spring: charged, discharged or unknown. The unknown status is displayed when it is not possible to determine the status of the spring.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Discharged</td><td>Shows when the PV spring switch disconnector is discharged.</td></tr>
<tr><td>Charged</td><td>Shows when the PV spring switch disconnector is charged.</td></tr>
<tr><td>Unknown</td><td>Shows when the PV spring switch disconnector is unknown.</td></tr>
</table></td></tr>
</table>

EN

3.1.5. Subgroup SV1.5: Digital I/O status

This group shows information about the status of the digital inputs and outputs.

SV1.5.1: uC central

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.1.1-Status UP RJ5 = 000000</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ5 of central control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DO Spring load BESS</td><td>Command to load the BESS spring switch disconnector.</td></tr>
<tr><td>1</td><td>DO Open breaker BESS</td><td>Command to open the BESS spring switch disconnector.</td></tr>
<tr><td>2</td><td>DO Close breaker BESS</td><td>Command to close the BESS spring switch disconnector.</td></tr>
<tr><td>3</td><td>DI feedback close breaker BESS</td><td>Feedback that the BESS switch disconnector has been closed.</td></tr>
<tr><td>4</td><td>DI feedback opened breaker BESS</td><td>Feedback that the BESS switch disconnector has been opened.</td></tr>
<tr><td>5</td><td>DI feedback spring loaded BESS</td><td>Feedback that the BESS spring switch disconnector has been loaded.</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.2-Status UP RJ6 = 000000</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ6 of central control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>1</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>2</td><td>DO Enable softcharge</td><td>Command to enable the softcharge</td></tr>
<tr><td>3</td><td>DI feedback softcharge BESS</td><td>Feedback that the softcharge has been enabled. NOTE: not works in sc target</td></tr>
<tr><td>4</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>5</td><td>Reserved</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.3-Status UP RJ7 = 000000</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ7 of central control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DO Spring load PV</td><td>Command to load the PV spring switch disconnector.</td></tr>
<tr><td>1</td><td>DO Open breaker PV</td><td>Command to open the PV spring switch disconnector.</td></tr>
<tr><td>2</td><td>DO Close breaker PV</td><td>Command to close the PV spring switch disconnector.</td></tr>
<tr><td>3</td><td>DI feedback close breaker PV</td><td>Feedback that the PV switch disconnector has been closed.</td></tr>
<tr><td>4</td><td>DI feedback opened breaker PV</td><td>Feedback that the PV switch disconnector has been opened.</td></tr>
<tr><td>5</td><td>DI feedback spring loaded PV</td><td>Feedback that the PV spring switch disconnector has been loaded.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.1.4-Status UP RJ8 = 000000</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ8 of central control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>Reserved.</td></tr>
<tr><td>1</td><td>Reserved</td><td>Reserved.</td></tr>
<tr><td>2</td><td>Reserved</td><td>Reserved.</td></tr>
<tr><td>3</td><td>Reserved</td><td>Reserved.</td></tr>
<tr><td>4</td><td>Reserved</td><td>Reserved.</td></tr>
<tr><td>5</td><td>Reserved</td><td>Reserved.</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.5-Status UP RJ9 = 000000</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ9 of central control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>Reserved.</td></tr>
<tr><td>1</td><td>Reserved</td><td>Reserved.</td></tr>
<tr><td>2</td><td>Reserved</td><td>Reserved.</td></tr>
<tr><td>3</td><td>Main inductance thermal</td><td>Feedback of main inductance thermal.</td></tr>
<tr><td>4</td><td>DI Feedback SPD auxiliary circuit</td><td>Feedback of Surge Protector Device auxiliary circuit.</td></tr>
<tr><td>5</td><td>DI Feedback SPD DC</td><td>Feedback of Surge Protector Device DC-</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.6-Status UP RJ10 = 000000</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ10 of central control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DO LED run</td><td>Command to activate the LED run.</td></tr>
<tr><td>1</td><td>DO Relay general fault</td><td>Command to relay the general fault.</td></tr>
<tr><td>2</td><td>DO LED fault / warning</td><td>Command to activate the LED fault / warning.</td></tr>
<tr><td>3</td><td>DI Selector key</td><td>Shows the position of selector key.</td></tr>
<tr><td>4</td><td>DI Emergency button</td><td>Shows the position of emergency button.</td></tr>
<tr><td>5</td><td>DI LOTO selector</td><td>Shows the position of LOTO selector.</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.7-Status UP RJ11 = 000000</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ11 of central control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>1</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>2</td><td>Internal fault</td><td>Command to activate output when happened internal fault</td></tr>
<tr><td>3</td><td>DI Feedback therm aux transformer</td><td>Feedback of auxiliar thermal transformer, activate when over temperature happen in the transformer</td></tr>
<tr><td>4</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>5</td><td>DI External fault</td><td>An external fault has occurred.</td></tr>
</table></td></tr>
</table>

EN

3.1.6. Subgroup SV1.6: Temperature

SCREEN	UNITS	DESCRIPTION
SV1.6.1-Internal Tº = 35 ºC	ºC	Shows the current internal temperature of PCB.
SV1.6.2-Max IGBT Tº = 0.00 ºC	ºC	Shows the maximum internal temperature of IGBTs.
SV1.6.3-Admission Tº = 0 ºC	ºC	Shows the temperature measured by the sensor located at the air inlet in the cabinet.
SV1.6.4-Internal humidity = 50 %	%	Shows the central control board humidity.
SV1.6.5-Module Tª = 0.00ºC	ºC	Shows the current internal temperature of PCB module.
SV1.6.6-Ind PT100 (1) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the LC filter.
SV1.6.7-Ind PT100 (2) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the LC filter.
SV1.6.8-Ind PT100 (3) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the LC filter.

3.1.7. Subgroup SV1.7: Internal registers

Note: These parameters are displayed if [G1.3-User] ≥ Power electronics staff.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.7.1-FPGA reg TF vis = 000000</td><td>NONE</td><td>Visualization of bit status of header frame TF (up to FPGA).
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>5</td><td>Msg ID</td><td>Frame ID</td></tr>
<tr><td>4</td><td>Blackbox</td><td>Blackbox bit request flag</td></tr>
<tr><td>3</td><td>Start</td><td>Start signal to get into RUN status</td></tr>
<tr><td>2</td><td>Softcharge</td><td>Softcharge request flag</td></tr>
<tr><td>1</td><td>Enable</td><td>Enable flag for module</td></tr>
<tr><td>0</td><td>Remote upgrade</td><td>Remote upgrade request flag</td></tr>
</table></td></tr>
<tr><td>SV1.7.2-FPGA reg TFV vis = 00000000</td><td>NONE</td><td>Visualization of bit status of header frame TFV (FPGA to up).
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>5</td><td>Msg ID</td><td>Frame ID</td></tr>
<tr><td>4</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>3</td><td>Enable SC board</td><td>Enable the softcharge board</td></tr>
<tr><td>2</td><td>Emergency stop</td><td>Emergency stop flag</td></tr>
<tr><td>1</td><td>Close PV breaker</td><td>Close PV breaker flag</td></tr>
<tr><td>0</td><td>Close BESS breaker</td><td>Close BESS breaker flag</td></tr>
</table></td></tr>
<tr><td>SV1.7.3-uP-FPGA status = 0000000000000000</td><td>NONE</td><td>Visualization of internal status bits of FPGA sended to Up.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>9</td><td>Autodiagnosis</td><td>Shows that there is a self-test to be performed</td></tr>
<tr><td>8</td><td>Polarization</td><td>Allows to perform the polarization maneuver.</td></tr>
<tr><td>7</td><td>External soft charge</td><td>Shows that the external soft charge mode is enabled</td></tr>
<tr><td>6</td><td>Emergency soft charge</td><td>Shows that the emergency soft charge mode is enabled</td></tr>
<tr><td>5</td><td>Standalone/ Inverter</td><td>0 - Standalone; 1-Inverter</td></tr>
<tr><td>4</td><td>Emergency stop</td><td>Emergency stop flag</td></tr>
<tr><td>3</td><td>BESS breaker state</td><td>0 – Open state; 1 – Close state</td></tr>
<tr><td>2</td><td>PV breaker state</td><td>0 – Open state; 1 – Close state</td></tr>
<tr><td>1</td><td>Discharge</td><td>Discharge request state</td></tr>
<tr><td>0</td><td>Reset</td><td>Reset fault request flag</td></tr>
</table></td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.7.4-uP status internal = 0000000000000000</td><td>NONE</td><td>Visualization of internal status bits of Up.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Selector</td><td>Selector status 1 ->Enable, 0 -> Disable</td></tr>
<tr><td>1</td><td>Loto DC</td><td>Loto DC status 1 ->Enable, 0 -> Disable</td></tr>
<tr><td>2</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>3</td><td>Special fault</td><td>Current fault is a special fault</td></tr>
<tr><td>4</td><td>Pending softcharge</td><td>Indicates that a soft load maneuver is pending.</td></tr>
<tr><td>5</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>6</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>7</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>8</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>9</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>10</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>11</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>12</td><td>IGBTs off</td><td>Status of IGBTS Enable -> 0 Disable -> 1</td></tr>
<tr><td>13</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>14</td><td>Reserved</td><td>Reserved</td></tr>
<tr><td>15</td><td>Fault</td><td>Device is in fault</td></tr>
</table></td></tr>
</table>

3.2. Group SV2: Measures

This group shows the information corresponding to the electrical measurements (power, voltage and current) of the equipment.

3.2.1. Subgroup SV2.1: PV bus

SCREEN	UNITS	DESCRIPTION
SV2.1.1-Vpv + int= 0 V	V	Shows the internal positive voltage in the PV side.
SV2.1.2-Vpv - int= 0 V	V	Shows the internal negative voltage in the PV side.
SV2.1.3-Vpv int= 0 V	V	Shows the internal voltage in the PV side.
SV2.1.4-PV bus voltage exter = 0.0 V	V	Shows the external voltage in the PV side.
SV2.1.5-PV power = 0.0 kW	kW	Shows the power in the PV side.
SV2.1.6-PV current = 0 A	A	Shows the current in the PV side.

3.2.2. Subgroup SV2.2: BESS bus

SCREEN	UNITS	DESCRIPTION
SV2.2.1-Vbess + in t= 0 V	V	Shows the internal positive voltage in the BESS side.
SV2.2.2-Vbess – int = 0 V	V	Shows the internal negative voltage in the BESS side.
SV2.2.3-Vbess int = 0 V	V	Shows the internal voltage in the BESS side.
SV2.2.4-BESS bus voltage exter = 0.0 V	V	Shows the external voltage in the BESS side.
SV2.2.5-BESS power = 0.0 kW	kW	Shows the power in the BESS side.
SV2.2.6-BESS current = 0 A	A	Shows the current in the BESS side.

3.2.3. Subgroup SV2.3: Inductances

SCREEN	UNITS	DESCRIPTION
SV2.3.1-I inductance 1 = 0	A	Shows the current inductance 1.
SV2.3.2-I inductance 2 = 0	A	Shows the current inductance 2.
SV2.3.3-I inductance 3 = 0	A	Shows the current inductance 3.

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3.3. Group SV3: Limits

3.3.1. Subgroup SV3.1: Deratings

SCREEN	UNITS	DESCRIPTION
SV3.1.1-Max power available = 1200 kW	kW	Shows the maximum power available.
SV3.1.2-Max I available = 1200 A	A	Shows the maximum current available.

3.4. Group SV4: Algorithms

3.4.1. Subgroup SV4.1-Auto diagnosis

Note: Role PE Staff or higher is required to modify or display these parameters.

SCREEN	UNITS	DESCRIPTION
SV4.1.1-Auto diagnosis status = Idle	Idle Softcharge Switching PV Switching BESS Discharge Fault	Visualization of the internal status of the self-diagnostics.

3.5. Group SV5: Counters

3.5.1. Subgroup SV5.1: Charged energy

SCREEN	UNITS	DESCRIPTION
SV5.1.1-Daily charged = 0.0	kWh	Shows the daily charged of energy obtained.
SV5.1.2-Yesterday charged = 0.0	kWh	Shows the charged of energy obtained yesterday.
SV5.1.3-Month charged = 0.0	kWh	Shows the monthly charged of energy obtained.
SV5.1.4-Last month charged = 0.0	kWh	Shows the charged of energy obtained in the last month.
SV5.1.5-Total charged = 0.0	kWh	Shows the total charged of energy obtained.

3.5.2. Subgroup SV5.2: Discharged energy

SCREEN	UNITS	DESCRIPTION
SV5.2.1-Daily discharged = 0.0	kWh	Shows the daily discharged of energy realized.
SV5.2.2-Yesterday discharged = 0.0	kWh	Shows the discharged of energy realized yesterday.
SV5.2.3-Month discharged = 0.0	kWh	Shows the monthly discharged of energy realized.
SV5.2.4-Last month discharged = 0.0	kWh	Shows the discharged of energy realized in the last month.
SV5.2.5-Total discharged = 0.0	kWh	Shows the total discharged of energy realized.

3.5.3. Subgroup SV5.3: Run time counters

SCREEN	UNITS	DESCRIPTION
SV5.3.1-Total run time years = 0	NONE	Shows the total run time years counter.
SV5.3.2-Total run time days = 0	NONE	Shows the total run time days counter.
SV5.3.3-Total run time hours = 0	NONE	Shows the total run time hours counter.
SV5.3.4-Partial run time year = 0	NONE	Shows the partial run time years counter.
SV5.3.5-Partial run time days = 0	NONE	Shows the partial run time days counter.
SV5.3.6-Partial run time hour = 0	NONE	Shows the partial run time hours counter.

3.5.4. Subgroup SV5.4: Stop emergency

SCREEN	UNITS	DESCRIPTION
SV5.4.1-Num stop emergency = 0	NONE	Shows the total times of emergency stop has been activated.
SV5.4.2-Num remote stop emer. = 0	NONE	Shows the total times of remote emergency stop has been activated.
SV5.4.3-Num extern stop emer. = 0	NONE	Shows the total times of external emergency stop has been activated.

3.5.5. Subgroup SV5.5 Breakers opened

SCREEN	UNITS	DESCRIPTION
SV5.5.1-Counter breaker open = 0	NONE	Shows the total times of switch disconnectors have been opened.

3.6. Group SV6: Comms visualization

This group shows information about the communications status in the converter components. Converters are composed of several components or devices that communicate with each other to provide proper operation.

3.6.1. Subgroup SV6.2: Central Board

This subgroup allows to visualize the internal communications status at central board.

Note: Role PE Staff or higher is required to modify or display these parameters.

SV6.2.2: Ethernet

SCREEN	UNITS	DESCRIPTION
SV6.2.2.1-Reset counters = 0	NONE	Shows the reset counters.
SV6.2.2.2-Total Processed = 0.00 %	%	Shows the process.
SV6.2.2.3-Ethernet Socket1 Status= Offline	NONE	Shows the status of current connection socket.
SV6.2.2.4-Socket1 IP = 0.0.0.0	NONE	Shows the IP connected to this socket.
SV6.2.2.5-Ethernet Socket1 Number = 0	NONE	Shows the number of sockets in use.
SV6.2.2.6-Ethernet Socket1 Port = 0	NONE	Shows the number of ports in use.
SV6.2.2.7-Ethernet Socket1 Packets = 0	NONE	Shows number of packets received. **Note:** This parameter is not used currently.
SV6.2.2.8-Ethernet Socket1 Pack Process= 0	NONE	Shows number of packet process. **Note:** This parameter is not used currently.
SV6.2.2.9-Ethernet Socket2 Status = Offline	NONE	Shows status of current connection socket.
SV6.2.2.10-Ethernet Socket2 Ip 1 = 0	NONE	Shows the IP connected to this socket.
SV6.2.2.11-Ethernet Socket2 Number = 0	NONE	Shows the number of sockets in use
SV6.2.2.12-Ethernet Socket2 Port = 0	NONE	Shows the number of ports in use.
SV6.2.2.13- Ethernet Socket2 Packets = 0	NONE	Shows number of packets received. **Note:** This parameter is not used currently.
SV6.2.2.14- Ethernet Socket2 Pack Process= 0	NONE	Shows number of packet process. **Note:** This parameter is not used currently.
SV6.2.2.15- Ethernet Socket3 Status = Offline	NONE	Shows status of current connection socket.
SV6.2.2.16- Ethernet Socket3 Ip 1 = 0.0.0.0	NONE	Shows the IP connected to this socket.
SV6.2.2.17- Ethernet Socket3 Number = 0	NONE	Shows the number of sockets in use
SV6.2.2.18- Ethernet Socket3 Port = 0	NONE	Shows the number of ports in use.
SV6.2.2.19- Ethernet Socket3 Packets = 0	NONE	Shows number of packets received. **Note:** This parameter is not used currently.
SV6.2.2.20- Ethernet Socket3 Pack Process = 0	NONE	Shows number of packet process. **Note:** This parameter is not used currently.

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SCREEN	UNITS	DESCRIPTION
SV6.2.2.21-Ethernet Socket4 Status= Offline	NONE	Shows status of current connection socket.
SV6.2.2.22-Ethernet Socket4 Ip 1= 0	NONE	Shows the IP connected to this socket.
SV6.2.2.23-Ethernet Socket4 Number = 0	NONE	Shows the number of sockets in use
SV6.2.2.24-Ethernet Socket4 Port= 0	NONE	Shows the number of ports in use.
SV6.2.2.25-Ethernet Socket4 Packets = 0	NONE	Shows number of packets received. **Note:** This parameter is not used currently.
SV6.2.2.26-Ethernet Socket4 Pack Process= 0	NONE	Shows number of packet process. **Note:** This parameter is not used currently.
SV6.2.2.27- Ethernet Socket5 Status= Offline	NONE	Shows status of current connection socket.
SV6.2.2.28- Ethernet Socket5 Ip 1 = 0	NONE	Shows the IP connected to this socket.
SV6.2.2.29- Ethernet Socket5 Number = 0	NONE	Shows the number of sockets in use
SV6.2.2.30- Ethernet Socket5 Port = 0	NONE	Shows the number of ports in use.
SV6.2.2.31- Ethernet Socket5 Packets = 0	NONE	Shows number of packets received. **Note:** This parameter is not used currently.
SV6.2.2.32- Ethernet Socket5 Pack Process = 0	NONE	Shows number of packet process. **Note:** This parameter is not used currently.
SV6.2.2.33-Ethernet Socket6 Status = Offline	NONE	Shows status of current connection socket.
SV6.2.2.34-Ethernet Socket6 Ip 1 = 0	NONE	Shows the IP connected to this socket.
SV6.2.2.35-Ethernet Socket6 Number = 0	NONE	Shows the number of sockets in use
SV6.2.2.36- Ethernet Socket6 Port = 0	NONE	Shows the number of ports in use.
SV6.2.2.37- Ethernet Socket6 Packets = 0	NONE	Shows number of packets received. **Note:** This parameter is not used currently.
SV6.2.2.38-Ethernet Socket6 Pack Process = 0	NONE	Shows number of packet process. **Note:** This parameter is not used currently.
SV6.2.2.39-Socket7 Status = Offline	NONE	Shows status of current connection socket.
SV6.2.2.40-Socket7 IP = 0	NONE	Shows the IP connected to this socket.
SV6.2.2.41-Socket7 Number = 0	NONE	Shows the number of sockets in use
SV6.2.2.42-Socket7 Port = 0	NONE	Shows the number of ports in use.
SV6.2.2.43-Socket7 Packets = 0	NONE	Shows number of packets received. **Note:** This parameter is not used currently.
SV6.2.2.44-Socket7 Processed = 0	NONE	Shows number of packet process. **Note:** This parameter is not used currently.

SV6.2.4: uP-FPGA

This group shows the status of the communication between the uP and the central FPGA of the central control board.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV6.2.4.1-Successful frames = 0</td><td>NONE</td><td>Indicates the number of frames that have been successfully processed between uP FPGA and isolation monitoring device.</td></tr>
<tr><td>SV6.2.4.2-Error Frames = 0</td><td>NONE</td><td>Indicates the number of erroneous frames between uP FPGA and isolation monitoring device.</td></tr>
<tr><td>SV6.2.4.3-Status = Disabled</td><td>NONE</td><td>Shows the FPGA Communications status.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Disabled</td><td>There is no communication between the uP and the central FPGA.</td></tr>
<tr><td>Comms OK</td><td>Communication between the uP and the host FPGA is correct.</td></tr>
<tr><td>Timeout</td><td>The time allowed to establish communication between the uP and the central FPGA has expired. Communications fault.</td></tr>
<tr><td>Error CRC</td><td>It is not possible to establish communication between the uP and the central FPGA, due to an error in the frame.</td></tr>
<tr><td>Error Driver</td><td>It is not possible to establish communication between the uP and the central FPGA due to a failure in the reception of the frame shared between the uP and the central FPGA.</td></tr>
</table></td></tr>
<tr><td>SV6.2.4.4-Frames/sec OK = 0</td><td>NONE</td><td>Counter that displays the number of correct frames per second between the two devices.</td></tr>
<tr><td>SV6.2.4.5-Frames/sec Timeout = 0</td><td>NONE</td><td>Counter that displays the number of frames that have failed per timeout per second between both devices.</td></tr>
<tr><td>SV6.2.4.6-Frames/sec CRC = 0</td><td>NONE</td><td>Counter that displays the number of frames that have failed due to CRC (communications frame error) per second between both devices.</td></tr>
<tr><td>SV6.2.4.7-Frames/sec Driver = 0</td><td>NONE</td><td>Counter that displays the number of frames that have failed due to an error in the reception of the frame shared between uP and central FPGA.</td></tr>
</table>

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SV6.2.5: Fans

For the ventilation layout, please consult "SV11: Fans" section.

SCREEN	UNITS	DESCRIPTION
SV6.2.5.1.1-Fan 1 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control board and fan 1.
SV6.2.5.1.2-Fan 1 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control board and fan 1.
SV6.2.5.2.1-Fan 2 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control board and fan 2.
SV6.2.5.2.2-Fan 2 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control board and fan 2.
SV6.2.5.3.1-Fan 3 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 3.
SV6.2.5.3.2-Fan 3 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control board and fan 3.

3.7. Group SV7: Faults and warnings

3.7.1. Subgroup SV7.1: Faults

SV7.1.1: Faults status

Parameters within this group show the state of each fault (0 = Not active; X = Active) in groups of 16 faults (being the lowest number the bit to the upmost right).

Note: from version 2.0.0 the parameters of this subgroup are read-only parameters.

SCREEN	UNITS	DESCRIPTION
SV7.1.1.1-Fault 1 to 16 = 0000000000000000	NONE	Displays the fault state 1 to 16.
SV7.1.1.2-Fault 17 to 32 = 0000000000000000	NONE	Displays the fault state 17 to 32.
SV7.1.1.3-Fault 33 to 48 = 0000000000000000	NONE	Displays the fault state 33 to 48.
SV7.1.1.4-Fault 49 to 64 = 0000000000000000	NONE	Displays the fault state 49 to 64.
SV7.1.1.5-Fault 65 to 80 = 0000000000000000	NONE	Displays the fault state 65 to 80.
SV7.1.1.6-Fault 81 to 96 = 0000000000000000	NONE	Displays the fault state 81 to 96.
SV7.1.1.7-Fault 97 to 112 = 0000000000000000	NONE	Displays the fault state 97 to 112.
SV7.1.1.8-Fault 113 to 128 = 0000000000000000	NONE	Displays the fault state 113 to 128.
SV7.1.1.9-Fault 129 to 144 = 0000000000000000	NONE	Displays the fault state 129 to 144.
SV7.1.1.10-Fault 145 to 160 = 0000000000000000	NONE	Displays the fault state 145 to 160.
SV7.1.1.11-Fault 161 to 176 = 0000000000000000	NONE	Displays the fault state 161 to 176.
SV7.1.1.12-Fault 177 to 192 = 0000000000000000	NONE	Displays the fault state 177 to 192.
SV7.1.1.13-Fault 193 to 208 = 0000000000000000	NONE	Displays the fault state 193 to 208.
SV7.1.1.14-Fault 209 to 224 = 0000000000000000	NONE	Displays the fault state 209 to 224.
SV7.1.1.15-Fault 225 to 240 = 0000000000000000	NONE	Displays the fault state 225 to 240.
SV7.1.1.16-Fault 241 to 256 = 0000000000000000	NONE	Displays the fault state 241 to 256.
SV7.1.1.17-Fault 257 to 272 = 0000000000000000	NONE	Displays the fault state 257 to 272.

SCREEN	UNITS	DESCRIPTION
SV7.1.1.18-Fault 273 to 288 = 0000000000000000	NONE	Displays the fault state 273 to 288.
SV7.1.1.19-Fault 289 to 304 = 0000000000000000	NONE	Displays the fault state 289 to 304.
SV7.1.1.20-Fault 305 to 320 = 0000000000000000	NONE	Displays the fault state 305 to 320.
SV7.1.1.21-Fault 321 to 336 = 0000000000000000	NONE	Displays the fault state 321 to 336.
SV7.1.1.22-Fault 337 to 352 = 0000000000000000	NONE	Displays the fault state 337 to 352.
SV7.1.1.23-Fault 353 to 368 = 0000000000000000	NONE	Displays the fault state 353 to 368.
SV7.1.1.24-Fault 369 to 384 = 0000000000000000	NONE	Displays the fault state 369 to 384.
SV7.1.1.25-Fault 385 to 400 = 0000000000000000	NONE	Displays the fault state 385 to 400.
SV7.1.1.26-Fault 401 to 416 = 0000000000000000	NONE	Displays the fault state 401 to 416.
SV7.1.1.27-Fault 417 to 432 = 0000000000000000	NONE	Displays the fault state 417 to 432.
SV7.1.1.28-Fault 433 to 448 = 0000000000000000	NONE	Displays the fault state 433 to 448.
SV7.1.1.29-Fault 449 to 464 = 0000000000000000	NONE	Displays the fault state 449 to 464.
SV7.1.1.30-Fault 465 to 480 = 0000000000000000	NONE	Displays the fault state 465 to 480.
SV7.1.1.31-Fault 481 to 496 = 0000000000000000	NONE	Displays the fault state 481 to 496.
SV7.1.1.32-Fault 497 to 512 = 0000000000000000	NONE	Displays the fault state 497 to 512.

EN

SV7.1.2: Faults history

This subgroup allows to visualize the list of the last ten faults of the system in chronological order. The parameters that register the date contain two Modbus addresses corresponding to the date and time of the fault history register.

The following is an example of how the date and time registers for each fault are interpreted internally.

The parameters return decimal numbers recording the date and time of the fault occurred. This information must be transformed into binary and then converted back to decimal. The process is as follows. Example:

- Date: 5/9/2037
  - Day (5): it is transformed into binary by shifting 11 bits to the left obtaining 0010 1000 0000 0000 0000. The decimal value is 10240.
  - Month (9): transformed into binary by shifting 7 bits to the left to obtain 0100 1000 0000. The decimal value is 1152.
  - Year (2037): 2000 must be subtracted from the year. The decimal value is 37.

Once the decimal values of the date have been obtained, they are added together to obtain 11429. This value is read by Modbus. For the time the process is similar:

- Time: 16:11
  - Time (16): it is transformed into binary by shifting 8 bits to the left, obtaining 0001 0000 0000 0000 0000. The decimal value is 4096.
  - Minutes (11): The decimal value is 11.

When adding the decimals that have come out of the hour and minute conversion, the value is 4107. This value is read by Modbus.

The parameters corresponding to this subgroup are shown below.

SCREEN	DESCRIPTION	FUNCTION
SV7.1.2.1-Last fault = No faults	Register number 1 of fault history.	Displays a list of the last ten converter faults, in chronological order. The most recent fault is shown first and stored. Whenever a new fault occurs, it will be placed in the top of the list and older fault messages will drop a position, deleting the last fault message. The converter is reset by pressing the STOP-RESET button on the display or using an external reset if it is available. Certain faults may rearm automatically using the Auto Reset.
SV7.1.2.2a-Date = 01/01/2000 00:00	Date and time of the register number 1 of fault history.	
SV7.1.2.3-Ninth fault = No faults	Register number 2 of fault history.	
SV7.1.2.4a-Date = 01/01/2000 00:00	Date and time of the register number 2 of fault history.	
SV7.1.2.5-Eighth fault = No faults	Register number 3 of fault history.	
SV7.1.2.6a-Date = 01/01/2000 00:00	Date and time of the register number 3 of fault history.	
SV7.1.2.7-Seventh fault = No faults	Register number 4 of fault history.	
SV7.1.2.8a-Date = 01/01/2000 00:00	Date and time of the register number 4 of fault history.	
SV7.1.2.9-Sixth fault = No faults	Register number 5 of fault history.	
SV7.1.2.10a-Date = 01/01/2000 00:00	Date and time of the register number 5 of fault history.	
SV7.1.2.11-Fifth fault = No faults	Register number 6 of fault history.	
SV7.1.2.12a-Date = 01/01/2000 00:00	Date and time of the register number 6 of fault history.	
SV7.1.2.13-Fourth fault = No faults	Register number 7 of fault history.	
SV7.1.2.14a-Date = 01/01/2000 00:00	Date and time of the register number 7 of fault history.	
SV7.1.2.15-Third fault = No faults	Register number 8 of fault history.	
SV7.1.2.16a-Date = 01/01/2000 00:00	Date and time of the register number 8 of fault history.	
SV7.1.2.17-Second fault = No faults	Register number 9 of fault history.	
SV7.1.2.18a-Date = 01/01/2000 00:00	Date and time of the register number 9 of fault history.	
SV7.1.2.19-First fault = No faults	Register number 10 of fault history.	
SV7.1.2.20a-Date = 01/01/2000 00:00	Date and time of the register number 10 of fault history.	

SV7.1.3: Faults type

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.1.3.0-Fault type filter Select = Abrupt stop</td><td>NONE</td><td>Shows the type of fault to get the information. This is a configuration parameter.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Special</td><td>Open the switch disconnector and isolate the converter. To manual reset, Power Electronics password is required.</td></tr>
<tr><td>External Emergency</td><td>Open the switch disconnector and isolate the converter. To manual reset, Advanced client password is required.</td></tr>
<tr><td>Abrupt stop</td><td>The equipment is stopped as soon as possible by not switching the IGBTs</td></tr>
</table>
Note: Status Abrupt stop and Controlled stop are only displayed when there is a module fault.</td></tr>
<tr><td>SV7.1.3.1-Select page = 1</td><td>NONE</td><td>Shows the page of the faults to be consulted. This is a configuration parameter.</td></tr>
<tr><td>SV7.1.3.2a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.1.3.2b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the switch disconnector and isolate the converter. To manual reset, Power Electronics password is required.</td></tr><tr><td>External Emergency</td><td>Open the switch disconnector and isolate the converter. To manual reset, Qualified customer password is required.</td></tr><tr><td>Abrupt stop</td><td>The equipment is stopped as soon as possible by not switching the IGBTs</td></tr></table></td></tr>
<tr><td>SV7.1.3.3a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.3b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.4a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.4b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.5a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.5b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.6a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.6b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.7a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.7b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.8a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.8b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.9a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.9b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.10a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.10b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.11a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.11b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.12a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.12b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.13a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.13b-Fault type = Crowbar + Special</td><td>NONE</td><td>Shows the type of the fault. See SV7.1.3.2b list.</td></tr>
<tr><td>SV7.1.3.14a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
</table>

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SCREEN	UNITS	DESCRIPTION
SV7.1.3.14b-Fault type = Crowbar + Special	NONE	Shows the type of the fault. See SV7.1.3.2b list.
SV7.1.3.15a-Fault number = No faults	NONE	Shows the number of the fault.
SV7.1.3.15b-Fault type = Crowbar + Special	NONE	Shows the type of the fault. See SV7.1.3.2b list.
SV7.1.3.16a-Fault number = No faults	NONE	Shows the number of the fault.
SV7.1.3.16b-Fault type = Crowbar + Special	NONE	Shows the type of the fault. See SV7.1.3.2b list.
SV7.1.3.17a-Fault number = No faults	NONE	Shows the number of the fault.
SV7.1.3.17b-Fault type = Crowbar + Special	NONE	Shows the type of the fault. See SV7.1.3.2b list.

3.7.2. Subgroup SV7.2: Warnings

SV7.2.1: Warning status

Parameters within this group show the state of each warning (0 = Not active; X = Active) in groups of 16 warnings (being the lowest number the bit to the upmost right).

Note: from version 2.0.0 the parameters of this subgroup are read-only parameters.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.2.1.1-Warning 1 to 16 = 0000000000000000</td><td>NONE</td><td>Displays the warning private state 1 to 16.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Not used.</td><td>8</td><td>W9: I limit CCL / DCL</td></tr>
<tr><td>1</td><td>Not used.</td><td>9</td><td>Not used.</td></tr>
<tr><td>2</td><td>Not used.</td><td>10</td><td>Not used.</td></tr>
<tr><td>3</td><td>Not used.</td><td>11</td><td>W12: Derating temperature IBGTS</td></tr>
<tr><td>4</td><td>W5: No start conditions</td><td>12</td><td>W13: Derating temperature admission.</td></tr>
<tr><td>5</td><td>Not used.</td><td>13</td><td>Not used.</td></tr>
<tr><td>6</td><td>W7: P limit</td><td>14</td><td>Not used.</td></tr>
<tr><td>7</td><td>Not used.</td><td>15</td><td>Not used.</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.2-Warning 17 to 32 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.3-Warning 33 to 48 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.4-Warning 49 to 64 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.5-Warning 65 to 80 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.6-Warning 81 to 96 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.7-Warning 97 to 112 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.8-Warning 113 to 128 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.9-Warning 129 to 144 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.2.1.10-Warning 145 to 160 = 0000000000000000</td><td>NONE</td><td>Displays the warning private state 145 to 160.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Not used</td><td>8</td><td>Not used</td></tr>
<tr><td>1</td><td>Not used</td><td>9</td><td>Not used</td></tr>
<tr><td>2</td><td>Not used</td><td>10</td><td>Not used</td></tr>
<tr><td>3</td><td>Not used</td><td>11</td><td>Not used</td></tr>
<tr><td>4</td><td>Not used</td><td>12</td><td>Not used</td></tr>
<tr><td>5</td><td>W150: No SD</td><td>13</td><td>Not used</td></tr>
<tr><td>6</td><td>Not used</td><td>14</td><td>Not used</td></tr>
<tr><td>7</td><td>W152: PPC comms</td><td>15</td><td>Not used</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.11-Warning 161 to 176 = 0000000000000000</td><td>NONE</td><td>Displays the warning private state 161 to 176.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Not used</td><td>8</td><td>Not used</td></tr>
<tr><td>1</td><td>Not used</td><td>9</td><td>W170: Overvoltage DC</td></tr>
<tr><td>2</td><td>Not used</td><td>10</td><td>W171: No RTC</td></tr>
<tr><td>3</td><td>Not used</td><td>11</td><td>Not used</td></tr>
<tr><td>4</td><td>W165: Fan motor status</td><td>12</td><td>Not used</td></tr>
<tr><td>5</td><td>W166: Temp sensor read</td><td>13</td><td>Not used</td></tr>
<tr><td>6</td><td>Not used</td><td>14</td><td>Not used</td></tr>
<tr><td>7</td><td>Not used</td><td>15</td><td>Not used</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.12-Warning 177 to 192 = 0000000000000000</td><td>NONE</td><td>Displays the warning private state 177 to 192.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Not used</td><td>8</td><td>Not used</td></tr>
<tr><td>1</td><td>W178: FTP_OPEN</td><td>9</td><td>W186: Supply low</td></tr>
<tr><td>2</td><td>Not used</td><td>10</td><td>Not used</td></tr>
<tr><td>3</td><td>W180:SD problem</td><td>11</td><td>Not used</td></tr>
<tr><td>4</td><td>W181:SPD auxiliary circuit</td><td>12</td><td>Not used</td></tr>
<tr><td>5</td><td>Not used</td><td>13</td><td>Not used</td></tr>
<tr><td>6</td><td>Not used</td><td>14</td><td>Not used</td></tr>
<tr><td>7</td><td>Not used</td><td>15</td><td>Not used</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.13-Warning 193 to 208 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.14-Warning 209 to 224 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.15-Warning 225 to 240 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.16-Warning 241 to 256 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.17-Warning 257 to 272 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.18-Warning 273 to 288 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.19-Warning 289 to 304 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.20-Warning 305 to 320 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.21-Warning 321 to 336 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.22-Warning 337 to 352 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.23-Warning 353 to 368 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.24-Warning 369 to 384 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.25-Warning 385 to 400 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.2.1.26-Warning 401 to 416 = 0000000000000000</td><td>NONE</td><td>Displays the warning private state 401 to 416.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W401: Stop conditions</td><td>8</td><td>Not used</td></tr>
<tr><td>1</td><td>W402: High voltage PV</td><td>9</td><td>Not used</td></tr>
<tr><td>2</td><td>W403: High voltage BESS</td><td>10</td><td>Not used</td></tr>
<tr><td>3</td><td>W404: Low voltage PV</td><td>11</td><td>Not used</td></tr>
<tr><td>4</td><td>W405: Low voltage BESS</td><td>12</td><td>Not used</td></tr>
<tr><td>5</td><td>W406: Derating thermal current</td><td>13</td><td>Not used</td></tr>
<tr><td>6</td><td>W407:Blackbox download enable</td><td>14</td><td>Not used</td></tr>
<tr><td>7</td><td>Not used</td><td>15</td><td>Not used</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.27-Warning 417 to 432 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.28-Warning 433 to 448 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.29-Warning 449 to 464 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.30-Warning 465 to 480 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.31-Warning 481 to 496 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
<tr><td>SV7.2.1.32-Warning 497 to 512 = 0000000000000000</td><td>NONE</td><td>Not used.</td></tr>
</table>

SV7.2.1a: Warning private status

These parameters are only displayed if [G1.3 = Power electronics staff] or [G1.3 = Developer].

SV7.2.1a to SV7.2.1.32a has symmetric parameters as SV7.2.1 to SV7.2.1.32 but shows private warnings. See description in previous section.

Note: from version 2.0.0 these are read-only parameters.

SV7.2.2: Warnings history

Note: See section "SV7.1.2: Faults history" for more information about the internal interpretation procedure of dates and times for warning history.

SCREEN	DESCRIPTION	FUNCTION
SV7.2.2.1-Last warning = 0	Register number 1 of the warning history.	Displays a list of the last ten warnings, in chronological order. When a warning occurs, it will be displayed on the status bar flashing with the current state. It will remain displayed until the problem is fixed. The most recent warning is shown first and stored. Whenever a new fault occurs, it will be placed in the top of the list and older fault messages will drop a position, deleting the last warning message.
SV7.2.2.2-Date = 01/01/2000 00:00	Date and time of the register number 1 of warning history.	
SV7.2.2.3-Ninth warning = 0	Register number 2 of the warning history.	
SV7.2.2.4-Date = 01/01/2000 00:00	Date and time of the register number 2 of warning history.	
SV7.2.2.5-Eighth warning = 0	Register number 3 of the warning history.	
SV7.2.2.6-Date = 01/01/2000 00:00	Date and time of the register number 3 of warning history.	
SV7.2.2.7-Seventh warning = 0	Register number 4 of the warning history.	
SV7.2.2.8-Date = 01/01/2000 00:00	Date and time of the register number 4 of warning history.	
SV7.2.2.9-Sixth warning = 0	Register number 5 of the warning history.	

SCREEN	DESCRIPTION	FUNCTION
SV7.2.2.10-Date = 01/01/2000 00:00	Date and time of the register number 5 of warning history.	
SV7.2.2.11-Fifth warning = 0	Register number 6 of the warning history.	
SV7.2.2.12-Date = 01/01/2000 00:00	Date and time of the register number 6 of warning history.	
SV7.2.2.13-Fourth warning = 0	Register number 7 of the warning history.	
SV7.2.2.14-Date = 01/01/2000 00:00	Date and time of the register number 7 of warning history.	
SV7.2.2.15-Third warning = 0	Register number 8 of the warning history.	
SV7.2.2.16-Date = 01/01/2000 00:00	Date and time of the register number 8 of warning history.	
SV7.2.2.17-Second warning = 0	Register number 9 of the warning history.	
SV7.2.2.18-Date = 01/01/2000 00:00	Date and time of the register number 9 of warning history.	
SV7.2.2.19-First warning = 0	Register number 10 of the warning history.	
SV7.2.2.20-Date = 01/01/2000 00:00	Date and time of the register number 10 of warning history.	

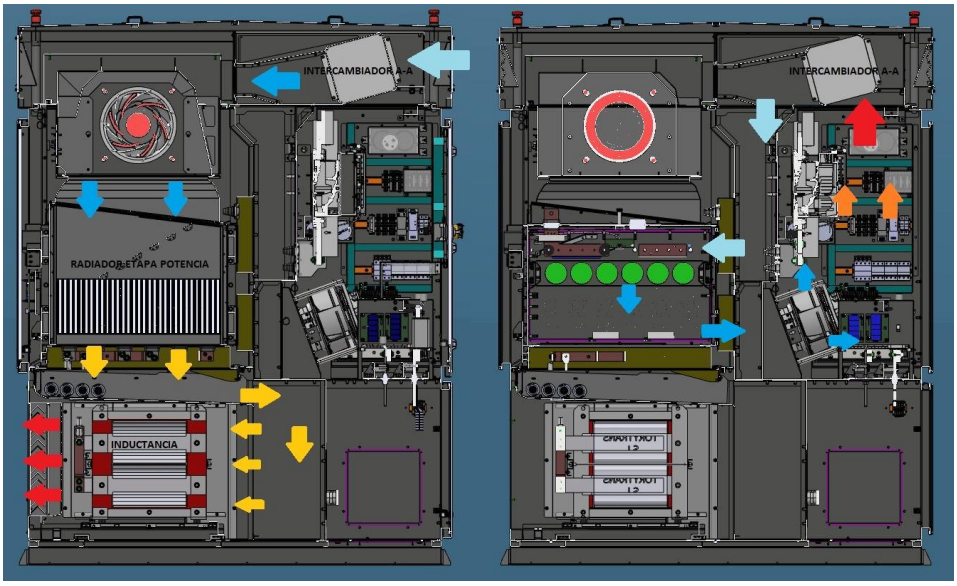
SV7.2.3: Private warn history

These parameters are only displayed if [G1.3 = Power electronics staff] or [G1.3 = Developer].

SV7.2.3.1 to SV7.2.3.20 has symmetric parameters as SV7.2.2.1 to SV7.2.1.2.20 but shows private warnings history. See description in previous section.

3.8. Group SV11: Fans

This group allows to visualize the information provided by each of the fans installed in the equipment for monitoring purposes. This product can support the control of up to 3 fans that communicate with the central control board via RS485. The information varies depending on the fan's supplier. Converter has two different cooling circuits. Next figure shows both circuits: dirty circuit on the left and clean circuit on the right.



3.8.1. Subgroup SV11.1: Fan 1

These parameters are only displayed if [G11.9-Availables Fan Nº] ≥ 1 (should be always). This fan is in DC zone.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.1.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the fan.
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Fan 1 is disabled.</td></tr><tr><td>OK fan</td><td>Communication between the uP and the fan 1 is correct.</td></tr><tr><td>Comms Error</td><td>uP is unable to communicate with the fan 1.</td></tr><tr><td>Error</td><td>Fan 1 is in error.</td></tr></table></td></tr>
<tr><td>SV11.1.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.1.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan.
Note: This parameter is only displayed if [G11.2.1.3-Type]= Emb-papst.</td></tr>
<tr><td>SV11.1.4-Fan zone = G11.2.1.1-Asigned Zone</td><td>NONE</td><td>Shows the zone assigned to the fan.
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Clean zone</td><td>Fan 1 is assigned to clean zone</td></tr><tr><td>Dirty zone</td><td>Fan 1 is assigned to dirty zone</td></tr></table></td></tr>
<tr><td>SV11.1.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Unknown request</td></tr><tr><td>Check ID</td><td>The ID of the fan with which the converter is communicating is checked for correctness.</td></tr><tr><td>Configuration 1</td><td rowspan="7">The converter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Converter sends a reset to the communicating fan.</td></tr><tr><td>Configuration 2</td></tr><tr><td>Configuration 3</td></tr><tr><td>Configuration 4</td></tr><tr><td>Configuration 5</td></tr><tr><td>Configuration 6</td></tr><tr><td>Configuration 7</td></tr><tr><td>Read serial number</td><td>The converter reads the serial number of the fan</td></tr><tr><td>Reset</td><td>The converter sends a reset to the communicating fan</td></tr><tr><td>Get status</td><td>The converter asks the fan to indicate its internal status to the inverter</td></tr><tr><td>Get speed</td><td>The converter asks to the fan to get the current fan speed.</td></tr><tr><td>Get temperature</td><td>The converter asks to the fan to get the internal temperature of the fan.</td></tr><tr><td>Get power</td><td>The converter asks to the fan to get the current power consumption of the fan.</td></tr><tr><td>Set setpoint</td><td>The converter sets the speed setpoint of the fan configured in the parameter “Setpoint” included in the group [G11.3: Config. clean zone] or [G11.4: Config. dirty zone].</td></tr><tr><td>Special request</td><td>Special request.</td></tr></table></td></tr>
<tr><td>SV11.1.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.1.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.</td></tr>
<tr><td>SV11.1.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.1.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>Output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>Fault RS485 communication with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Error in fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.1.3-Type] = Emb-papst.</td></tr>
<tr><td>SV11.1.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fault fan.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.1.3-Type]= Tesoer.</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.1.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronic temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>Line voltage high.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>Active the shedding function.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.1.3-Type] = Emb-papst.</td></tr>
<tr><td>SV11.1.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus undervoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL fault.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.1.3-Type] = Tesoer.</td></tr>
</table>

SCREEN	UNITS	DESCRIPTION
SV11.1.11-Temperature motor = 0 ºC	ºC	Shows the temperature of the fan motor.
SV11.1.12-Temperature module = 0 ºC	ºC	Shows the temperature of the module motor.
SV11.1.13-Temp. electronic = 0 ºC	ºC	Shows the temperature of the electronics of the motor. **Note:** This parameter is only displayed if [G11.2.1.3 Type = Emb-papst].
SV11.1.14-Current power = 0 W	W	Shows the current active power of the motor.
SV11.1.15-Fan type = G11.2.1.3-Type	NONE	Shows the fan type configured in G11.2. **STATUS** / **DESCRIPTION** Emb-Papst / Emb-Papst fan supplier. Tesoer / Tesoer fan supplier. Delta / Delta fan supplier.
SV11.1.16-Curr. setpoint RPM = 0 rpm	rpm	Shows current value of setpoint.

3.8.2. Subgroup SV11.2: Fan 2

These parameters are only displayed if [G11.9-Availables Fan Nº] ≥ 2 (should be always). This fan is in BESS zone.

SV11.2.1 to SV11.2.16 has symmetric parameters as SV11.1.1 to SV11.1.16 but shows configuration for fan 2. See description in previous section.

3.8.3. Subgroup SV11.3: Fan 3

These parameters are only displayed if [G11.9-Availables Fan Nº] ≥ 3 (should be always). This fan is in PV zone.

SV11.3.1 to SV11.3.16 has symmetric parameters as SV11.1.1 to SV11.1.16 but shows configuration for fan 2. See description in previous section.

EN

3.9. Group SV12: Modules Information

3.9.1. Subgroup SV12.1: Module

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV12.1.1-Status = Idle</td><td>NONE</td><td>Shows the status of the FPGA.
<table>
<tr><th>OPT.</th><th>STATUS</th><th>OPT.</th><th>STATUS</th></tr>
<tr><td>0</td><td>Idle</td><td>5</td><td>Stop</td></tr>
<tr><td>1</td><td>Ready</td><td>6</td><td>Discharge</td></tr>
<tr><td>2</td><td>Softcharge</td><td>7</td><td>Polarization</td></tr>
<tr><td>3</td><td>Ready</td><td>8</td><td>Auto diagnosis</td></tr>
<tr><td>4</td><td>Wait</td><td>9</td><td>Fault</td></tr>
</table></td></tr>
<tr><td>SV12.1.2-Fault = No faults</td><td>NONE</td><td>Shows the fault.</td></tr>
<tr><td>SV12.1.3-Status softcharge = Idle</td><td>NONE</td><td>Shows the current status of the softcharge.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Idle</td><td>The equipment is ready for initializing the softcharge.</td></tr>
<tr><td>1</td><td>SC BESS</td><td>The BESS softcharge is initializing.</td></tr>
<tr><td>2</td><td>SC BESS done</td><td>The BESS softcharge is done.</td></tr>
<tr><td>3</td><td>SC PV</td><td>The PV softcharge is initializing.</td></tr>
<tr><td>4</td><td>SC PV done</td><td>The PV softcharge is done.</td></tr>
</table></td></tr>
<tr><td>SV12.1.4-Working mode = Buck</td><td>NONE</td><td>Shows the working mode of the equipment.
<table>
<tr><th>OPTION</th><th>STATUS</th></tr>
<tr><td>0</td><td>Buck</td></tr>
<tr><td>1</td><td>Buck Boost</td></tr>
<tr><td>2</td><td>Boost</td></tr>
</table></td></tr>
<tr><td>SV12.1.5-Humidity = 50 %</td><td>%</td><td>Shows the humidity of the module board measured internally.</td></tr>
<tr><td>SV12.1.6-Feedfordward = 0</td><td>NONE</td><td>Shows the feedfordward status.</td></tr>
<tr><td>SV12.1.7-Warning register 1 = 0000</td><td>NONE</td><td>Shows the register of the warnings.</td></tr>
<tr><td>SV12.1.8-Vcc PCB 24V = 0.00 V</td><td>V</td><td>Shows 24V module card power supply.</td></tr>
<tr><td>SV12.1.9-Vcc PV gate drive = 0.00 V</td><td>V</td><td>Shows PV-side gate drive power supply.</td></tr>
<tr><td>SV12.1.10-Vcc PCB 15V = 0.00 V</td><td>V</td><td>Shows 15V module card power supply generated for gate drives.</td></tr>
<tr><td>SV12.1.11-Vcc BESS gate drive = 0.00 V</td><td>V</td><td>Shows BESS-side gate drive power supply.</td></tr>
</table>

3.9.2. Subgroup SV12.2: Electrical Measures

SCREEN	UNITS	DESCRIPTION
SV12.2.1-Current L1= 0.0	A	Shows the phase current.
SV12.2.2-Current L2 = 0.0	A	Shows the phase current.
SV12.2.3-Current L3 = 0.0	A	Shows the phase current.
SV12.2.4-V half PV (+) = 0	V	Shows the positive voltage in the PV side.
SV12.2.5-V half PV (-) = 0	V	Shows the negative voltage in the PV side.
SV12.2.6-V half BESS (+) = 0	V	Shows the positive voltage in the BESS side.
SV12.2.7-V half BESS (+) = 0	V	Shows the negative voltage in the BESS side.

SCREEN	UNITS	DESCRIPTION
SV12.2.8-P = 0.0 W	W	Shows the active power of the module.
SV12.2.10-Duty 1 = 0	V	Shows the raw duty cycle of inductance 1
SV12.2.11-Duty 2 = 0	V	Shows the raw duty cycle of inductance 2
SV12.2.12-Duty 3 = 0	V	Shows the raw duty cycle of inductance 3
SV12.2.13-P PV = 0.0	kW	Shows the active power in the PV side.
SV12.2.14-P BESS = 0.0	kW	Shows the active power in the BESS side.
SV12.2.15-Current BESS = 0	A	Shows the current in the BESS side.
SV12.2.16-Current PV = 0	A	Shows the current in the PV side

3.9.3. Subgroup SV12.3: Temp. Measures

SCREEN	UNITS	DESCRIPTION
SV12.3.1-Ambient Temp = 0	ºC	Shows the ambient temperature that is measuring the module.
SV12.3.2-Max temperature = 0.00	ºC	Shows the maximum temperature of the module IGBT.
SV12.3.3-Temp. IGBT PV S1 = 0.00	ºC	Shows the temperature of the PV IGBT.
SV12.3.4-Temp. IGBT PV S2 = 0.00	ºC	
SV12.3.5-Temp. IGBT PV S5 = 0.00	ºC	
SV12.3.6-Temp. IGBT PV S6 = 0.00	ºC	
SV12.3.7-Temp. IGBT PV S9 = 0.00	ºC	
SV12.3.8-Temp. IGBT PV S10 = 0.00	ºC	
SV12.3.9-Temp. IGBT BESS S3 = 0.00	ºC	Shows the temperature of the BESS IGBT.
SV12.3.10-Temp. IGBT BESS S4 = 0.00	ºC	
SV12.3.11-Temp. IGBT BESS S7 = 0.00	ºC	
SV12.3.12-Temp. IGBT BESS S8 = 0.00	ºC	
SV12.3.13-Temp. IGBT BESS S11 = 0.00	ºC	
SV12.3.14-Temp. IGBT BESS S12 = 0.00	ºC	

3.9.4. Subgroup SV12.4: Versions

SCREEN	UNITS	DESCRIPTION
SV12.4.1-SW version = 0.0.0-0	NONE	Shows the SW version. **Note:** Role PE Staff or higher is required to modify or display these parameters.

3.9.5. Subgroup SV12.5: Debug

SCREEN	UNITS	DESCRIPTION
SV12.5.1-Debug 0 = 0	NONE	Used by developer.
SV12.5.2-Debug 1 = 0	NONE	
SV12.5.3-Debug 2 = 0	NONE	
SV12.5.4-Debug 3 = 0	NONE	
SV12.5.5-Debug 4 = 0	NONE	
SV12.5.6-Debug 5 = 0	NONE	

4. PROGRAMMING PARAMETERS

4

Programming parameters allow setting and modifying the converter operating conditions and general parameters.

NOTICE

Depending on their role, user will have read and write or read only permissions.

EN

4.1. Group G1: General

In this group, user can enter the password to change the user mode and adjust date, time and UTC offset.

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G1.1-Init parameters = No</td><td>No
Yes</td><td>Initializes all parameters when is set to “Yes”.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>No</td><td>Not initialize.</td></tr>
<tr><td>Yes</td><td>Initializes configuration group parameters with following exceptions:
• G11 - Fans
• G6 - Communications
• G1.5 - RTC
• G14 – Custom modbus
• SV1.1 Current fault
• SV7 fault and warning history
• SV5 Energy counters
• Config Faults and warnings</td></tr>
</table>
Note: This parameter is only displayed if [G1.3 User ≥ Admin].</td><td>No</td></tr>
<tr><td>G1.2-Access code = 0</td><td>0 to 65535</td><td>Shows the current user mode access code.</td><td>Yes</td></tr>
<tr><td>G1.3-User = Customer</td><td>Customer
Commissioning
Qualified customer
Power electronics staff
Developer</td><td>Shows the current user mode:
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Customer</td><td>Basic customer access.</td></tr>
<tr><td>Commissioning</td><td>Not available.</td></tr>
<tr><td>Qualified customer</td><td>Advanced customer access.</td></tr>
<tr><td>Power electronics staff</td><td>Power electronics staff access.</td></tr>
<tr><td>Developer</td><td>Developer access.</td></tr>
</table></td><td>Yes</td></tr>
</table>

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G1.5-Date/time = 01/01/2016 0:00:00	0 to 59 0 to 59 0 to 23 1 to 31 1 to 12 2000 to 2159	Sets current date and time. Format DD/MM/YYYY H:M:S Note: See section "SV7.1.2: Faults history" for more information on the internal interpretation procedure of the date and time.	Yes
G1.6 -UTC offset = 0 min	-720 to 840 min	Sets the difference, in minutes, from UTC (Universal Time Coordinated) for a particular place, if needed.	Yes
G1.7-DCDC name = DCDC1	32 characters	Displays the converter's name. This parameter is set up with a string of 32 characters. **Note:** Only users with role superior to PE Staff can display and modify this parameter.	Yes
G1.8-Facility name = Planta1	32 characters	Displays the name of the facility to identify the location. This parameter is set up with a string of 32 characters. **Note:** Only users with role superior to PE Staff can display and modify this parameter.	Yes

4.2. Group G2: Running conditions

This group covers all the definition concerning running, stop and reconnection conditions of the equipment.

Note: Role Qualified Customer or higher is required to modify or display these parameters.

4.2.1. Subgroup G2.1: Start

Parameters to start the converter can be set in this subgroup.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.1.1-PV Vdc low = 850V	850 to G2.1.2-PV Vdc high V	Allows setting the minimum DC voltage in the PV side (positive bus) to start the equipment.	Yes
G2.1.2-PV Vdc high = 1500 V	G2.1.1-PV Vdc low to 1500 V	Allows setting the maximum DC voltage in the PV side (positive bus) to start the equipment.	Yes
G2.1.3-BESS Vdc low = 850 V	850 to G2.1.4-BESS Vdc high V	Allows setting the minimum DC voltage in the BESS side (positive bus) to start the equipment.	Yes
G2.1.4-BESS Vdc high = 1500 V	G2.1.3-BESS Vdc low to 1500 V	Allows setting the maximum DC voltage in the BESS side (positive bus) to start the equipment.	Yes
G2.1.5-Start delay = Off s	0 to 600 s	Allows setting the delay applied since the DC grid reconnection conditions of voltage and frequency are met.	Yes
G2.1.6-Enable polarization = No	No Yes	Allows the polarization algorithm. Recommended only for ground pole plants.	

4.2.2. Subgroup G2.2: Stop

Parameters to stop the converter can be set in this group.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.2.1-Vdc low = 850 V	850 to 1500 V	Allow configuring the automatic converter shutdown depending on the minimum DC voltage.	Yes
G2.2.2-Stop delay = Off s	Off to 600 s	Allows setting the automatic stop relay.	Yes

4.2.3. Subgroup G2.3: Softcharge

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.3.1-Softcharge mode = Automatic	Conventional Emergency External (Only visible if [G1.3-User] = Developer) Automatic	Defines the softcharge mode. **Notes:** - To modify this parameter, the PE Staff role is required. - To deactivate the selector, the converter must not be in ON. It must first be stopped and then deactivate the selector, otherwise fault F447 will be triggered.	No

4.2.4. Subgroup G2.4: Selectors

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.4.1-General Selector = Off	Off On	When the converter is in ready status this parameter must be set to one. The converter will start working if all conditions are met. **Note:** This parameter is displayed if [SG1.1.11-Selector config] = 1 or [SG1.1.11-Selector config] =2.	Yes

EN

4.2.5. Subgroup G2.5: Discharge

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.5.1-Bus discharge = Off	Off On	This parameter is obsolete and has no function. Will be deleted in new releases	Yes

4.3. Group G3: Limits

This group is intended for the configuration of the limits of the equipment, both power and deratings implemented in the equipment, equipment currents and batteries.

Note: Role Qualified Customer or higher is required to modify these parameters.

4.3.1. Subgroup G3.1: Voltage limits

This group allows setting the current limits due to voltage of BESS and PV field.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.1.1-Enable voltage limit = Yes	No Yes	Allows the current limited by voltage to be activated or deactivated. **Note**: Role PE Staff or higher is required to display and modify these parameters.	Yes
G3.1.2-Upper voltage BESS = 1500 V	850 to 1500 V	Allows setting the upper voltage limit in the BESS side. **Note:** This parameter is only displayed if [G3.1.1] = Yes.	Yes
G3.1.3-Lower voltage BESS = 850 V	850 to 1500 V	Allows setting the lower voltage limit in the BESS side. **Note:** This parameter is only displayed if [G3.1.1] = Yes.	Yes
G3.1.4-Upper voltage PV = 1500 V	850 to 1500 V	Allows setting the upper voltage limit in the PV side. **Note:** This parameter is only displayed if [G3.1.1] = Yes.	Yes
G3.1.5-Lower voltage PV = 850 V	850 to 1500 V	Allows setting the lower voltage limit in the PV side. **Note:** This parameter is only displayed if [G3.1.1] = Yes.	Yes

4.3.2. Subgroup G3.2: Batteries limits

Parameters to configure the current and active power of the batteries.

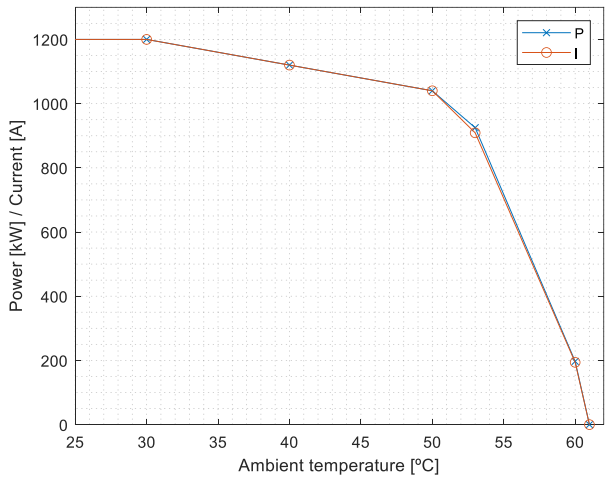
SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.2.1-I charge (CCL) = [SG10.1.6-Idc max] A	0 to [SG10.1.6-Idc max] A	Allows setting the charge current limit expressed in ampers.	Yes
G3.2.2-I discharge (DCL) = [SG10.1.6-Idc max] A	0 to [SG10.1.6-Idc max] A	Allows setting the discharge current limit expressed in ampers.	Yes
G3.2.3-Limit P charge = [SG10.3.2-Max P limit] %	0.00 to [SG10.3.2-Max P limit] %	Allows setting the charge active power.	Yes
G3.2.4-Limit P discharge = [SG10.3.2-Max P limit] %	0.00 to [SG10.3.2-Max P limit] %	Allows setting the discharge active power.	Yes
G3.2.5-Limit P charge KW = 1200 kW	0 to 1200 kW	Shows the active power to charge available.	Yes
G3.2.6-Limit P discharge KW = 1200 kW	0 to 1200 kW	Allows setting the discharge active power.	Yes

4.3.3. Subgroup G3.3: Derating Tª env.

G3.3.1: Power

Note: Only users with role Power Electronics staff or higher can modify or display the parameters [G3.3.1.0] a [G3.3.1.14] from this subgroup.

Power and current limit with ambient temperature are implemented in parallel. Power derating can be configured at this group, but current derating is forced. Power derating is designed for this converter and it is not recommended to change defalt configuration. Next figure shows both deratings:



SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.3.1.0-Set default curve = No	No Yes	Adjusts the curve reduction temperature to the default value. **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.1-Temperature 1 = -35 °	-40 to [G3.3.1.3-Temperature 2] °	Temperature setting at point 1 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.2-Power 1 = 1200kW	0 to 1200kW	Power setting at point 1 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.3-Temperature 2 = 30 °	[G3.3.1.1-Temperature 1] to [G3.3.1.5-Temperature 3]°	Temperature setting at point 2 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.4-Power 2 = 1200kW	0 to 1200kW	Power setting at point 2 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.5-Temperature 3 = 40 °	[G3.3.1.3-Temperature 2] to [G3.3.1.7-Temperature 4]	Temperature setting at point 3 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.6-Power 3 = 1120kW	0 to 1200kW	Power setting at point 3 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.7-Temperature 4 = 50 °	[G3.3.1.5-Temperature 3] to [G3.3.1.9-Temperature 5]	Temperature setting at point 4 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.8-Power 4 = 1040 kW	0 to 1200kW	Power setting at point 4 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.9-Temperature 5 = 53 °	[G3.3.1.7-Temperature 4] to [G3.3.1.11-Temperature 6]	Temperature setting at point 5 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.10-Power 5 = 925 kW	0 to 1200kW	Power setting at point 5 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.11-Temperature 6 = 60 °	[G3.3.1.9-Temperature 5] to [G3.3.1.13-Temperature 7]	Temperature setting at point 6 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.12-Power 6 = 197 kW	0 to 1200kW	Power setting at point 6 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.13-Temperature 7 = 61 °	[G3.3.1.11-Temperature 6] to 70 °	Temperature setting at point 7 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.14-Power 7 = 0 kW	0 to 1200kW	Power setting at point 7 on the curve **Note**: Role PE Staff or higher is required to display and modify this parameter.	Yes
G3.3.1.15-Max power available = 0 kW	0 to 1200kW	Maximum power available after applying ambient temperature conditions. This parameter is only for visualization.	Yes
G3.3.1.16-Max I available = 0 A	0 to 1200A	Maximum current available after applying ambient temperature conditions. This parameter is only for visualization.	Yes

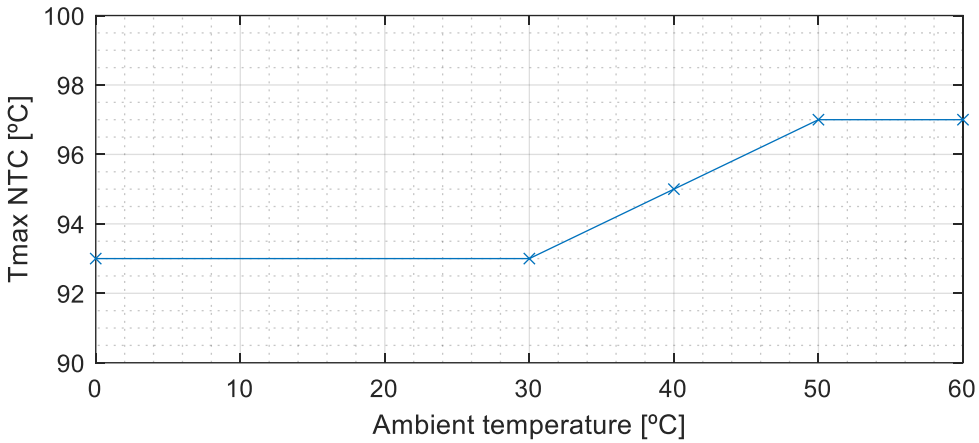
EN

4.3.4. Subgroup G3.4: Derating Tª IGBTs

G3.4.1-Curve Tª adm vs NTC

Note: Only users with role Power electronics staff can modify and display this subgroup.

Next figure shows default configuration for this group. It shows maximum expected NTC temperature as ambient temperature function.



Once this temperature is computed, fan boost (see [G3.4.1.11]), temperature fault (0.5ºC fixed over maximum expected NTC temperature) and derating (see [G3.4.1.12]) are computed with delta.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.4.1.0-Set default curve = No	No Yes	Set IGBTs curve derating to defaults value	Yes
G3.4.1.1-Temp. 1 env. IGBT = -40º	-50º to [G3.4.1.3-Temp. 2 env. IGBT]	Temperature setting at point 1 on the curve	Yes
G3.4.1.2-Temp 1 NTC max IGBT = 99º	-50º to 100 º	Temperature NTC setting at point 1 on the curve	Yes
G3.4.1.3-Temp. 2 env. IGBT = 30º	[G3.4.1.1-Temp. 1 env. IGBT] to [G3.4.1.5-Temp. 3 env. IGBT]	Temperature setting at point 2 on the curve	Yes
G3.4.1.4-Temp 2 NTC max IGBT = 99º	-50º to 100 º	Temperature NTC setting at point 2 on the curve	Yes
G3.4.1.5-Temp. 3 env. IGBT = 40º	[G3.4.1.3-Temp. 2 env. IGBT] to [G3.4.1.7-Temp. 4 env. IGBT]	Temperature setting at point 3 on the curve	Yes
G3.4.1.6-Temp 3 NTC max IGBT = 99º	-50º to 100º	Temperature NTC setting at point 3 on the curve	Yes
G3.4.1.7-Temp. 4 env. IGBT = 50º	[G3.4.1.5-Temp. 3 env. IGBT] to [G3.4.1.9-Temp. 5 env. IGBT]	Temperature setting at point 4 on the curve	Yes
G3.4.1.8-Temp 4 NTC max IGBT = 99º	-50º to 100º	Temperature NTC setting at point 4 on the curve	Yes
G3.4.1.9-Temp. 5 env. IGBT = 100º	[G3.4.1.7-Temp. 4 env. IGBT] to 100º	Temperature setting at point 5 on the curve	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.4.1.10-Temp 5 NTC max IGBT = 99º	-50º to 100º	Temperature NTC setting at point 5 on the curve	Yes
G3.4.1.11-Delta init boosting = 6º	0º to 20º	Adjustment of the activation of the Boosting referred to the maximum admissible temperature in the NTC (Tª_Ini_Boost= Max Tª NTC – Delta_Ini_Boost). Can be checked at G3.4.3	Yes
G3.4.1.12-Delta init derating = 0º	0º to [G3.4.1.11-Delta init boosting]	Adjustment of the start of the Derating referred to the maximum admissible temperature in the NTC (Tª_Ini_Der= Max Tª NTC – Delta_Ini_Der). Can be checked at G3.4.4	Yes
G3.4.1.13-Delta fault = 8º	4º to 8º	Difference between NTC maximum temperature and failure temperature.	Yes

G3.4.2-Curve Tª NTC vs I (%)

Note: Only users with role Power electronics staff can modify and display this subgroup.

Only [G3.4.2.1-Temp. 1 env. IGBT] and [G3.4.2.6-Current (%) P3] can be configured. All other parameters are only visualization that depends on G3.4.1 curve and [G3.4.1.12-Delta init derating].

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.4.2.0-Set default curve = No	No Yes	Set curve derating igbts to defaults value	Yes
G3.4.2.1-Temp. 1 env. IGBT = -40.00º	-40.00º to -40.00º	Temperature setting at point 1 on the curve Tª NTC vs I (%)	Yes
G3.4.2.2-Current (%) P1 = 100%	100 to 100%	Current (%) at point 1 of curve I vs Tª NTC	Yes
G3.4.2.3-Temp. 2 env. IGBT = G3.4.4-Temp init derating º	[G3.4.4-Temp init derating] to [G3.4.4-Temp init derating]	Temperature setting at point 2 on the curve Tª NTC vs I (%)	Yes
G3.4.2.4-Current (%) P2 = 100%	100 to 100%	Current (%) at point 2 of curve I vs Tª NTC	Yes
G3.4.2.5-Temp. 3 env. IGBT = G3.4.5-Act. limit NTC IGBTsº	[G3.4.5-Act. limit NTC IGBTs] to [G3.4.5-Act. limit NTC IGBTs]	Temperature setting at point 3 on the curve Tª NTC vs I (%)	Yes
G3.4.2.6-Current (%) P3 = 100%	100 to 100%	Current (%) at point 3 of curve I vs Tª NTC	Yes
G3.4.2.7-Temp. 4 env. IGBT = G3.4.6-Act. fault NTC IGBTs º	[G3.4.6-Act. fault NTC IGBTs] to [G3.4.6-Act. fault NTC IGBTs]	Temperature setting at point 4 on the curve Tª NTC vs I (%)	Yes
G3.4.2.8-Current (%) P4 = 0%	0 to 0%	Current (%) at point 4 of curve I vs Tª NTC	Yes

G3.4 - Others

Note: Only users with role Power electronics staff can modify and display this subgroup.

Next parameters are only for visualization

SCREEN	RANGE	DESCRIPTION
G3.4.3-Temp init boosting = 80 º	60º to 100º	Show Temperature activation of the Boosting referred to the maximum admissible temperature in the NTC (Tª_Ini_Boost= Max Tª NTC – Delta_Ini_Boost)
G3.4.4-Temp init derating = 97.00 º	70.00º to 120.00º	Show temperature start of the Derating referred to the maximum admissible temperature in the NTC (Tª_Ini_Der= Max Tª NTC – Delta_Ini_Der)
G3.4.5-Act. limit NTC IGBTs = 93.00 º	-40.00º to 120.00º	Show actual limit of temperature to igbts NTC
G3.4.6-Act. fault NTC IGBTs = 93.00 º	-40.00º to 120.00º	Show actual fault of temperature to igbts NTC
G3.4.7-Output limit device = 100.00 %	0.00 to 100.00%	Show power output limit before applying derating temperature of igbts

4.4. Group G4: Algorithms

4.4.1. Subgroup G4.3: Gradients

This algorithm allows the converter's behavior to be ramped. Disabling the ramps means that the converter will have a stepped behavior.

Note: Role Qualified Customer or higher is required to modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.3.1.1-Enable = No	No Yes	Enable the rise start gradient	Yes
G4.3.1.2-Rise start = 10.0%/s[1]	0.1 to 3000.0 %/s	Rise start gradient slope value. Start gradient is applied during time equivalent to reach 100% of reference.	Yes
G4.3.2.1-Enable = No	No Yes	Enable the rise start gradient	Yes
G4.3.2.2-Rise stop = 10.0%/s[2]	0.1 to 3000.0%/s	Rise stop gradient slope value. Stop gradient applies at stopping process only.	Yes
G4.3.3.1-Enable = No	No Yes	Enable operation gradients	Yes
G4.3.3.2-Rise ramp = 10.0%/s[3]	0.1 to 3000.0%/s	Upwards operation gradient value. Rise gradient applies once start gradient time is over.	Yes
G4.3.3.3-Down ramp = 10.0%/s[3]	0.1 to 3000.0%/s	Downwards operation gradient value. Down gradient applies once start gradient time is over.	Yes

[1] This parameter is only displayed if [G4.3.1.1-Enable = Yes].
[2] This parameter is only displayed if [G4.3.2.1-Enable = Yes].
[3] This parameter is only displayed if [G4.3.3.1-Enable = Yes].

4.5. Group G5: Protections

This group allows configuring (enable / disable, set limits) the different available protections.

4.5.1. Subgroup G5.1: Voltage

G5.1.1-Undervoltage

Note: Role Power Electronics staff or higher is required to display and modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.1.2-Threshold PV = 830 V	800 to 900 V	Allows setting the threshold of the protection level in PV undervoltage protection.	Yes
G5.1.1.3-Threshold BESS = 830 V	800 to 900 V	Allows setting the threshold of the protection level in BESS undervoltage protection.	Yes
G5.1.1.4-Hysteresis = 10 V	0 to 150 V	Allows defining the hysteresis to clear the undervoltage fault.	Yes

G5.1.2-Overvoltage

Note: Role Power Electronics staff or higher is required to display and modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.2.2-Threshold PV = 1520 V	1000 to 1600 V	Allows setting the threshold of the protection level in PV overvoltage protection.	Yes
G5.1.2.3-Threshold BESS = 1520 V	1000 to 1600 V	Allows setting the threshold of the protection level in BESS overvoltage protection.	Yes
G5.1.2.4-Hysteresis = 10 V	0 to 150 V	Allows defining the hysteresis to clear the overvoltage fault.	Yes

G5.1.3-Semibus overvoltage

Note: Role Power Electronics staff or higher is required to display and modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.3.2-Threshold PV = 785 V	500 to 785 V	Allows setting the threshold of the protection level in PV semibus overvoltage protection.	Yes
G5.1.3.3-Threshold BESS = 785 V	500 to 785 V	Allows setting the threshold of the protection level in BESS semibus overvoltage protection.	Yes
G5.1.3.4-Hysteresis = 25 V	0 to 150 V	Allows defining the hysteresis to clear the semibus overvoltage fault.	Yes

EN

G5.1.4-Batteries

Note: Role Power Electronics staff or higher is required to display and modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.4.1- Thresh dis stop V/us = 4	0-300 V/s	Open battery detection condition at not running condition	Yes
G5.1.4.2- Thresh dis start V/us = 36	0-300 V/us	Open battery detection condition at running condition	Yes
G5.1.4.3- Fault delay = 64	0-100 us	Open battery detection delay	Yes

4.5.2. Subgroup G5.2: Unbalanced

Note: Role Power Electronics staff or higher is required to display and modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.2.2- Max. PV Unbalance = 50 V	10 to 50 V	Allows setting the permissible maximum voltage unbalance in the PV bus.	Yes
G5.2.3-Max. BESS Unbalance V = 50 V	10 to 50 V	Allows setting the permissible maximum voltage unbalance in the BESS bus.	Yes

4.5.3. Subgroup G5.3: Overcurrent

Note: Role Power Electronics staff or higher is required to display and modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.3.1-Overcurrent threshold = Internal	600 to SG10.2.1-Overcurrent threshold A	Setting of the overcurrent protection of the stage and inductances	No
G5.3.2-Enable in inputs = Yes	No Yes	Enable or disable overcurrent protection in inputs PV and BESS	No

4.5.4. Subgroup G5.4: Temperature

Parameters to configure the temperature protections.

Note: Role Commissioning or higher is required to display and modify these parameters.

G5.4.1- Power stage

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.1.1-IGBTs T fault enable = Enabled	Disabled Enabled	Allows to enable / disable the IGBTs temperature protection. **Note:** from version 2.0.0, this parameter is not available.	Yes
G5.4.1.2-IGBTs temp thresh flt = 100 ºC	70 to [SG10.2.2-Igbts temp threshold] ºC	Sets the IGBT temperature protection threshold applied when the equipment is in discharge. **Note:** This parameter is only displayed if [G5.4.1.1 = Enabled]. **Note**: from version 2.0.0, this parameter is not available.	Yes
G5.4.1.3-IGBTsTempHysteresis = 5 ºC	0 to 25 ºC	Configures the hysteresis of the IGBT temperature, below which the fault is cleared. **Note:** This parameter is only displayed if [G5.4.1.1 = Enabled].	Yes

G5.4.2- PCB

G5.4.2.1-Module

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.2.1.1-Mod T fault enable = Enabled	Disabled Enabled	Allows to enable / disable the internal module central control board temperature protection. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G5.4.2.1.2-ModuleTempThresFlt = 70 ºC	20 to 85 ºC	Sets the module central control board protection threshold. **Note:** This parameter is only displayed if [G5.4.2.1 = Enabled]. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G5.4.2.1.3-ModulTempHysteresi = 10 ºC	0 to 20 ºC	Configures the hysteresis of the module central control board temperature, below which the fault is cleared. **Note:** This parameter is only displayed if [G5.4.2.1 = Enabled]. **Note**: To modify this parameter, the PE Staff role is required.	Yes

EN

G5.4.2.2-Central

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.2.2.1-Inter T fault ena = Enabled	Disabled Enabled	Allows to enable / disable the internal module central control board temperature protection. **Note**: To modify this parameter, the PE Staff role is required.	Yes
G5.4.2.2.2-Main control board = 70 ºC	Disabled 20 to 85 ºC	Temperature protection threshold for the main control board. When the threshold is exceeded, Fault 40 is displayed. **Note:** This parameter is only displayed if [G5.4.2.2.1 = Enabled]. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G5.4.2.2.3-CentTempHysteresis = 10 ºC	0 to 20 ºC	Configures the hysteresis of the control central board temperature, below which the fault is cleared. **Note:** This parameter is only displayed if [G5.4.2.2.1 = Enabled]. **Note**: To modify this parameter, the PE Staff role is required.	Yes

G5.4.4- Inductance

Note: Role Power Electronics staff or higher is required to display and modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.4.1-Ind T fault enable = Enabled	Disabled Enabled	Allows to enable / disable the inductance temperature protection.	Yes
G5.4.4.2-Ind Threshold = 140 ºC	100 to 145 ºC	Defines the inductance protection threshold. **Note:** This parameter is only displayed if [G5.4.4.1 = Enabled].	Yes
G5.4.4.3-Ind delay = 10 s	5 to 30 s	Defines the time that the fault waits until it is triggered. **Note:** This parameter is only displayed if [G5.4.4.1 = Enabled].	Yes
G5.4.4.4-Ind crash T enable = Enabled	Disabled Enabled	Allows to enable / disable the inductance critical temperature protection.	Yes
G5.4.4.5-IndTcriticThld = 140 ºC	130 to 145 ºC	Defines the inductance critical protection threshold. **Note:** This parameter is only displayed if [G5.4.4.1 = Enabled].	Yes
G5.4.4.6-Ind Temp Hysteresis = 10 ºC	0 to 50 ºC	Defines the hysteresis required to clear the fault automatically.	Yes

4.5.5. Subgroup G5.5: Config Protections

Note: Role commissioning is required to display this parameter.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.5.2-SPD main circuit = Warning	Warning Fault	Allows setting as warning or fault SPD main circuit protection. **Note:** To modify this parameter, the PE Staff role is required.	No

4.5.6. Subgroup G5.7: Faults/Warnings config

Note: Role Power Electronics staff or higher is required to display and modify these parameters.

G5.7.1- Faults

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G5.7.1.0-Time to Zero State = 3600 s</td><td>Off
Off to 10800 s</td><td>Allows the time to attempt to clear all fault counters and set the converter's autoreset mechanism to zero state only applies if the converter is not faulted at the time of expiration.</td><td>Yes</td></tr>
<tr><td>G5.7.2.1-Fault selected = F1: Watchdog</td><td>Check complete list in tab Faults</td><td>Allows fault selected of enum to configure.</td><td>Yes</td></tr>
<tr><td>G5.7.1.2-Fault type = Special</td><td>Special
Abrupt stop
External emergency</td><td>Allows fault of type can be special DU abrupt stop or controlled stop.
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the switch disconnector and isolate the converter. To manual reset, Power Electronics password is required.</td></tr><tr><td>Abrupt stop</td><td>The equipment is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>External emergency</td><td>Open the switch disconnector and isolate the converter. To manual reset, Power Electronics password is not required.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G5.7.1.3-Type of clean = Manual</td><td>Manual
Autoreset
Measure</td><td>Allows the type of clean can be never or autoreset.
Note: This parameter is only displayed if [G5.7.1.2 = Abrupt stop].</td><td>Yes</td></tr>
<tr><td>G5.7.1.4-Number of retries = 1</td><td>1
2
5
10
20
Infinite</td><td>Allows number of retries of autoreset.
Note: This parameter is only displayed if [G5.7.1.2 = Abrupt stop] and [G5.7.1.3-Type of clean = Measure or Manual].</td><td>Yes</td></tr>
<tr><td>G5.7.1.5-Time between retries = 1 s</td><td>1
2
5
10
20</td><td>Allows number of retries of autoreset.
Note: This parameter is only displayed if [G5.7.1.2 = Abrupt stop] and [G5.7.1.3-Type of clean = Measure or Manual].</td><td>Yes</td></tr>
<tr><td>G5.7.1.6-User Manual Reset = Customer</td><td>Customer
Comissioning
Qualified customer
Power electronics staff
Developer</td><td>User Role that is allowed to perform a reset of the current fault.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Customer</td><td>Resetting the fault can be done by a customer role or higher.</td></tr><tr><td>Comissioning</td><td>Resetting the fault can be done by a commissioning role or higher.</td></tr><tr><td>Qualified customer</td><td>Resetting the fault can be done by a qualified customer role or higher.</td></tr><tr><td>Power electronics staff</td><td>Resetting the fault can be done by a PEE Staff role or higher.</td></tr><tr><td>Developer</td><td>Resetting the fault can be done by a developer role.</td></tr></table></td><td>Yes</td></tr>
</table>

EN

G5.7.2- Warnings

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.7.2.1-Warning selected = W1:Input high voltage	Check complete list in tab Warnings	Allows to configure the warning modules selected.	Yes
G5.7.2.2-Source = UP	UP DSP Module	Sets the warning source. **Note**: from version 2.0.0, this parameter is hidden.	Yes
G5.7.2.3-Private = No	No Yes	Allows the set if warning is private or not.	Yes

G5.7.3- Clean History

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.7.3.1-Faults = No	No Yes	Allows the clean faults history.	Yes
G5.7.3.2-Warnings = No	No Yes	Allows the clean warning history.	Yes

4.6. Group G6: Communications

This group allows the configuration of the Modbus, Ethernet and FTP communication properties of the equipment, as well as the definition of the communications with the external PPC / SCADA.

4.6.1. Subgroup G6.1: Modbus

Note: Role Commissioning or higher is required to display or modify the parameters in this subgroup.

G6.1.1-Master 1

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.1.1.1-Modbus baudrate = 19200 bps	9600 bps 19200 bps 57600 bps 115200 bps 230400 bps 460800 bps 921600 bps	Adjusts the baudrate (bits per second) of Master 1.	Yes
G6.1.1.2-Modbus parity = None	None Odd Even	Adjusts the parity of Master 1.	Yes
G6.1.1.3-Modbus stop bits = 1 bit	1 bit 2 bits	Adjusts the stop bits number (per byte) of Master 1.	Yes
G6.1.1.4-Modbus error CRC = 0	0 to 65535	Shows the number of frames that have been lost of Master 1. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.1.5-Modbus OK frames = 0	0 to 65535	Shows the number of successful frames that have been sent of Master 1. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.1.6-Modbus recover = No	No Yes	This parameter should only be used by Developers. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.1.7-Modbus Timeout = 0.6 s	0.2 to 5.0 s	This parameter should only be used by Developers. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.1.8-Period = 0.250 s	0.010 to 1.000 s	This parameter should only be used by Developers. **Note:** To modify this parameter, the PE Staff role is required.	Yes

G6.1.2-Master 2

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.1.2.1-Modbus baudrate = 9600 bps	9600 bps 19200 bps 57600 bps 115200 bps 230400 bps 460800 bps 921600 bps	Adjusts the baudrate (bits per second) of Master 2.	Yes
G6.1.2.2-Modbus parity = None	None Odd Even	Adjusts the parity of Master 2.	Yes
G6.1.2.3-Modbus stop bits = 1 bit	1 bit 2 bits	Adjusts the stop bits number (per byte) of Master 2.	Yes

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.1.2.4-Modbus error CRC = 0	0 to 65535	Shows the number of frames that have been lost of Master 2. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.2.5-Modbus OK frames = 0	0 to 65535	Shows the number of successful frames that have been sent of Master 2. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.2.6-Modbus recover = No	No Yes	This parameter should only be used by Developers. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.2.7-Modbus Timeout 2 = 0.6 s	0.2 to 5.0 s	This parameter should only be used by Developers. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.2.8-Period = 0.250 s	0.010 to 1.000 s	This parameter should only be used by Developers. **Note:** To modify this parameter, the PE Staff role is required.	Yes

G6.1.3-Master 3

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.1.3.1-Modbus baudrate = 115200 bps	9600 bps 19200 bps 57600 bps 115200 bps 230400 bps 460800 bps 921600 bps	Adjusts the baudrate (bits per second) of Master 3.	Yes
G6.1.3.2-Modbus parity = None	None Odd Even	Adjusts the parity of Master 3.	Yes
G6.1.3.3-Modbus stop bits = 1 bit	1 bit 2 bits	Adjusts the stop bits number (per byte) of Master 3.	Yes
G6.1.3.4-Modbus error CRC = 0	0 to 65535	Shows the number of frames that have been lost of Master 3. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.3.5-Modbus OK frames = 0	0 to 65535	Shows the number of successful frames that have been sent of Master 3. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.3.6-Modbus recover = No	No Yes	This parameter should only be used by Developers. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.3.7-Modbus Timeout 3 = 0.6 s	0.2 to 5.0 s	This parameter should only be used by Developers. **Note:** To modify this parameter, the PE Staff role is required.	Yes
G6.1.3.8-Period = 0.100 s	0.010 to 1.000 s	This parameter should only be used by Developers. **Note:** To modify this parameter, the PE Staff role is required.	Yes

4.6.2. Subgroup G6.2: Ethernet

Note: from version 2.0.0, Power Electronics staff or higher is required to display and modify the parameters [G6.2.3-Assigned subnet], [G6.2.4-Assigned gateway] and [G6.2.8-MAC address].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.2.1-Automatic IP = No	No Yes	Allows selecting the IP assignment method of the module: **OPTION** / **DESCRIPTION** No: The converter will take the IP address, subnet mask and gateway set by the user. Yes: DHCP protocol is enabled and the network server will automatically set the IP address, subnet mask and gateway.	Yes
G6.2.2-Assigned IP = 0.0.0.0	0.0.0.0 to 255.255.255.255	Shows the IP address assigned to the module when DHCP is active.	Yes
G6.2.3-Assigned subnet = 0.0.0.0	0.0.0.0 to 255.255.255.255	Sets the IP address assigned to the module when DHCP is active.	Yes
G6.2.4-Assigned gateway = 0.0.0.0	0.0.0.0 to 255.255.255.255	Sets the gateway address assigned to the module.	Yes
G6.2.5-IP address = 192.168.204.202	0.0.0.0 to 255.255.255.255	Sets the IP address in the user's local network, according to the guidelines given by the network Admin. IP address format is: A.B.C.D. To adjust this address, introduce a value in each of the four parameters that compose it.	Yes
G6.2.6-Subnet Mask = 255.255.255.0	0.0.0.0 to 255.255.255.255	Sets the subnet mask address in the user's local network, according to the guidelines given by the network Admin. Subnet mask address format is: A.B.C.D. To adjust this address, introduce a value in each of the four parameters that compose it.	Yes
G6.2.7-Gateway = 0.0.0.0	0.0.0.0 to 255.255.255.255	Sets the gateway address in the user's local network so that the converter is able to communicate to an external network. This address must follow the guidelines given by the network Admin. Gateway's address format is: A.B.C.D. To adjust this address, introduce a value in each of the four parameters that compose it.	Yes
G6.2.8-MAC address = 21-1B-78-81-E9-44	00-00-00-00-00-00 to FF-FF-FF-FF-FF-FE	Set the MAC address. This address is unique and exclusive, and it is related with the LAN / inverter board. It must be given by Power Electronics. MAC address format is: A.B.C.D.E.F. To adjust this address, introduce a value in each of the six parameters that compose it.	Yes
G6.2.9-Primary DNS = 8.8.8.8	0.0.0.0 to 255.255.255.255	Sets the IP to connect to the remote log.	Yes
G6.2.10-Secondary DNS = 8.8.4.4	0.0.0.0 to 255.255.255.255	Sets the IP to connect to the remote log.	Yes
G6.2.11-Number sockets by IP = 2	0-6	Defines the maximum number of connections available for the same IP.	Yes
G6.2.12-Enable UDP response = Disabled	Disabled = 0 Enabled = 1	Enables or disables response when UDP message is sent.	Yes
G6.2.13-IP address golden = 0.0.0.0	0.0.0.0 to 255.255.255.255	IP that always guarantees access to a socket on the computer.	Yes

EN

4.6.3. Subgroup G6.3: [WIP] Ethernet IPv6

Note: These parameters can be seen with Customer role and can be modified with a role equal or superior to Admin (qualified customer).

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.3.1-Automatic IPv6 = No	No Yes	Protocol under development	Yes
G6.3.2a-Assigned IPv6 Addr 1 = 0x00	0x00 to 0xFF	Protocol under development	Yes
G6.3.2b-Assigned IPv6 Addr 2 = 0x00	0x00 to 0xFF	Protocol under development	Yes
G6.3.2c-Assigned IPv6 Addr 3 = 0x00	0x00 to 0xFF	Protocol under development	Yes
G6.3.2d-Assigned IPv6 Addr 4 = 0x00	0x00 to 0xFF	Protocol under development	Yes
G6.3.2e-Assigned IPv6 Addr 5 = 0x00	0x00 to 0xFF	Protocol under development	Yes
G6.3.2f-Assigned IPv6 Addr 6 = 0x00	0x00 to 0xFF	Protocol under development	Yes
G6.3.2g-Assigned IPv6 Addr 7 = 0x00	0x00 to 0xFF	Protocol under development	Yes
G6.3.2h-Assigned IPv6 Addr 8 = 0x00	0x00 to 0xFF	Protocol under development	Yes
G6.3.3-Assigned Ipv6 prefix = 0	1 to 128	Protocol under development	Yes
G6.3.4a-Assig Ipv6 gateway 1 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.4b-Assig Ipv6 gateway 2 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.4c-Assig Ipv6 gateway 3 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.4d-Assig Ipv6 gateway 4 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.4e-Assig Ipv6 gateway 5 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.4f-Assig Ipv6 gateway 6 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.4g-Assig Ipv6 gateway 7 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.4h-Assig Ipv6 gateway 8 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.5a-IPv6 address 1 = 0xFE80	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.5b-IPv6 address 2 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.5c-IPv6 address 3 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.5d-IPv6 address 4 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.5e-IPv6 address 5 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.5f-IPv6 address 6 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.3.5g-IPv6 address 7 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.5h-IPv6 address 8 = 0x21	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.6-IPv6 prefix len = 64	1 to 128	Protocol under development	Yes
G6.3.7a-IPv6 Gateway 1 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.7b-IPv6 Gateway 2 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.7c-IPv6 Gateway 3 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.7d-IPv6 Gateway 4 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.7e-IPv6 Gateway 5 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.7f-IPv6 Gateway 6 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.7g-IPv6 Gateway 7 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes
G6.3.7h-IPv6 Gateway 8 = 0x00	0x00 to 0xFFFF	Protocol under development	Yes

EN

4.6.4. Subgroup G6.4: FTP Guest

Note: Role Admin (qualified customer) or higher is required to display the parameters in this subgroup.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.4.1-Enable FTP Guest = Yes	No Yes	Enables or disables the FTP Guest functionality. **Note:** Only users with role superior to PE Staff can modify this parameter.	Yes
G6.4.2-Guest Username = guest	10 characters	Sets the type of user: User 1. **Note:** Only users with role Admin (qualified customer) or higher can modify this parameter.	Yes
G6.4.3-Guest Password = guest	10 characters	Configures password for user 1. **Note:** Only users with role Admin (qualified customer) or higher can modify this parameter.	Yes

4.6.5. Subgroup G6.5: PPC

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.5.1-Home refs timeout = 4s	Off = 0 Off to 60s	PPC Timeout Home References. **Note:** Only users with role equal or higher than Admin. (qualified customer) can modify these parameters.	Yes
G6.5.2-Comm fault timeout = 60s	Off = 0 Off to 600s	PPC Timeout to show fault of communication Lost. **Note:** Only users with role Power Electronics staff can display and modify this parameter.	Yes
G6.5.3-Home ref = 0.00%	-100.00 to 100.00%	PPC Home Reference. **Note:** Only users with role equal or higher than Admin. (qualified customer) can modify these parameters.	Yes
G6.5.4-PPC Heartbeat Reg coun = 0	0 to 65535	Is a counter of PPC Heartbeat register received. **Note:** Only users with role equal or higher than Admin. (qualified customer) can modify these parameters.	Yes
G6.5.5-Comms state = First heartbeat	First heartbeat Communicating No Communicating	PPC Communication state. **Note:** Only users with role equal or higher than Admin. (qualified customer) can modify these parameters.	Yes
G6.5.6-Ramp selector = Device ramps	No Ramps Device ramps Custom ramps	PPC Ramp Selector. **Note:** Only users with role equal or higher than Admin. (qualified customer) can modify these parameters.	Yes
G6.5.7-Custom ramp op rise = 10.0%/s	0.1 to 3000.0%/s	PPC custom ramp operation rise **Note:** Only users with role equal or higher than Admin. (qualified customer) can modify these parameters.	Yes
G6.5.8-Custom ramp op down = 10.0%/s	0.1 to 3000.0%/s	PPC custom ramp operation down **Note:** Only users with role equal or higher than Admin. (qualified customer) can modify these parameters.	Yes

4.7. Group G8: Control

4.7.1. Subgroup G8.1: References

Note: Only users with role Commissioning or higher can display or modify the parameters in this subgroup.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.1.1-P control mode = I	I P	Allows setting the control mode for the active power. There are two control modes, as shown in the table below: **OPT.** / **STATUS** / **DESCRIPTION** 0 / I / The converter maintains constant the output of active current set on [G8.1.3 Id reference]. 1 / P / The converter maintains constant the output of active power set on [G8.1.2 P reference].	No

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G8.1.2-P reference = 0.00 %</td><td>-100.00 to 100.00 %</td><td rowspan="2">The reference for the active power depends on which control mode has been chosen in [G8.1.1-P control mode].
Thus, parameter G8.1.3 will vary as shown in the following table:
<table><tr><th>G8.1.1</th><th>G8.1.3</th><th>DESCRIPTION</th></tr><tr><td>I</td><td>Id reference</td><td>Sets an active current reference in percentage.</td></tr><tr><td>P</td><td>P reference</td><td>Sets an active power reference in percentage.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G8.1.3-Id reference = 100.00 %</td><td>-100.00 to 100.00 %</td><td>Yes</td></tr>
<tr><td>G8.1.4.-Regulation side = Output</td><td>BESS
PV
Output</td><td>Allows choosing the drive side to apply the references.</td><td>Yes</td></tr>
</table>

4.7.2. Subgroup G8.2: Advanced config

Note: Only users with role Power electronics staff or higher can modify the parameters in this subgroup.

EN

G8.2.1- Current Loops FPGA

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.1.1-Kp I FPGA = 0.0 %	0.0 to 100.0 %	Sets the current loop proportional constant of FPGA. **Note:** Only users with role equal or higher than PE Staff can display and modify this parameter.	Yes
G8.2.1.1.2-Ki I FPGA = 0.0 %	0.0 to 100.0 %	Sets the current loop integral constant of FPGA. **Note:** Only users with role equal or higher than PE Staff can display and modify this parameter.	Yes

G8.2.2- Current Loops uP

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.2.1-Kp I uP = 2.13	0.00 to 1.50	Sets the current loop proportional constant of uP. **Note:** Only users with role equal or higher than PE Staff can display and modify this parameter.	Yes
G8.2.2.2-Ki I uP = 0.001	0.00 to 1.00	Sets the current loop integral constant of uP. **Note:** Only users with role equal or higher than PE Staff can display and modify this parameter.	Yes
G8.2.2.3-Max PI uP = 100.00 %	0.00 to 1.00	Sets the maximum active power loop integral constant of uP. **Note:** Only users with role equal or higher than PE Staff can display and modify this parameter.	Yes

G8.2.3- Gain balance

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.3-Gain balance = 10.0 %	0.0 to 100.0 %	Configuration of FPGA gain balance. **Note:** Only users with role equal or higher than PE Staff can display and modify this parameter.	Yes

4.8. Group G11: Fans

The DCDC GEN3 converter is compatible with the following fans Ebm-Papst, Tesoer and Delta.

When the communication between central control board and the fans is lost a fault will be triggered, depending on the fan zone F449 or F450.

For the ventilation layout, please consult "SV11: Fans" section.

Note: Role PE Staff or higher is required to modify these parameters.

4.8.1. Subgroup G11.1: Detection

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.1.1a-Zone Search = None	None Clean zone Dirty zone	Allows to choose which fan zone needs to be searched. **Note:** This parameter is only displayed if [G11.1.4 = Emb-Papst] and [G11.1.3 = Automatic + zone].	Yes
G11.1.1ab-Fan to search = Fan clean 1	Fan clean zone 1 Fan dirty zone 1 Fan dirty zone 2	Allows to choose which fan needs to be searched. **Note:** This parameter is only displayed if [G11.1.3 = Automatic + individual].	Yes
G11.1.1b-Start Search = No	No Yes	Starts the automatic search algorithm	Yes
G11.1.2-Search State = Disabled	Disabled Not detected Detected Searching Automatically Searching Manually	Indicates the status of the fan search.	Yes
G11.1.3-Detection Mode = Automatic + zone	Automatic + zone Automatic + individual Manual	Allows to choose if the process to search the fans is manual or automatic. **Note:** This parameter is only displayed if [G11.1.4 = Emb-Papst].	Yes
G11.1.4-Fan to detect = Emb-Papst	Emb-Papst Tesoer Delta	Allows which fans provider is going to be used: TESOER, EBM PAPST or DELTA.	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G11.1.5-Search Request = Unknown</td><td>Unknown
Update Master config
Set password 1
Set password 2
Set password 3
Set default config
Reset to apply config
Broadcast id 1
Reset
Search by S/N
Reset by S/N
Check ID
Check default ID
Get configuration
Set default configuration
Iterative id search
Set new ID
Check new ID
Set new Delta baudrate
Check Delta default ID
Check Delta ID
Set new Delta ID</td><td>Sets the search request.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Unknown status in the search process.</td></tr><tr><td>Update Master config</td><td>Command to update the Master configuration.</td></tr><tr><td>Set password 1</td><td>Command to set the password 1.</td></tr><tr><td>Set password 2</td><td>Command to set the password 2.</td></tr><tr><td>Set password 3</td><td>Command to set the password 3.</td></tr><tr><td>Set default config</td><td>Command to set the default configuration.</td></tr><tr><td>Reset to apply config</td><td>Reset to apply</td></tr><tr><td>Broadcast id 1</td><td>Search fan by broadcast.</td></tr><tr><td>Reset</td><td>Command to reset fan.</td></tr><tr><td>Search by S/N</td><td>Search fan by serial number.</td></tr><tr><td>Reset by S/N</td><td>Command to reset the fan by its serial number.</td></tr><tr><td>Check ID</td><td>Check ID fan.</td></tr><tr><td>Check default ID</td><td>Check ID of the fan which are communicating.</td></tr><tr><td>Get configuration</td><td>Command to get the fan configuration.</td></tr><tr><td>Iterative id search</td><td>Iterative search for the fan by its ID</td></tr><tr><td>Set new ID</td><td>Assign new ID to the fan</td></tr><tr><td>Check new ID</td><td>Check that the ID has been correctly assigned to the fan</td></tr><tr><td>Set new Delta baudrate</td><td>Assign the Delta baudrate.</td></tr><tr><td>Check Delta default ID</td><td>Check ID of the Delta fan which are communicating.</td></tr><tr><td>Check Delta ID</td><td>Check ID Delta fan.</td></tr><tr><td>Set new Delta ID</td><td>Assign new ID to the Delta fan</td></tr></table></td><td>Yes</td></tr>
<tr><td>G11.1.6-Fan 1 SN = 0000000000</td><td>0000000000 to 995900ZZZZ</td><td>Sets the serial number of the fan 1.
Note: This parameter is only displayed if [G11.1.9 ≥ 1] and ([G11.1.4 = Emb-Papst] or [G11.1.3 = Manual]).</td><td>Yes</td></tr>
<tr><td>G11.1.7-SN 1 Found = Not detected</td><td>Not detected
Searching
Detected</td><td>Sets if the fan 1 has been detected.
Note: This parameter is only displayed if [G11.1.9 ≥ 1]</td><td>Yes</td></tr>
<tr><td>G11.1.8-Fan 2 SN = 0000000000</td><td>0000000000 to 995900ZZZZ</td><td>Sets the serial number size of fan 2.
Note: This parameter is only displayed if [G11.1.9 ≥ 2] and ([G11.1.4 = Emb-Papst] or [G11.1.3 = Manual]).</td><td>Yes</td></tr>
<tr><td>G11.1.9-SN 2 Found = Not detected</td><td>Not detected
Searching
Detected</td><td>Sets if the fan 2 has been detected.
Note: This parameter is only displayed if (G11.1.1a-Zone Search = Dirty zone) and (G11.1.4-Fan to detect = Emb-Papst))</td><td>Yes</td></tr>
<tr><td>G11.1.10-Fan 3 SN = 0000000000</td><td>0000000000 to 995900ZZZZ</td><td>Allows to set the serial number for fan 3.
Note: This parameter is only displayed if [G11.1.4-Fan to detect] ≠ Tesoer or [G11.1.3-Detection Mode]] = Manual and [G11.9-Availables Fan Nº] ≥ 3.</td><td>Yes</td></tr>
</table>

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.1.11-SN 3 Found = Not detected	Not detected Searching Detected	Indicates whether fan 3 has been detected. **Note:** this parameter is only displayed if any of the following conditions are met. ([G11.1.1a- Zone Search] ≠ Dirty zone) or ([G11.1.1a- Zone Search] ≠ Clean zone) and ([G11.1.3-Detection Mode] = Automatic + Zone) or ([G11.1.4-Fan to detect] = Emb-Papst).	Yes

4.8.2. Subgroup G11.2: Zone assign

G11.2.1-Fan 1

These parameters are only displayed if [G11.9 ≥ 1].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.1.1-Asigned Zone = DU zone	Clean zone Dirty zone	Sets the assigned zone.	Yes
G11.2.1.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK	No
G11.2.1.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: - Ebm-papst. - Tesoer. - Delta.	Yes

G11.2.2-Fan 2

These parameters are only displayed if [G11.9-Availables Fan Nº] ≥ 2.

The group of parameters from [G11.2.2.1] to [G11.2.2.3] are symmetric to the group of parameters from [G11.2.1.1] to [G11.2.1.3] but shows configuration for fan 2. See description in the previous section for fan 1.

G11.2.3-Fan 3

These parameters are only displayed if [G11.9-Availables Fan Nº] = 3.

The group of parameters from [G11.2.3.1] to [G11.2.3.3] are symmetric to the group of parameters from [G11.2.1.1] to [G11.2.1.3] but shows configuration for fan 3. See description in the previous section for fan 1.

4.8.3. Subgroup G11.3: Config clean zone

G11.3.0-Curve temperature

These parameters are only displayed if [G11.3.1-Mode] = Temperature.

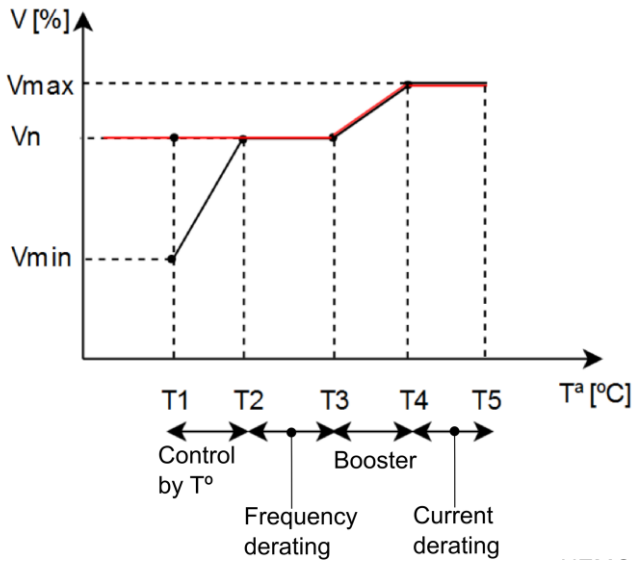
Note: Only users with role Power electronics staff or higher can modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.3.0.0-Set default curve = No	No Yes	Set fans temperature curve to defaults value	Yes
G11.3.0.1-Temp. point 1 = -40ºC	-40 to [G11.3.0.4-Speed point 2] ºC	Set value of temperature at this point of curve	Yes
G11.3.0.2-Speed point 1 = 20%	0 to 100%	Set max speed of fans at this point of curve	Yes
G11.3.0.3-Temp. point 2 = 20ºC	[G11.3.0.1-Temp. point 1] to [G11.3.0.5-Temp. point 3] ºC	Set value of temperature at this point of curve	Yes
G11.3.0.4-Speed point 2 = 20%	[G11.3.0.2-Speed point 1] to 100%	Set max speed of fans at this point of curve	Yes
G11.3.0.5-Temp. point 3 = 58ºC	[G11.3.0.3-Temp. point 2] to [G11.3.0.7-Temp. point 4] ºC	Set value of temperature at this point of curve	Yes
G11.3.0.6-Speed point 3 = 80%	[G11.3.0.4-Speed point 2] to 100%	Set max speed of fans at this point of curve	Yes
G11.3.0.7-Temp. point 4 = 65ºC	[G11.3.0.5-Temp. point 3] to [G11.3.0.9-Temp. point 5] ºC	Set value of temperature at this point of curve	Yes
G11.3.0.8-Speed point 4 = 100%	[G11.3.0.6-Speed point 3] to 100%	Set max speed of fans at this point of curve	Yes
G11.3.0.9-Temp. point 5 = 150ºC	[G11.3.0.7-Temp. point 4] to 150ºC	Set value of temperature at this point of curve	Yes
G11.3.0.10-Speed point 5 = 100%	[G11.3.0.8-Speed point 4] to 100%	Set max speed of fans at this point of curve	Yes

Others

The next picture shows the operation. Black line is temperature and red one is fixed.

EN



SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.3.1-Mode = Temperature	Disabled Fixed Temperature	Configures the fans mode to start working (fixed or temperature mode). **STATUS** / **BIT** / **DESCRIPTION** Disabled / 0 / Disable the zone ventilation. Fixed / 2 / Fans rotate at all times, independent of the equipment status. Temperature / 3 / The fans rotate when the temperature conditions are given, which are configured with the following parameters, following a straight line between the minimum and maximum values of speed and temperature.	Yes
G11.3.2-Setpoint = 80%	0 to 100 %	Configures the speed of the fans. **Note:** This parameter is only displayed if [G11.3.1-Mode = Fixed].	Yes
G11.3.3-Delay off after run = 1200 s	0 to 3600 s	Configures the time after which the fans should stop when they work with Temperature mode. **Note:** This parameter is only displayed if [G11.3.1-Mode = Temperature].	Yes
G11.3.4-Error type = Warning	Fault Warning	Sets the action when the fans fault.	Yes

4.8.4. Subgroup G11.4: Config dirty zone

Note: Only users with role Power electronics staff or higher can modify these parameters.

[G11.4.0.0] to [G11.4.3] has symmetric parameters as [G11.3.0.0] to [G11.3.4] but shows config for dirty zone. See description in previous section.

4.8.5. Subgroup G11.7: Global config

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.7.9-Ramp-up time = 10 s	10 to 65535 s	Defines the time to reach a higher setpoint for Delta and Emb-Papst fans.	Yes
G11.7.10-Ramp-down time = 60 s	60 to 65535 s	Defines the time to reach a lower setpoint for Delta and Emb-Papst fans.	Yes

4.8.6. Subgroup G11.8: Additional Reg Read

This group allows to consult the fan Modbus addresses that are not currently displayed in Power Electronics' remote management tool in groups [G11-Fans] and / or [SV11-Fans].

Note: This subgroup is only displayed if [G1.3-User] ≥ Power electronics staff.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.8.1-Enable = No	No Yes	Allows to enable or disable the fan custom address functionality.	Yes
G11.8.2-Type = Input Register	Input Register Holding Register	Allows to configure the type of register. OPTION: Input Register – Type of register to be consulted is read. OPTION: Holding register – Type of register to be consulted is read / write. **Note:** This parameter is only displayed if [G11.8.1 = Yes].	Yes
G11.8.3-Address = 0x01 Hex	0x00 to 0xFFFF Hex	Allows to configure the address of the fan Modbus map that needs to be asked for. **Note:** This parameter is only displayed if [G11.8.1 = Yes].	Yes
G11.8.4-Register Fan 1 = 0	Not available = 6666 0 to 65535	Sets the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1 = Yes] and [G11.9 ≥ 1].	Yes
G11.8.5-Register Fan 2 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1 = Yes] and [G11.9 ≥ 2].	Yes
G11.8.6-Register Fan 3 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1 = Yes] and [G11.9 ≥ 3].	Yes

4.8.7. Subgroup G11: Others

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.9-Avaliables Fan Nº = 3	0 to 3	Shows the number of the impulsion fans of the converter.	Yes
G11.11-Fans 485 Enable = No	No Yes	Enables or disables the functionality of the fans connected with RS485. **Note:** this parameter has been reorganized in version 2.0.0. As of that version it is [SG4.6-Fans 485 Enable].	Yes
G11.12-Setpoint deviation err = 20 %	10 to 30 %	Indicates the maximum allowed difference between fan setpoint and actual speed. **Note:** Role PE Staff or higher is required to display and modify this parameter.	Yes

4.9. Group G12: SD card

Note: This subgroup is only displayed if [G1.3-User] ≥ Power electronics staff.

All the equipment is equipped with an SD card that is physically connected to the central control board. The main function of the SD card is to store information for further post-processing and to analyse the behaviour of the equipment.

In this group, some general properties of the SD can be configured (such as its enablement), as well as some important data on the card status. In addition, it is possible to configure the custom log. It consists in a log that stores the data of a series of previously configured parameters, as well as everything related to the Blackbox (logs supplied by the module). The following table shows the content of each log.

LOG NUMBER	DESCRIPTION
Log 1	This log saves information about grid current, grid voltage, reactive and active power, limits, and frequency. It is stored every 0,5 seconds. Information is obtained from the following parameters: SV2.1.6-PV current SV2.2.6-BESS current SV2.1.3-PV voltage internal SV2.2.3-BESS voltage internal SV2.1.4-PV voltage externa SV2.2.4-BESS voltage external SV2.1.5-PV power SV2.2.5-BESS power G3.2.2-Discharge I limit (DCL) G3.2.1-Charge I limit (CCL) G3.2.4-P discharge limit G3.2.3-P charge limit G8.1.1-P control mode G8.1.2-P reference G8.1.3-Id reference G8.1.4-Regulation side
Log 2	This log saves information about grid current. Variables from this log will be saved per module. It is stored every 1 seconds. Information is obtained from the following parameters: SV1.1.6-Working mode SV2.1.1-PV half voltage (+) internal SV2.1.2-PV half voltage (-) internal SV2.2.1-BESS half voltage (+) internal SV2.2.2-BESS half voltage (-) internal SV2.3.1-I inductance 1 SV2.3.2-I inductance 2 SV2.3.3-I inductance 3

LOG NUMBER	DESCRIPTION
Log 3	This log saves information about the temperature measurement of the main inductance sensors, ambient temperature of each module PCB, internal temperature of the PCB, air intake temperature of the air entering the equipment, humidity of the PCB of each module, humidity measured on the central control board and temperature of the medium voltage transformer. It is stored every 30 seconds. Information is obtained from the following parameters: SV1.6.6-Inductance PT100 1 SV1.6.7-Inductance PT100 2 SV1.6.8-Inductance PT100 3 SV1.6.5-Module temperature SV1.6.1-Internal temperature SV1.6.3-Admission temperature SV12.1.5-Humidity SV1.6.4-Internal humidity SV11.1.1-Fan 1 coms status SV11.2.1-Fan 2 coms status SV11.3.1-Fan 3 coms status G11.3.2-Fans setpoint / G11.3.2b-Fans setpoint temperature G11.4.2-Fans setpoint / G11.4.2b-Fans setpoint temperature SV11.1.7-Fan 1 speed SV11.2.7-Fan 2 speed SV11.3.7-Fan 3 speed
Log 5	This log saves information about the temperature of each IGBT in each module. It is stored every 10 seconds. Information is obtained from the following parameters: SV12.3.3-Temperature IGBT PV S1 SV12.3.4-Temperature IGBT PV S2 SV12.3.5-Temperature IGBT PV S5 SV12.3.6-Temperature IGBT PV S6 SV12.3.7-Temperature IGBT PV S9 SV12.3.8-Temperature IGBT PV S10 SV12.3.9-Temperature IGBT BESS S3 SV12.3.10-Temperature IGBT BESS S4 SV12.3.11-Temperature IGBT BESS S7 SV12.3.12-Temperature IGBT BESS S8 SV12.3.13-Temperature IGBT BESS S11 SV12.3.14-Temperature IGBT BESS S12

EN

LOG NUMBER	DESCRIPTION
Events Log	The value of these variables is not monitored but the configuration change that occur in these variables. When these variables are modified, the previous value, the new value and the person who modified them are displayed. This log saves information about the current faults and warnings, FPGA status, P and I control mode. Information is obtained from the following parameters: G8.1.1-P control mode SG1.1.11-Selector configuration G2.4.1-Selector status G5.1.1.1-Undervoltage enable. **Note:** from version 1.0.0, this parameter is not available. G5.1.1.2-Threshold PV r G5.1.1.3-Threshold BESS G5.1.1.4-Hysteresis G5.1.2.1-Overvoltage enable. **Note:** from version 1.0.0, this parameter is not available. G5.1.2.2-Threshold PV G5.1.2.3-Threshold BESS G5.1.2.4-Hysteresis G5.3.1-Overcurrent threshold protection G5.4.1.1-Igbts Temp Enable G5.4.1.2-IGBTs temperature prot threshold G5.4.1.3-Temperature IGBTs hysteresis G5.4.2.1.1-Module Temp Enable G5.4.2.1.2-Temperature module threshold fault G5.4.2.1.3-Temperature module hysteresis G5.4.2.2.1-Internal Temp Enable G5.4.2.2.2-Internal temperature fault threshold G5.4.2.2.3-Internal temperature hysteresis G5.4.4.1-Inductance Temp Enable G5.4.4.2-Inductance temp threshold G5.4.4.3-Inductance temp delay G5.4.4.4-Inductance Temp Cri Enable G5.4.4.5-Inductance crash temp threshold G5.4.4.6-Inductance Temp Hysteresis SV1.5.1.1-Up RJ5 status SV1.5.1.2-Up RJ6 status SV1.5.1.3-Up RJ7 status SV1.5.1.4-Up RJ8 status SV1.5.1.5-Up RJ9 status SV1.5.1.6-Up RJ10 status SV1.5.1.7-Up RJ11 status
Custom Log	This log saves information about Modbus address. It is stored every 1 second. Information is obtained from the following parameters: - G12.1.2-Modbus address 1 - G12.1.3-Modbus address 2 - G12.1.4-Modbus address 3 - G12.1.5-Modbus address 4 - G12.1.6-Modbus address 5 - G12.1.7-Modbus address 6 - G12.1.8-Modbus address 7 - G12.1.9-Modbus address 8 - G12.1.10-Modbus address 9 - G12.1.11-Modbus address 10 - G12.1.12-Modbus address 11 - G12.1.13-Modbus address 12 - G12.1.14-Modbus address 13 - G12.1.15-Modbus address 14 - G12.1.16-Modbus address 15

Note: Consult Modbus addresses in the associated MCF.

4.9.1. Subgroup G12.0: General

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G12.0.1-SD card status = OK</td><td>OK
Error
Not formatted
Not mounted
Not initialized
Settings loaded
Settings partially loaded
Error loading settings
Formatted SD</td><td>Displays the status of the SD card.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>OK</td><td>SD card is ready.</td></tr><tr><td>Error</td><td>SD card has an error.</td></tr><tr><td>Not formatted</td><td>SD card is not formatted.</td></tr><tr><td>Not mounted</td><td>SD card is not mounted.</td></tr><tr><td>Not initialized</td><td>SD card has not been initialized.</td></tr><tr><td>Settings loaded</td><td>SD card settings have been loaded.</td></tr><tr><td>Settings partially loaded</td><td>SD card settings have been partially loaded.</td></tr><tr><td>Error loading settings</td><td>Error loading SD card settings.</td></tr><tr><td>Formatted SD</td><td>SD card has been formatted.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G12.0.2-SD operation = Init SD card</td><td>Init SD card
Format SD card
Update status
Disable SD</td><td>Sets the task to execute with the SD card.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Init SD card</td><td>Inits the SD card.</td></tr><tr><td>Format SD card</td><td>Formats the SD card partition.</td></tr><tr><td>Update status</td><td>Updates the status of the card; if it is mounted and accessible, OK will appear.</td></tr><tr><td>Disable SD</td><td>Disables the SD card</td></tr></table></td><td>Yes</td></tr>
<tr><td>G12.0.3-Apply changes = No</td><td>No
Yes</td><td>Allows applying or discarding the modifications made to the SD card.</td><td>Yes</td></tr>
<tr><td>G12.0.4-Logger enable = Yes</td><td>No
Yes</td><td>Besides having the possibility to store a customized set of parameters, the SD card will always keep a record of the following information:
Log 1
Log 2
Log 3
Log 5
Log events</td><td>Yes</td></tr>
<tr><td>G12.0.5-SD test stress = No</td><td>No
Yes</td><td>This parameter is used for development tests.</td><td>Yes</td></tr>
<tr><td>G12.0.6-Used Capacity = 0.00 %</td><td>SD Problem = 100.01; 0.00 to SD Problem %</td><td>Displays the % used capacity of the SD card that is installed in the central control board.</td><td>Yes</td></tr>
<tr><td>G12.0.7-Days to keep files = 30 days</td><td>15 to 90 days</td><td>Sets the number of days that information is to be stored on the SD card. When the set number of days is reached, the oldest files on the card are deleted.</td><td>Yes</td></tr>
<tr><td>G12.0.8-Supervisor time = 15000 ms</td><td>500 to 15000 ms</td><td>Indicates the maximum time allowed for the SD card to sequence correctly. If this time is exceeded, the warning “W180:SD corrupted” is generated.</td><td>Yes</td></tr>
<tr><td>G12.0.9-Supervisor max time = 0 ms</td><td>0 to 65535 ms</td><td>Displays the maximum time used by the SD to perform its sequence.</td><td>Yes</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G12.0.10-SD Task of maximeter = Idle</td><td>Unknown
Idle
RTX timer
Net core
Net eth0
PE osal
Menu Debug
Debug Task Serve
Master Serial
Master Serial
Master Serial
GPIO Tools
Periodic
EEPROM
Network
Autotest
Autotest Polling
Fault Manager
Fault Obs Sub
SD Manager
SD Log Msg
SD Delete
I2C
Event Manager
Warning Manager
Temp Inductances
FPGA Comms
SD supervisor
SD even
SD enable
Control loops
Blackbox
SD Log Sys
Recovery manager</td><td>Displays the most CPU consuming task writing the SD.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Unknown task.</td></tr><tr><td>Idle</td><td>Idle task.</td></tr><tr><td>RTX timer</td><td>RTX time task.</td></tr><tr><td>Net core</td><td>Net core task.</td></tr><tr><td>Net eth0</td><td>Net eth0 task.</td></tr><tr><td>PE osal</td><td>PE osal task.</td></tr><tr><td>Menu Debug</td><td>Menu Debug task.</td></tr><tr><td>Debug Task Serve</td><td>Debug Task Serve task.</td></tr><tr><td>Master Serial 1</td><td>Master Serial 1 task.</td></tr><tr><td>Master Serial 2</td><td>Master Serial 2 task.</td></tr><tr><td>Master Serial 3</td><td>Master Serial 3 task.</td></tr><tr><td>GPIO Tools</td><td>GPIO Tools task.</td></tr><tr><td>Periodic</td><td>Periodic task.</td></tr><tr><td>EEPROM</td><td>EEPROM task.</td></tr><tr><td>Network</td><td>Network task.</td></tr><tr><td>Autotest</td><td>Autotest task.</td></tr><tr><td>Autotest Polling</td><td>Autotest Polling task.</td></tr><tr><td>Fault Manager</td><td>Fault Manager task.</td></tr><tr><td>Fault Obs Sub</td><td>Fault Obs Sub task.</td></tr><tr><td>SD Manager</td><td>SD Manager task.</td></tr><tr><td>SD Log Msg</td><td>SD Log Msg task.</td></tr><tr><td>SD Delete</td><td>SD Delete task.</td></tr><tr><td>I2C</td><td>I2C task.</td></tr><tr><td>Event Manager</td><td>Event Manager task.</td></tr><tr><td>Warning Manager</td><td>Warning Manager task.</td></tr><tr><td>Temp Inductances</td><td>Temp Inductances task.</td></tr><tr><td>FPGA Comms</td><td>FPGA Comms task.</td></tr><tr><td>SD supervisora</td><td>SD supervisora task.</td></tr><tr><td>SD event</td><td>SD evento task.</td></tr><tr><td>SD enable</td><td>SD enable task.</td></tr><tr><td>Control loops</td><td>Control loops task</td></tr><tr><td>Blackbox</td><td>Blackbox task.</td></tr><tr><td>SD Log Sys</td><td>SD Log Sys task.</td></tr><tr><td>Recovery manager</td><td>Recovery manager task.</td></tr></table></td><td>Yes</td></tr>
</table>

4.9.2. Subgroup G12.1: Parameters logger

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G12.1.1-Enable = Yes	No Yes	Enable or disable the converter custom log. The custom log is saved on the SD card. The content of this log is the one configures in this group. Up to 15 variables information can be saved, configured as follows. In each of parameters that appear below, indicate the Modbus address of the parameter. There are some defined by default and others that are free.	Yes
G12.1.2-Modbus address 1 = 1000	Off Off to 9999	Shows the modbus address 1.	Yes
G12.1.3-Modbus address 2 = 1001	Off Off to 9999	Shows the modbus address 2.	Yes
G12.1.4-Modbus address 3 = 1002	Off Off to 9999	Shows the modbus address 3.	Yes
G12.1.5-Modbus address 4 = 1003	Off Off to 9999	Shows the modbus address 4.	Yes
G12.1.6-Modbus address 5 = 6505	Off Off to 9999	Shows the modbus address 5.	Yes
G12.1.7-Modbus address 6 = 6540	Off Off to 9999	Shows the modbus address 6.	Yes
G12.1.8-Modbus address 7 = 6575	Off Off to 9999	Shows the modbus address 7.	Yes
G12.1.9-Modbus address 8 = 6610	Off Off to 9999	Shows the modbus address 8.	Yes
G12.1.10-Modbus address 9 = 2345	Off Off to 9999	Shows the modbus address 9.	Yes
G12.1.11-Modbus address 10 = 2346	Off Off to 9999	Shows the Modbus address 10.	Yes
G12.1.12-Modbus address 11 = 2347	Off Off to 9999	Shows the modbus address 11.	Yes
G12.1.13-Modbus address 12 = 2348	Off Off to 9999	Shows the modbus address 12.	Yes
G12.1.14-Modbus address 13 = 2351	Off Off to 9999	Shows the modbus address 13.	Yes
G12.1.15-Modbus address 14 = 2352	Off Off to 9999	Shows the modbus address 14.	Yes
G12.1.16-Modbus address 15 = 1336	Off Off to 9999	Shows the modbus address 15.	Yes
G12.1.17-Modbus address 16 = 1337	Off = 0 Off to 9999	Shows the modbus address 16.	Yes
G12.1.18-Modbus address 17 = 1338	Off = 0 Off to 9999	Shows the modbus address 17.	Yes
G12.1.19-Modbus address 18 = 1339	Off = 0 Off to 9999	Shows the modbus address 18.	Yes
G12.1.20-Modbus address 19 = 1340	Off = 0 Off to 9999	Shows the modbus address 19.	Yes
G12.1.21-Modbus address 20 = Off	Off = 0 Off to 9999	Shows the modbus address 20.	Yes

4.9.3. Subgroup G12.2: Blackbox

This functionality consists of a logger that feeds FPGA (, which contains information about its electrical measures and other interesting data that can be used to analyze its behavior. This group allows to configure and check this functionality and its status.

The blackbox automatic download allows that when a type of event occurs in the equipment (a fault, module status, etc), the module blackbox or all the blackboxes are automatically downloaded and the result log is saved in the SD card.

In the G12.2 subgroup, users can access to command a manual download and can check the status of the download process.

Note: This subgroup is displayed if [G1.3-User] ≥ Power electronics staff.

EN

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G12.2.1.1-Flt Sys Trigger En = Yes</td><td>No
Yes</td><td>Enables the automatic download functionality of the blackbox when a system fault is triggered.
If enabled, when the type of fault configured in the following parameters occurs in the equipment, then all the blackboxes will be automatically downloaded and the information will be saved in the SD memory.</td><td>Yes</td></tr>
<tr><td>G12.2.1.2-Level Fault Sys Typ = Special</td><td>Special
External emergency
Abrupt stop</td><td>In case automatic download is enabled, this parameter allows to select with which minimum fault type, this functionality is launched.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the switch disconnector and isolate the converter (opens DC and AL low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>External emergency</td><td>Open the switch disconnector and isolate the converter. To manual reset, Qualified client password is required.</td></tr><tr><td>Abrupt stop</td><td>The equipment is stopped as soon as possible by not switching the IGBTs</td></tr></table></td><td>Yes</td></tr>
<tr><td>G12.2.2-Blackbox download = No</td><td>No
Module 1</td><td>Blackbox functionality consists on sending part of the FPGA information to the microprocessor in order to restore it in case of fault.
To consult this information, the user must download the csv file that is stored in the SD.
Configuring the following options defines the data download in the Blackbox:<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>No</td><td>No download.</td></tr><tr><td>Module 1</td><td>Select module to download.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G12.2.3-Download Status = Download Idle</td><td>Download Idle
Download Ongoing</td><td>Indicates the status of the download.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Download Idle</td><td>The download is idle.</td></tr><tr><td>Download Ongoing</td><td>The download is in process.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G12.2.4-Last/Actual Download = All Mod by Manual Download</td><td>0 to 20</td><td>Shows which is the last blackbox that has been downloaded, or which is the blackbox that is downloading, in case the system is in process of downloading.<table><tr><th>OPT.</th><th>STATUS</th></tr><tr><td>0</td><td>No</td></tr><tr><td>1</td><td>All by Manual Download</td></tr><tr><td>2</td><td>Module 1 by Manual Download</td></tr><tr><td>3</td><td>All by Automatic Download</td></tr><tr><td>4</td><td>Module 1 by Automatic Download</td></tr></table></td><td>Yes</td></tr>
<tr><td>G12.2.5-Download Progress M1 = 0 %</td><td>Error Timeout =101;
0 to 100 %</td><td>Shows the status of the download progress of the module inverter 1.</td><td>Yes</td></tr>
</table>

4.9.4. Subgroup G12.3: Vcc protection

This subgroup implements a new protection that disables the SD card when the voltage conditions are not sufficient to ensure that the SD card can function properly.

Note: This subgroup is displayed if [G1.3-User] ≥ Power electronics staff.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G12.3.1-VCC prot thresh = 21.50 V	20.00 to 23.00 V	Configures the minimum voltage below which plus a hysteresis [G12.3.2-VCC prot hyste], the SD card would be disable.	Yes
G12.3.2-VCC prot hyste = 0.50 V	0.10 to 0.50 V	Configures the hysteresis. If the power supply voltage of the central control board is below the sum of this hysteresis and the voltage configured in the previous parameter, the SD card will be disabled.	Yes

EN

4.10. Group G14: Custom modbus

This group allows accessing 120 variables consecutively (even if they are not at consecutive Modbus addresses). This is useful for SCADA that need to read or write certain variables in particular, consecutively.

To configure Modbus addresses, a password must first be set:

- In parameter [SG1.1.12-Custom modbus key], by setting the password 1710, the group is enabled to configure the Modbus addresses. When the password is removed, then the G14 parameters show the actual values of the parameters previously configured. Those can be read if they are read or write and those can be written if they are read / write. Those that are read/write, can be modified from this group and from the corresponding group where they are placed.
- To prevent all roles from accessing and modifying these Modbus addresses, in [SG14.2-Custom Modbus Level] it is configured which role can access this G14 and modify the parameters configured here.

Note: These parameters can be modified only with a role equal or superior to Admin (qualified customer).

4.10.1. Subgroup G14.1: Custom modbus 1

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G14.1.1-Custom 1 = 0	0 to 65535	Defines addresses 1 to 60 of the customized Modbus map. **Note**: Only users with a role equal or higher than Admin (qualified customer) can modify these parameters.	Yes
G14.1.2-Custom 2 = 0			
G14.1.3-Custom 3 = 0			
G14.1.4-Custom 4 = 0			
G14.1.5-Custom 5 = 0			
G14.1.6-Custom 6 = 0			
G14.1.7-Custom 7 = 0			
G14.1.8-Custom 8 = 0			
G14.1.9-Custom 9 = 0			
G14.1.10-Custom 10 = 0			
…			
G14.1.55-Custom 55 = 0			
G14.1.56-Custom 56 = 0			
G14.1.57-Custom 57 = 0			
G14.1.58-Custom 58 = 0			
G14.1.59-Custom 59 = 0			
G14.1.60-Custom 60 = 0			

4.10.2. Subgroup G14.2: Custom modbus 2

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G14.2.1-Custom 61 = 0	0 to 65535	Defines addresses 61 to 120 of the customized Modbus map. **Note:** Only users with a role equal or higher than Admin (qualified customer) can modify these parameters.	Yes
G14.2.2-Custom 62 = 0			
G14.2.3-Custom 63 = 0			
G14.2.4-Custom 64 = 0			
G14.2.5-Custom 65 = 0			
G14.2.6-Custom 66 = 0			
G14.2.7-Custom 67 = 0			
G14.2.8-Custom 68 = 0			
G14.2.9-Custom 69 = 0			
G14.2.10-Custom 70 = 0			
...			
G14.2.55-Custom 115 = 0			
G14.2.56-Custom 116 = 0			
G14.2.57-Custom 117 = 0			
G14.2.58-Custom 118 = 0			
G14.2.59-Custom 119 = 0			
G14.2.60-Custom 120 = 0			

EN

5. SERVICE GROUP PARAMETERS

5

5.1. Group SG1: Service

5.1.1. Subgroup SG1.1: uP service

This group includes variables mainly used by the uP to configure certain functionalities that depend primarily on this device.

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.1.1-New modbus map = No	No Yes	Sets a new Modbus map.	Yes
SG1.1.2-MAC protection key = 0	0 to 65535	Sets the MAC protection key.	Yes
SG1.1.3-Init factory defaults = 0	0 to 65535	Sets all parameters to factory values. Init Factory: This should be avoided unless exceptional circumstances occur. This option deletes all data, including calibration values of currents, voltages. It sets all parameters to their default value, except the MAC address. If the new SW version changes the measurement system, it will be necessary to restore the calibrations and calibrate afterwards. If it does not change, avoid this step if the equipment is not to be calibrated.	Yes
SG1.1.4-Init service group = No	No Yes	Initialize service groups.	Yes
SG1.1.5-NTP IP address = 98.175.203.200	1.0.0.1 to 255.255.255.254	NTP, Network Time Protocol is a protocol for clock synchronization between different equipment. This parameter allows defining the IP address of the NTP server that will be used for time synchronization. IP address format is: A.B.C.D. To adjust this address, introduce a value in each of the four parameters that compose it.	Yes
SG1.1.5b-NTP time request = 5 min	1 to 30 min	Allows to configure a delay for a periodic synchronization through GPS.	Yes
SG1.1.6-System reboot = 0	0 to 65535	Reboot the system once a password is entered. Similar to turning off and on the control.	Yes
SG1.1.10-TCP logger = No	No Yes	This parameter is reserved for future use.	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.1.11-Selector config = Key</td><td>Key
Remote
Key + Remote</td><td>Allows configuring when the bus charge should start (connection to Disconnection Unit), depending on the selector status.
There are two inputs that can be used as a selector: The physical key or a virtual selector that is configured in SG1.1.11 according to the following options.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Key</td><td>It will work according to the position of the physical key. The selector key must be placed and in ON position. The bus will charge as soon as charge conditions are detected.</td></tr><tr><td>Remote</td><td>Regardless the position of the key, bus will charge if will instead function according to the setting of SG1.4.2. [SG1.4.2-Selector status EEPR] = On.</td></tr><tr><td>Key + remote</td><td>It is a combination of the previous options.
The selector key must be placed and in ON position. Besides, [SG1.4.2-Selector status EEPR] must be “On”. Then, the bus charge Will start.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.1.12-Custom modbus key = 0</td><td>0 to 65535</td><td>Password to edit the customized Modbus map.</td><td>Yes</td></tr>
<tr><td>Bootloader_update_action</td><td>0 to 0xFFFF</td><td>Reserved parameter</td><td>Yes</td></tr>
<tr><td>SG1.1.19-Watchdog[4] time = 10000 ms</td><td>500 to 60000 ms</td><td>Sets the watchdog time.</td><td>Yes</td></tr>
<tr><td>SG1.1.20-Watchdog FTP time = 30000 ms</td><td>500 to 60000 ms</td><td>Sets the watchdog FTP time.</td><td>Yes</td></tr>
<tr><td>SG1.1.21-Active watchdog = Regular</td><td>Regular
FTP</td><td>Sets the watchdog active type.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Regular</td><td>Normal watchdog operation. Applies the timeout from parameter [SG1.1.19-Watchdog time].</td></tr><tr><td>FTP</td><td>Special watchdog operation. Applies the timeout from parameter [SG1.1.20-Watchdog FTP time]. Different time allowed before watchdog is skipped when downloading files from SD via FTP</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.1.22-Watchdog max retries = 1000</td><td>0 to 65535</td><td>Number of consecutives retries before restarting in those where the result is wrong.</td><td>Yes</td></tr>
<tr><td>SG1.1.23-Watchdog period = 10 ms</td><td>0 to 65535 ms</td><td>Sets the watchdog period in ms.</td><td>Yes</td></tr>
<tr><td>SG1.1.24-Recovery network = No</td><td>No
Yes</td><td>Allows to detect possible watchdog associated to network tasks and avoid watchdog fault destroying and creating again network tasks.</td><td>Yes</td></tr>
<tr><td>SG1.1.25-Delay network = 0 ms</td><td>0 to 100 ms</td><td>Network delay to apply to slow down the communication.</td><td>Yes</td></tr>
<tr><td>SG1.1.26-Retries IRQ blocked = 1</td><td>1 to 10</td><td>Max number of retries before block system when some IRQ has been blocked.</td><td>Yes</td></tr>
<tr><td>SG1.1.27-Time IRQ blocked = 5 ms</td><td>1 to 10 ms</td><td>Specifies the value that must exceed the difference between the current system tick and the previous system tick.</td><td>Yes</td></tr>
</table>

EN

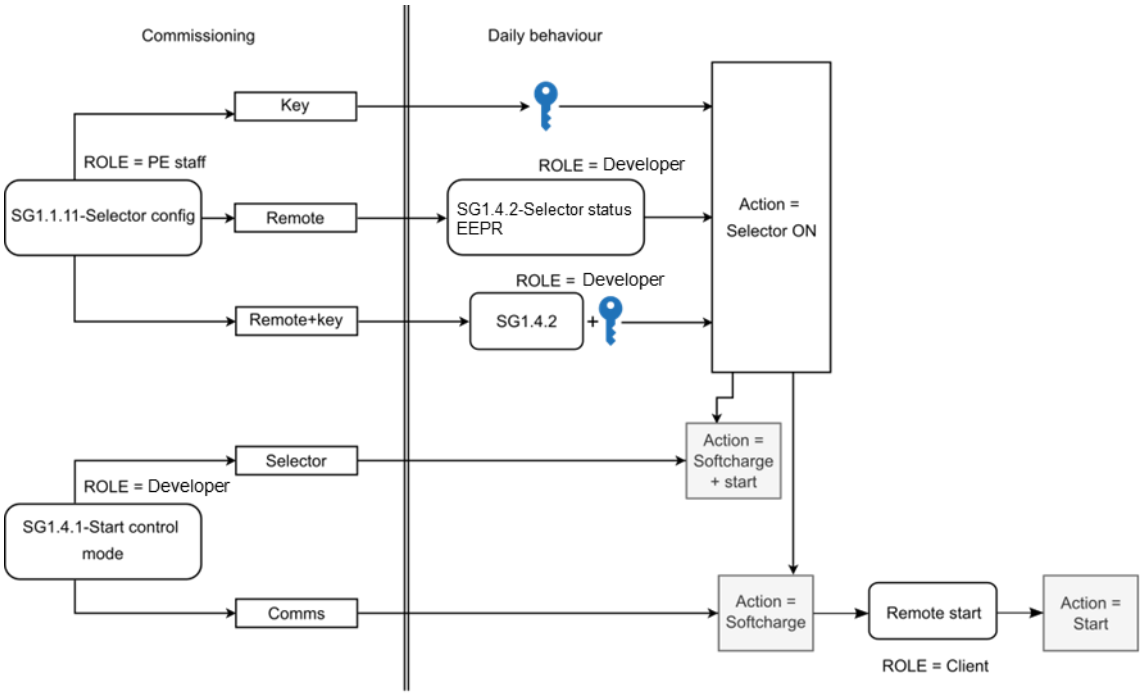
[4] A watchdog is a peripheral that periodically checks that the state of the microcontroller is correct. If not, it restarts it.

5.1.3. Subgroup SG1.4: Start control

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.4.1-Start control mode = Selector</td><td>Selector
Comms</td><td>Selects the mode for start control (selector or communications) and will execute the softcharge.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Selector</td><td>The converter will start automatically once start conditions are fulfilled.
This is the automatic start mode (only by switching SG1.4.2 to ON, if the start conditions are met (G2.X) the converter would reach RUN or ON status.</td></tr><tr><td>Comms</td><td>The converter will start if start conditions are met and once it has received the START command either from the display or from communications.
This is the manual mode, once the selector (SG1.4.2=ON) is activated, the converter performs only the soft load. Then, in order to start injecting power, the StartLocalControl command (register XXXX) or the green button on the display must be applied. the converter performs only the soft load. Then, in order to start injecting power, the StartLocalControl command (register XXXX) or the green button on the display must be applied.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.4.2-Selector status EEPR = Yes</td><td>No
Yes</td><td>Defines whether the selector value should be stored in the EEPROM or not.
Note: This parameter will only be displayed if SG1.1.11 is set to ‘Remote’ or ‘Key + remote’.
Note: from version 2.0.0, this parameter is not available.</td><td>Yes</td></tr>
</table>

The following diagram shows the types of equipment startup indicating how the start-up operation should be performed on the DC/DC converter.



5.1.4. Subgroup SG1.5: Modules configuration

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.5.1-Module 1 state = Enabled	Disabled Enabled Lockable Reserved	Displays the status of the module.	Yes

5.1.5. Subgroup SG1.6: Faults initialisation

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.6.1-General Init = No	No Yes	Fault and warning initialization to default config	Yes
SG1.6.2-Faults Init = No	No Yes	Faults Initialization to default config	Yes
SG1.6.3-Warnings Init = No	No Yes	Warnings Initialization to default config	Yes

EN

5.1.6. Subgroup SG1.7: SD log time

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.7.2-Time LOG 2 = 500 ms	500 to 2000 ms	Configurable time SD log 1.	Yes
SG1.7.2-Time LOG 2 = 500 ms	500 to 2000 ms	Configurable time SD log 2.	Yes
SG1.7.3-Time LOG 4 = 500 ms	500 to 2000 ms	Configurable time SD log 4	Yes

5.1.7. Subgroup SG1.11: Auto diagnosis

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.11.1-Enable auto diag. = Yes	No Yes	Enables self-diagnosis, this parameter returns to default after 24h.	Yes
SG1.11.2-Disable time = 1440 min	1 to 1440 min	Time to reset SG1.11.1 configuration (for debugging only)	Yes

5.2. Group SG2: Debug

This group includes debug variables used internally to debug functionalities.

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

5.2.1. Subgroup SG2.1: uP debug

SG2.1.1: uP Debug vars

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG2.1.1.1-Micro debug 1 = 0	0 to 65535	Microcontroller debug variables for testing and programming.	Yes
SG2.1.1.2-Micro debug 2 = 0	0 to 65535		Yes
SG2.1.1.3-Micro debug 3 = 0	0 to 65535		Yes
SG2.1.1.4-Micro debug 4 = 0	0 to 65535		Yes
SG2.1.1.5-Micro debug 5 = 0	0 to 65535		Yes
SG2.1.1.6-Micro debug 6 = 0	0 to 65535		Yes
SG2.1.1.7-Micro debug 7 = 0	0 to 65535		Yes
SG2.1.1.8-Micro debug 8 = 0	0 to 65535		Yes
SG2.1.1.9-Micro Visual 1 = 0	0 to 65535	Microcontroller visual variables for testing and programming.	Yes
SG2.1.1.10-Micro Visual 2 = 0	0 to 65535		Yes
SG2.1.1.11-Micro Visual 3 = 0	0 to 65535		Yes
SG2.1.1.12-Micro Visual 4 = 0	0 to 65535		Yes
SG2.1.1.13-Micro Visual 5 = 0	0 to 65535		Yes
SG2.1.1.14-Micro Visual 6 = 0	0 to 65535		Yes
SG2.1.1.15-Micro Visual 7 = 0	0 to 65535		Yes
SG2.1.1.16-Micro Visual 8 = 0	0 to 65535		Yes

SG2.1.2: uP Power

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG2.1.2.1-Init micro mode 1 = No data</td><td>No data
Watchdog
Power on
Sw reset
Hardfault</td><td rowspan="3">Sets the initial mode of the microcontroller, according to the following table:

<table><tr><th>OPTION</th><th>NORMATIVE</th></tr><tr><td>No data</td><td>There is no information.</td></tr><tr><td>Watchdog</td><td>Start by watchdog.</td></tr><tr><td>Power on</td><td>Start by power on.</td></tr><tr><td>Sw reset</td><td>Start by software reset.</td></tr><tr><td>Hardfault</td><td>Start by hardfault</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG2.1.2.2-Init micro mode 2 = No data</td><td>No data
Watchdog
Power on
Sw reset
Hardfault</td><td>Yes</td></tr>
<tr><td>SG2.1.2.3-Init micro mode 3 = No data</td><td>No data
Watchdog
Power on
Sw reset
Hardfault</td><td>Yes</td></tr>
<tr><td>SG2.1.2.4-Hardfault LR = 0x00</td><td>0x00 to 0xFFFF0000</td><td>Sets LR (Link Register) register allocation.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.5-Hardfault PC = 0x00</td><td>0x00 to 0xFFFF0000</td><td>Sets PC (Program Counter) register allocation.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.6-Last Running Task = Watchdog</td><td>Unknow
Idle
RTX timer
Net core
Net eth0
Watchdog
Menu Debug
Debug Task Serve
Master Serial 1
Master Serial 2
Master Serial 3
GPIO Tools
Periodic
EEPROM
Network
Autotest
Autotest Polling
Fault Manager
Fault Obs Sub
SD Manager
SD Log Msg
SD Delete
I2C
Event Manager
Warning Manager
Temp Inductance
FPGA Comms
SD supervisor
SD event
SD enable
Broker
PE osal
Blackbox</td><td>This parameter is for internal use by the developer. What each of the options indicates is where the uP has been before jumping the uP watchdog.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.7-Last task ThrID = 0x00</td><td>0x00 to 0xFFFF0000</td><td>Sets the last running task ThreadID before the reset.</td><td>Yes</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG2.1.2.8-Rt Watchdog Enable = 0000000000000000</td><td>0000000000000000 to XXXXXXXXXXXXXXXX</td><td>Sets the Watchdog Enable Mask before Reset.

<table><tr><th>BIT</th><th>DESCRIPCTION</th></tr><tr><td>0</td><td>Modules Init</td></tr><tr><td>1</td><td>Network</td></tr><tr><td>2</td><td>Periodic</td></tr><tr><td>3</td><td>EEPROM</td></tr><tr><td>4</td><td>FPGA Comms</td></tr><tr><td>5</td><td>Reserved</td></tr><tr><td>6</td><td>Reserved</td></tr><tr><td>7</td><td>Diagnostic</td></tr><tr><td>8</td><td>Faults Sub Obs</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG2.1.2.9-Rt Watchdog Update = 0000000000000000</td><td>0000000000000000 to XXXXXXXXXXXXXXXX</td><td>Sets the tasks of the microcontroller that have updated their watchdog.

<table><tr><th>BIT</th><th>DESCRIPCTION</th></tr><tr><td>0</td><td>Modules Init</td></tr><tr><td>1</td><td>Network</td></tr><tr><td>2</td><td>Periodic</td></tr><tr><td>3</td><td>EEPROM</td></tr><tr><td>4</td><td>FPGA Comms</td></tr><tr><td>5</td><td>Reserved</td></tr><tr><td>6</td><td>Reserved</td></tr><tr><td>7</td><td>Diagnostic</td></tr><tr><td>8</td><td>Faults Sub Obs</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG2.1.2.10-Reset Modbus Addr = 0</td><td>0 to 65535</td><td>Sets the address from the last Modbus access before reset.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.11-Reset Modbus Value = 0</td><td>0 to 65535</td><td>Sets the value from the last Modbus access before reset.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.12-Reset Modbus IP = 0.0.0.0</td><td>1.0.0.0 to 255.255.255.255</td><td>Displays the last IP that accessed the equipment before a system reset was triggered.</td><td>Yes</td></tr>
</table>

SG2.1.3: uP Tasks

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG2.1.3.1-Selected Task = Unknow	Unknow Idle RTX timer Net core Net eth0 Watchdog Menu Debug Debug Task Serve Master Serial 1 Master Serial 2 Master Serial 3 GPIO Tools Periodic EEPROM Network Autotest Autotest Polling Fault Manager Fault Obs Sub SD Manager SD Log Msg SD Delete I2C Event Manager Warning Manager Temp Inductances FPGA Comms SD supervisor SD event SD enable Broker PE osal Blackbox	This parameter is for internal use by the developer. What each of the options indicates is where the uP has been before jumping the uP watchdog.	Yes
SG2.1.3.2-CPU Usage = 0.00 %	0.00 to 100.00 %	Sets selected task CPU Use.	Yes
SG2.1.3.3-Max CPU Usage = 0.00 %	0.00 to 100.00 %	Sets selected task CPU use maximum.	Yes
SG2.1.3.4-Reset Max CPU Usage = No	No Yes	Sets selected task CPU reset maximum.	Yes

EN

SG2.1.4: Up Log SYS

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG2.1.4.1-Debug logging level = ERROR log level</td><td>Disabled
DEBUG log level
INFO log level
WARNING log level
ERROR log level
CRITICAL log level</td><td>Allows to select the quantity of depuration frames stored on the SD card.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>This option does not store frames.</td></tr><tr><td>DEBUG log level</td><td>Maximum quantity of frames stored.</td></tr><tr><td>INFO log level</td><td>Frames from the CRITICAL, ERROR, WARNING and this log level are stored.</td></tr><tr><td>WARNING log level</td><td>Frames from the CRITICAL, ERROR and this log level are stored.</td></tr><tr><td>ERROR log level</td><td>Frames from the CRITICAL and this log level are stored.</td></tr><tr><td>CRITICAL log level</td><td>Minimum quantity of frames stored.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG2.1.4.2-Changes in MCF log = Enabled</td><td>Disabled
Enabled</td><td>Enables or disables traces in the SYS log of Modbus variable changes by external Modbus/TCP.</td><td>Yes</td></tr>
<tr><td>SG2.1.4.3-Lim frames/s Modbus = 20</td><td>1 to 100</td><td>Limit of traces/s that can be recorded in SYSLog, to avoid blocking the SD card.
When the limit is exceeded, it is indicated in SYSLog with the message "Max Modbus Events/s reached".</td><td>Yes</td></tr>
<tr><td>SG2.1.4.4-Maximeter frames/s = 0</td><td>0 to 65535</td><td>Value of frames/s reached updated every second.
Variable for debugging the load of Modbus changes that are occurring.</td><td>Yes</td></tr>
<tr><td>SG2.1.4.5-Fault code invalid = 0</td><td>0 to 65535</td><td>Shows fault code invalid.</td><td>Yes</td></tr>
</table>

5.2.2. Subgroup SG2.3: FPGA debug

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG2.3.1-Config FPGA 1 = 0	0 to 65535	FPGA debug variables for testing and programming.	Yes
SG2.3.2-Config FPGA 2 = 0	0 to 65535		Yes
SG2.3.3-Config FPGA 3 = 0	0 to 65535		Yes
SG2.3.4-Config FPGA 4 = 0	0 to 65535		Yes
SG2.3.5-Config FPGA 5 = 0	0 to 65535		Yes
SG2.3.6-Config FPGA 6 = 0	0 to 65535		Yes
SG2.3.13-Visual FPGA 1 = 0	0 to 65535	Displays FPGA debug variables.	Yes
SG2.3.15-Visual FPGA 2 = 0	0 to 65535		Yes
SG2.3.16-Visual FPGA 3 = 0	0 to 65535		Yes
SG2.3.17-Visual FPGA 4 = 0	0 to 65535		Yes
SG2.3.18-Visual FPGA 5 = 0	0 to 65535		Yes
SG2.3.19-Visual FPGA 6 = 0	0 to 65535		Yes

5.3. Group SG3: Remote

This group configures the remote log (Grafana) and enables this functionality. The remote log allows to monitor variables in real time and to be able to graph them to analyse the behaviour of the equipment over time. It is an alternative to the SD.

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

5.3.1. Subgroup SG3.1: Conexion

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG3.1.1-Enable log = Disabled	Disabled Enabled	Enable log.	Yes
SG3.1.2-TimeOut send log = 1 s	1 to 65535 s	Displays the timeout send log.	Yes
SG3.1.3-Server access mode = IP	IP Name	Displays the server access mode.	Yes
SG3.1.4-Port server = 8080	0 to 65535	Port server.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG3.1.5-Server name =	-	Sets the server name.	Yes
SG3.1.6-Subdomain name =	-	Sets the subdomain name.	Yes
SG3.1.7-IP log server = 0.0.0.0	0.0.0.0 to 255.255.255.255	Shows IP log server.	Yes
SG3.1.8-Log websocket status = Not connected	Not connected Connecting Sending header Waiting for response Processing connection Connected Timeout	Shows the log websocket status.	Yes
SG3.1.9-Log websocket timeout = 5 s	1 to 65535 s	Shows the log websocket timeout.	No
SG3.1.10-Local port = 0	0 to 65535	Shows the local port.	Yes

5.3.2. Subgroup SG3.2: Configurable variables

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG3.2.1a-ID_1 local PCB = 0	0 to 65535	Configure which extra addresses need to be displayed in Grafana, in addition to the ones you already have fixed.	Yes
SG3.2.1a-ID_2 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_3 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_4 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_5 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_6 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_7 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_8 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_9 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_10 local PCB = 0	0 to 65535		Yes

5.4. Group SG4: Test mode

This group includes parameters to configure certain conditions when not working with the real equipment, to simulate some states and/or measurements. Also, for testing purposes in during the validation stage.

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG4.3-Open loop = No	No Yes	Configure open loop control.	Yes
SG4.4-Open loop value = 0	0 to 4095	Configures open loop value. **Note:** This parameter will only appear if SG4.3 = Yes.	Yes
SG4.5-Forze auto diagnosis = No	No Yes	Force conditions to start auto diagnosis.	Yes
SG4.6-Fans 485 Enable= Yes	No Yes	Enables or disables the functionality of the fans connected with RS485. **Note:** this parameter, in versions prior to 2.0.0, used to be [G11.11-Fans 485 Enable].	Yes
SG4.7-Pwd Fans Extend ranges = Disabled	Disabled = 0 1 to 65535	Password to extend ventilation ranges. **Note:** This functionality should ONLY be used for testing, never on field.	Yes
SG4.10-Enable ext. loop = Yes	No Yes	Allows enable/disable outer loop.	No
SG4.11-Enable derating temp amb = Yes	No Yes	Allows enable/disable ambient derating. Could be dangerous.	Yes
SG4.12-Enable derating temp IGBTs = Yes	No Yes	Allows enable/disable IGBT temperature derating. Could be dangerous.	No

EN

5.5. Group SG6: Calibrations

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG6.1-Gain current L1 = 1.00	0.80 to 1.20	Gain for current measured in inductance L1	Yes
SG6.2-Gain current L2 = 1.00	0.80 to 1.20	Gain for current measured in inductance L2.	Yes
SG6.3-Gain current L3 = 1.00	0.80 to 1.20	Gain for current measured in inductance L3.	Yes
SG6.4-Gain current BESS = 1.00	0.80 to 1.20	Gain for current measured in BESS side.	Yes
SG6.5-Gain current PV = 1.00	0.80 to 1.20	Gain for current measured in PV side.	Yes
SG6.6-Gain V half(+) PV = 1.00	0.80 to 1.20	Gain for current measured in PV positive semibus.	Yes
SG6.7-Gain V half(-) PV = 1.00	0.80 to 1.20	Gain for current measured in PV negative semibus.	Yes
SG6.8-Gain V half(+) BESS = 1.00	0.80 to 1.20	Gain for current measured in BESS positive semibus.	Yes
SG6.9-Gain V half(-) BESS = 1.00	0.80 to 1.20	Gain for current measured in BESS negative semibus.	Yes
SG6.10-Gain V external bus PV = 1.00	0.80 to 1.20	Gain for external voltage in PV side.	Yes
SG6.11-Gain V ext. bus BESS = 1.00	0.80 to 1.20	Gain for external voltage in BESS side.	Yes
SG6.12-Calib. offset I L1 = 0A	-32768 to 32767A	Autocalibration offset for current sensor in inductance L1.	Yes
SG6.13-Calib. offset I L2 = 0A	-32768 to 32767A	Autocalibration offset for current sensor in inductance L2.	Yes
SG6.14-Calib. offset I L3 = 0A	-32768 to 32767A	Autocalibration offset for current sensor in inductance L3.	Yes
SG6.15-Calib. offset I BESS = 0A	-32768 to 32767A	Autocalibration offset for current sensor in BESS side.	Yes
SG6.16-Calib. offset I PV = 0A	-32768 to 32767A	Autocalibration offset for current sensor in PV side.	Yes

5.6. Group SG7: Diagnosis

This group allows to directly test certain functionalities of the uP to verify the correct functioning of its parts.

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

5.6.1. Subgroup SG7.0: Password

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.0.1-Password 1 = 0	0 to 65535	Sets the password 1.	Yes
SG7.0.2-Password 2 = 0	0 to 65535	Sets the password 2.	Yes

5.6.2. Subgroup SG7.1: EEPROM

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.1.0-Eeprom number = 0	0 to 1	Variables for EEPROM internal diagnostics, for debugging and verification. This parameter is used to configure which EEPROM is to be accessed for testing.	Yes
SG7.1.1-Eeprom offset = 0	0 to 4096	Variables for EEPROM internal diagnostics, for debugging and verification. This parameter is used to configure which EEPROM address is to be accessed for testing.	Yes
SG7.1.2-Eeprom value = 0	0 to 65535	Variables for EEPROM internal diagnostics, for debugging and verification. This parameter is used to write a value in range to the eeprom variable configured in the previous parameter.	Yes
SG7.1.3-Eeprom out = 0	0 to 65535	Returns the value stored in the EEPROM address configured in SG7.1.1.	Yes
SG7.1.4-Eeprom operations = Read	Read Write	Operations to make in EEPROM.	Yes

5.6.3. Subgroup SG7.2: TCP Client

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SG7.2.1- Create Task

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.2.1.0-Create TCP Task = No	No Yes	Create TCP tasks.	Yes
SG7.2.1.2-Ip = 192.168.1.143	1.0.0.0 to 255.255.255.255	TCP Client diagnostic group variable. This variable configures the IP of the server.	Yes
SG7.2.1.3-Puerto = 502	0 to 65535	TCP Client diagnostic group variable. This variable configures the server port.	Yes
SG7.2.1.4-Periodo = 1000 ms	0 to 65535 ms	Configures the read/write period.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.2.1.5-Timeout = 500 ms	0 to 65535 ms	Configures the timeout after request.	Yes
SG7.2.1.5-Retries = 1	0 to 5	Configures the number of attempts per request.	Yes
SG7.2.1.8-Slave ID = 1	0 to 255	Configures the ID identifier of the slave.	Yes
SG7.2.1.9-Creation Result = Correct	Correct Error: Full Error: Parameters	Sets the result of creation task.	Yes
SG7.2.1.10-Task ID Creada = 0	0 to 65535	Configures the task creation ID.	Yes

SG7.2.2- Watch Task

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.2.2.1-Task ID = 0	0 to 65535	Selected ID to display task.	Yes
SG7.2.2.2-Satus = 0	0 to 65535	Indicates the status of the task.	Yes
SG7.2.2.3-Value = 0	0 to 65535	Write/read value of the task.	Yes
SG7.2.2.4-Param copy = No	No Yes	Copy parameter	Yes
SG7.2.2.5-Eliminate Task = No	No Yes	Delete task.	Yes

5.6.4. Subgroup SG7.3: RS485

SG7.3.1- Config

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.3.1.1-Enable = No	No Yes	Enables the RS485 device.	Yes
SG7.3.1.2-Slave id 1 = 0	0 to 255	Configures ID RS485 of the slave 1.	Yes
SG7.3.1.3-Slave id 2 = 0	0 to 255	Configures ID RS485 of the slave 2.	Yes
SG7.3.1.4-Slave id 3 = 0	0 to 255	Configures ID RS485 of the slave 3.	Yes
SG7.3.1.5-Slave id 4 = 0	0 to 255	Configures ID RS485 of the slave 4.	Yes
SG7.3.1.6-Slave id 5 = 0	0 to 255	Configures ID RS485 of the slave 5.	Yes
SG7.3.1.7-Slave id 6 = 0	0 to 255	Configures ID RS485 of the slave 6.	Yes

SG7.3.2- Read/Write

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG7.3.2.0-Comms status 1 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 1.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.1-Comms status 2 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 2.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.2-Comms status 3 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 3.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.3-Comms status 4 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 4.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG7.3.2.4-Comms status 5 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 5.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.5-Comms status 6 = Disabled</td><td>Disable
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 6.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.6-Read device 1 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 1.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.7-Read device 2 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 2.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.8-Read device 3 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 3.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.9-Read device 4 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 4.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.10-Read device 5 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 5.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.11-Read device 6 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 6.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.12-Write device 1 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 1.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.13-Write device 2 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 2.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.14-Write device 3 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 3.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.15-Write device 4 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 4.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.16-Write device 5 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 5.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.17-Write device 6 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 6.</td><td>Yes</td></tr>
</table>

5.6.5. Subgroup SG7.4: Network

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.4.1-Create Log Msg = 0	0 to 65535	Sets the parameter to create the log message.	Yes

5.6.6. Subgroup SG7.5: SD

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.5.1-Enable SD ops = No	No Yes	Enable and mount SD card in target.	Yes
SG7.5.2-Write log version = No	No Yes	Sets the parameter to write log version and Ip in file.	Yes
SG7.5.3-Write log faults = No	No Yes	Sets the parameter to write log faults in SD file.	Yes
SG7.5.4-Write log warnings = No	No Yes	Sets the parameter to write log warning in SD file.	Yes
SG7.5.5-SD status operations = Disabled	Disabled Mounting sd SD correct Write file ok Write file error SD not inserted SD error SD formatting SD format ok SD format error	Shows the result of different operations in SD card. STATUS / DESCRIPTION Disabled: SD disabled. Mounting SD: The SD is being mounted (previous step to save logs on the SD). SD correct: SD is detected correctly and does not have any faults. Write file ok: The SD has been written successfully. Write file error: There has been an error writing to the SD. SD not inserted: No presence of SD has been detected. SD error: The SD has some kind of error. SD formatting: The SD is being formatted. SD format ok: The SD has been formatted correctly. SD format error: The SD has not been formatted correctly.	Yes

5.6.7. Subgroup SG7.7: FPGA Comms

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.7.1-Enable = No	No Yes	Enables or disenables he FPGA communications.	Yes
SG7.7.2-Period Tx = 100 ms	1 to 255 ms	Configures the Tx frame sending period for the diagnostic mode of uP to FPGA, uP to central FPGA communications.	Yes
SG7.7.3-Eco Variable Tx = 0	0 to 65535	Configures the variable to be sent from the uP to the central FPGA.	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG7.7.4-Eco Variable Rx = 0</td><td>0 to 65535</td><td>Returns the value of the variable sent from the uP to the FPGA. That is, what the central FPGA has received it sends to the uP.</td><td>Yes</td></tr>
<tr><td>SG7.7.6-Status = Disabled
Disabled
Comms OK
Timeout
Error CRC
Error Driver</td><td></td><td>Shows the current status of the communication status between uP and central FPGA, of the central control.
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Test has not been produced.</td></tr><tr><td>Comms OK</td><td>Frames between uP and central FPGA sent and received successfully.</td></tr><tr><td>Timeout</td><td>There has been a timeout in the communication between uP and central FPGA, the frame could not be processed by any of the devices.</td></tr><tr><td>Error CRC</td><td>The communication between- uP and central FPGA does not work correctly. CRC failed.</td></tr><tr><td>Error driver</td><td>The communication between- uP and central FPGA does not work correctly. driver failed.</td></tr></table></td><td>Yes</td></tr>
</table>

5.6.8. Subgroup SG7.8: Watchdog

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.8.1-Enable = No	No Yes	Sets if Watchdog is enabled or not.	Yes
SG7.8.2-Refresh = Yes	No Yes	Allows to force a uP reset by watchdog. Allows developer level debugging and verification of the behaviour of the uP.	Yes
SG7.8.3-Update Mask Reset = 1	0 to 1	Sets the task that was running before the watchdog was triggered.	Yes
SG7.8.4-Enable Mask Reset = 1	0 to 1	Sets the task that was running before the watchdog was triggered.	Yes

5.6.9. Subgroup SG7.9: CPU Usage

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.9.1-Current CPU usage = 0.0 %	0.0 to 100.0 %	Indicates the current CPU usage of the converter's uP.	Yes
SG7.9.2-Max CPU usage = 0.0 %	0.0 to 100.0 %	Indicates the maximum CPU usage of the converter's uP.	Yes
SG7.9.3-Reset max CPU usage = No	No Yes	Parameter to reset maximum allowed CPU usage in the converter uP (SG7.9.2).	Yes

5.7. Group SG8: Modbus Memory Access

This group allows to directly access the memory of the microcontroller and to check its correct operation.

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG8.1-Password 1 = 0	0 to 65535	Password 1 must be set to activate diagnostic mode.	Yes
SG8.2-Password 2 = 0	0 to 65535	Password 2 must be set to activate diagnostic mode.	Yes
SG8.4-Var 1 Address High = 0x00 Hex	0x00 to 0xFFFF Hex	Highest part of the memory address to be tested on.	Yes
SG8.4-Var 1 Address Low = 0x00 Hex	0x00 to 0xFFFF Hex	Lower part of the memory address to be tested on.	Yes
SG8.5-Var 1 Tipo = Uint8	Uint8 Uint16	Data type to be configured.	Yes
SG8.6-Var 1 Value = 0	0 to 65535	Value to be written to the memory address configured in the previous parameters (Var1 address high and Var1 address low).	Yes
SG8.7-Var 2 Address High = 0x2400 Hex	0x00 to 0xFFFF Hex	Parameter to be able to set the lower part of the memory address on which the test is to be performed.	Yes
SG8.8-Var 2 Address Low = 0x00 Hex	0x00 to 0xFFFF Hex	Parameter to be able to set the high part of the memory address on which the test is to be performed.	Yes
SG8.9-Var 2 Tipo = Uint8	Uint8 Uint16	Data type to be configured.	Yes
SG8.10-Var 2 Value = 0	0 to 65535	Value to be written to the memory address configured in the previous parameters (Var2 address high and Var2 address low).	Yes
SG8.11-Read memory address = No	No Yes	Action to be taken on the configured memory address.	Yes
SG8.12-Write memory address = No	No Yes	Action to be taken on the configured memory address.	Yes

EN

5.8. Group SG9: Reset counters

There are counters that allow to monitor certain properties of the equipment. From this group it is possible to reset these counters to 0 and start counting again.

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

5.8.1. Subgroup SG9.1: Rst energy counters

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG9.1.1- Rst energy counter = No	No Yes	Allows to reset the energy total counters to 0 by hand.	Yes

5.8.2. Subgroup SG9.2: Rst run time counters

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG9.2.1-Rst run time tot coun = No	No Yes	Allows to reset the run time total counters to 0 by hand.	Yes
SG9.2.2-Rst run time par coun = No	No Yes	Allows to reset the run time partial counters to 0 by hand.	Yes

5.8.3. Subgroup SG9.3: Rst emer stop counters

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG9.3.1-Reset emer stop conns = No	No Yes	Allows to reset the emergency stops counter to 0 by hand.	Yes

5.8.4. Subgroup SG9.4: Rst breakers counter

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG9.4.1-Rst breakers counter = No	No Yes	Allows to reset the breakers counter to 0 by hand.	Yes

5.9. Group SG10: Drive select

This group configures the hardware (from the nominal properties of the equipment, to the enabling of possible options installed, depending on the final application) of the equipment so that the software works according to the equipment characteristics configured.

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

5.9.1. Subgroup SG10.1: General

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.1.1-Password = 0	0 to 65535	With the right password this parameter allows the user to modify the rest of the parameters that are contained in the drive select section (SG10).	Yes
SG10.1.2-Force defaults = No	No Yes	It allows to modify all the parameters contained in the drive select section (SG10) to its default value.	Yes
SG10.1.3-Product type = DCDC GEN3	DCDC GEN 3	Defines the type of converter DCDC GEN 3.	No
SG10.1.4-Converter type = Standalone	Standalone Inverter	Defines the type of the inverter that it is been configured. OPTION: Standalone — DESCRIPTION: Shows when the equipment is controlled by standalone. OPTION: Inverter — DESCRIPTION: Shows when the equipment is controlled by inverter.	No
SG10.1.5-Vdc max = 1500 V	800 to 1500 V	Configures the maximum DC voltage at which the converter will work.	No
SG10.1.6-Idc max = 1130 A	300 to 1200 A	Configures the maximum DC current at which the converter will work.	No
SG10.1.7-Nominal P = 1200 kW	0 to 1200 kW	Configures the nominal active power at which the converter will work.	No
SG10.1.9-Hardware version = HW 1.0	HW 1.0 HW 2.0	Defines the version number of the hardware (there are differences between the hardware version defined).	No
SG10.1.10-Enable inductances = XXX	000 to XXX	Defines the number of inductances are enabled.	Yes

5.9.2. Subgroup SG10.2: Protections

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.2.1-Overcurrent threshol = 950 A	600 to 950 A	Sets the PDINT overcurrent limit for Vdc of the equipment.	No
SG10.2.2-Igbts disc temp thre = 125 ºC	20 to 150 ºC	Allows to set the temperature derating point of the IGBTs in discharge (injecting to mains). **Note:** from version 2.0.0, this parameter is not available.	No

5.9.3. Subgroup SG10.3: Limits

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.3.1-Max I limit = 1200 A	0 to 1200	Maximum inductive current per module.	No
SG10.3.2-Max P limit = 100 %	0 to 100	Maximum active power per module.	No

5.10. Group SG14: Access Settings

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

5.10.1. Subgroup SG14.1: Ethernet

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG14.1.1-White List IP 1 = 192.168.1.143	1.0.0.0 to 255.255.255.255	IP configured in list	Yes
SG14.1.2-User Level IP 1 = Customer	Customer Comissioning Qualified customer Power electronics staff Developer Black List	User level for IP	Yes
SG14.1.3-White List IP 2 = 192.168.1.143	1.0.0.0 to 255.255.255.255	IP configured in list	Yes
SG14.1.4-User Level IP 2 = Customer	Customer Comissioning Qualified customer Power electronics staff Developer Black List	User level for IP	Yes
SG14.1.5-White List IP 3 = 192.168.1.143	1.0.0.0 to 255.255.255.255	IP configured in list	Yes
SG14.1.6-User Level IP 3 = Customer	Customer Comissioning Qualified customer Power electronics staff Developer Black List	User level for IP	Yes
SG14.1.7-White List IP 4 = 192.168.1.143	1.0.0.0 to 255.255.255.255	IP configured in list	Yes
SG14.1.8-User Level IP 4 = Customer	Customer Comissioning Qualified customer Power electronics staff Developer Black List	User level for IP	Yes
SG14.1.9-White List IP 5 = 192.168.1.143	1.0.0.0 to 255.255.255.255	IP configured in list	Yes
SG14.1.10-User Level IP 5 = Customer	Customer Comissioning Qualified customer Power electronics staff Developer Black List	User level for IP	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG14.1.11-White List IP 6 = 192.168.1.143	1.0.0.0 to 255.255.255.255	IP configured in list	Yes
SG14.1.12-User Level IP 6 = Customer	Customer Comissioning Qualified customer Power electronics staff Developer Black List	User level for IP	Yes

5.10.2. Subgroup SG14.2: Custom Modbus

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG14.2.1-Custom Modbus Level = Power electronics staff	600 to 2200 A	Sets the PDINT overcurrent limit for Vdc of the equipment.	Yes

EN

5.11.Group SG15: Communications

This group allows to enable customized communications, the equipment will act as a master, allowing to make customized requests to the selected slaves. Currently it only supports the modbus protocol, in which the slave ID and the type of read request “read holding register” or “input register” can be configured, writing will always be of the “write single register” type. The initial modbus address and the number of consecutive registers to be read can also be configured.

Note: Only users with role Power electronics staff or higher can modify and display these parameters.

5.11.1. Subgroup SG15.0: Custom modbus master

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG15.0.0-Enable master modbus = Disabled</td><td>0 to 3</td><td>Enables or disables master communications.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Custom master requests disabled for all masters.</td></tr><tr><td>Custom master modbus 1 enable</td><td>Custom master requests are enabled for master 1.</td></tr><tr><td>Custom master modbus 2 enable</td><td>Custom master requests are enabled for master 2.</td></tr><tr><td>Custom master modbus 3 enable</td><td>Custom master requests are enabled for master 3.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG15.0.1-Slave id = 1</td><td>0 to 255</td><td>ID slave to device.</td><td>Yes</td></tr>
<tr><td>SG15.0.2-Comms status = Disabled</td><td>0 to 4</td><td>Shows device communication status.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG15.0.3-Function code = Read Holding Register</td><td>0 to 1</td><td>Type of function code to request device.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Read Holding Register</td><td>Type of active modbus request.</td></tr><tr><td>Read Input Register</td><td>Type of active modbus request.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG15.0.4-Modbus address = 0</td><td>0 to 65535</td><td>Modbus address direction to ask slave.</td><td>Yes</td></tr>
<tr><td>SG15.0.5-Number of registers = 1</td><td>1 to 20</td><td>Number of registers to read or write.</td><td>Yes</td></tr>
<tr><td>SG15.0.6-Register 1 = 0</td><td>0 to 65535</td><td>Value to read of register 1.</td><td>Yes</td></tr>
<tr><td>SG15.0.7-Register 2 = 0</td><td>0 to 65535</td><td>Value to read of register 2.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 2.</td><td>Yes</td></tr>
<tr><td>SG15.0.8-Register 3 = 0</td><td>0 to 65535</td><td>Value to read of register 3.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥3.</td><td>Yes</td></tr>
<tr><td>SG15.0.9-Register 4 = 0</td><td>0 to 65535</td><td>Value to read of register 4.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 4.</td><td>Yes</td></tr>
<tr><td>SG15.0.10-Register 5 = 0</td><td>0 to 65535</td><td>Value to read of register 5.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 5.</td><td>Yes</td></tr>
<tr><td>SG15.0.11-Register 6 = 0</td><td>0 to 65535</td><td>Value to read of register 6.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 6.</td><td>Yes</td></tr>
<tr><td>SG15.0.12-Register 7 = 0</td><td>0 to 65535</td><td>Value to read of register 7.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 7.</td><td>Yes</td></tr>
<tr><td>SG15.0.13-Register 8 = 0</td><td>0 to 65535</td><td>Value to read of register 8.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 8.</td><td>Yes</td></tr>
<tr><td>SG15.0.14-Register 9 = 0</td><td>0 to 65535</td><td>Value to read of register 9.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 9.</td><td>Yes</td></tr>
<tr><td>SG15.0.15-Register 10 = 0</td><td>0 to 65535</td><td>Value to read of register 10.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 10.</td><td>Yes</td></tr>
<tr><td>SG15.0.16-Register 11 = 0</td><td>0 to 65535</td><td>Value to read of register 11.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 11.</td><td>Yes</td></tr>
<tr><td>SG15.0.17-Register 12 = 0</td><td>0 to 65535</td><td>Value to read of register 12.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 12.</td><td>Yes</td></tr>
<tr><td>SG15.0.18-Register 13 = 0</td><td>0 to 65535</td><td>Value to read of register 13.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 13.</td><td>Yes</td></tr>
<tr><td>SG15.0.19-Register 14 = 0</td><td>0 to 65535</td><td>Value to read of register 14.
Note: this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 14.</td><td>Yes</td></tr>
</table>

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG15.0.20-Register 15 = 0	0 to 65535	Value to read of register 15. **Note:** this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 15.	Yes
SG15.0.21-Register 16 = 0	0 to 65535	Value to read of register 16. **Note:** this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 16.	Yes
SG15.0.22-Register 17 = 0	0 to 65535	Value to read of register 17. **Note:** this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 17.	Yes
SG15.0.23-Register 18 = 0	0 to 65535	Value to read of register 18. **Note:** this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 18.	Yes
SG15.0.24-Register 19 = 0	0 to 65535	Value to read of register 19. **Note:** this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 19.	Yes
SG15.0.25-Register 20 = 0	0 to 65535	Value to read of register 20. **Note:** this parameter is only displayed if [SG15.0.5-Number of registers] ≥ 20.	Yes

EN

6. MODBUS COMMUNICATION

6

6.1. Best practices for Modbus communication

Modbus TCP is a widely used industrial communication protocol that enables efficient data exchange over Ethernet networks. To ensure optimal performance, stability, and reliability, it is crucial to follow best practices when implementing Modbus TCP in an application. This section highlights key considerations and recommendations for using Modbus TCP efficiently.

1) Maintain a single persistent socket connection:

One of the most important best practices when using Modbus TCP is to maintain a single, persistent socket connection for all Modbus requests instead of opening and closing sockets for each transaction. This approach has several advantages:

- Reduces network overhead and improves performance.
- Avoids excessive connection establishment delays.
- Minimizes resource consumption on both client and server.

Implementation tip:

- Open a single TCP socket at the beginning of the communication session and reuse it for all subsequent Modbus requests.
- Only close the socket when the communication session is complete or if an error occurs that necessitates a reconnect.

2) Ensure a minimum delay of 50 milliseconds between requests

To prevent network congestion and avoid overloading the Modbus server, a minimum delay of 50 milliseconds should be maintained between consecutive Modbus requests. Rapid consecutive polling can lead to increased CPU load and potential data loss due to excessive traffic.

Implementation tip:

- Implement a request scheduler with a minimum inter-request delay of 50 milliseconds.
- Use a non-blocking approach to manage timing efficiently.

3) Optimize read operations for non-consecutive addresses

In applications where it is necessary to read multiple non-consecutive addresses periodically, issuing separate read requests for each address can lead to increased communication overhead and inefficiencies. Instead, the use of Custom Modbus mapping is recommended. This feature involves grouping non-consecutive registers into a contiguous range in the Modbus server so they can be read in a single request.

Implementation tip:

- Configure the Modbus server to map non-contiguous registers into a contiguous range.
- Use a single read request to fetch all required data, reducing the number of transactions.

4) Disable Nagle's algorithm for low latency communication

Nagle's Algorithm is designed to reduce the number of small packets sent over a network by buffering data before transmission. However, in Modbus TCP applications, this can introduce unwanted latency when multiple requests are sent in quick succession. To achieve real-time performance, it is recommended to disable Nagle's Algorithm.

Implementation tip:

- Enable the TCP_NODELAY flag on the socket to disable Nagle's Algorithm.
- This ensures that each Modbus request is transmitted immediately without delay.
- Sending requests immediately also prevents the CPU from receiving large packets containing multiple requests, reducing processing overhead and improving response times.
- Disabling Nagle's Algorithm is particularly important in time-sensitive industrial applications where low-latency communication is required.

5) Conclusion

By following these best practices, Modbus TCP applications can achieve improved efficiency, reliability, and reduced network load. Maintaining a persistent socket, ensuring appropriate timing between requests, and optimizing read operations for non-consecutive addresses will result in a more robust and stable Modbus TCP communication system.

For further technical assistance, refer to the Modbus specification or contact Power Electronics for support.

6.2. Summary of Modbus addresses

Please refer to associated MCF for the complete list of parameters and their configuration details.

6.3. Remote Control Functions

START	
Screen	-
Range	0 – 1
Modbus address	**40113**
Modbus range	0 to 1
Read / Write	Yes
Description	Manual start: Send the start command to the converter through the communications network.

STOP	
Screen	-
Range	0 – 1
Modbus address	**40114**
Modbus range	0 to 1
Read / Write	Yes
Description	Sends the stop command to the converter through the communications network.

RESET	
Screen	-
Range	0 – 1
Modbus address	**40115**
Modbus range	0 to 1
Read / Write	Yes
Description	Sends a reset command to the converter through the communications network.

EMERGENCY STOP	
Screen	-
Range	0 – 1
Modbus address	**43038**
Modbus range	0 to 1
Read / Write	Yes
Description	Sends an emergency stop command to the converter through the communications network. To perform a remote emergency stop, the following steps must be followed: 1. Enter the password 1492 at Modbus address 43038. 2. Execute the emergency stop. Write the value 1 to Modbus address 43039. 3. Once the emergency stop has been executed, the Modbus address 43038 will return the value 1571 as confirmation of the action. The device will also trigger the fault [F56 Remote emergency stop] corresponding to the remote emergency stop.

PPC HEARTBEAT	
Screen	-
Range	0 – 1
Modbus address	**41437**
Modbus range	0 to 65535
Read / Write	Yes
Description	Sends a heartbeat PPC command to the converter through the communications network.